Advanced scheduling

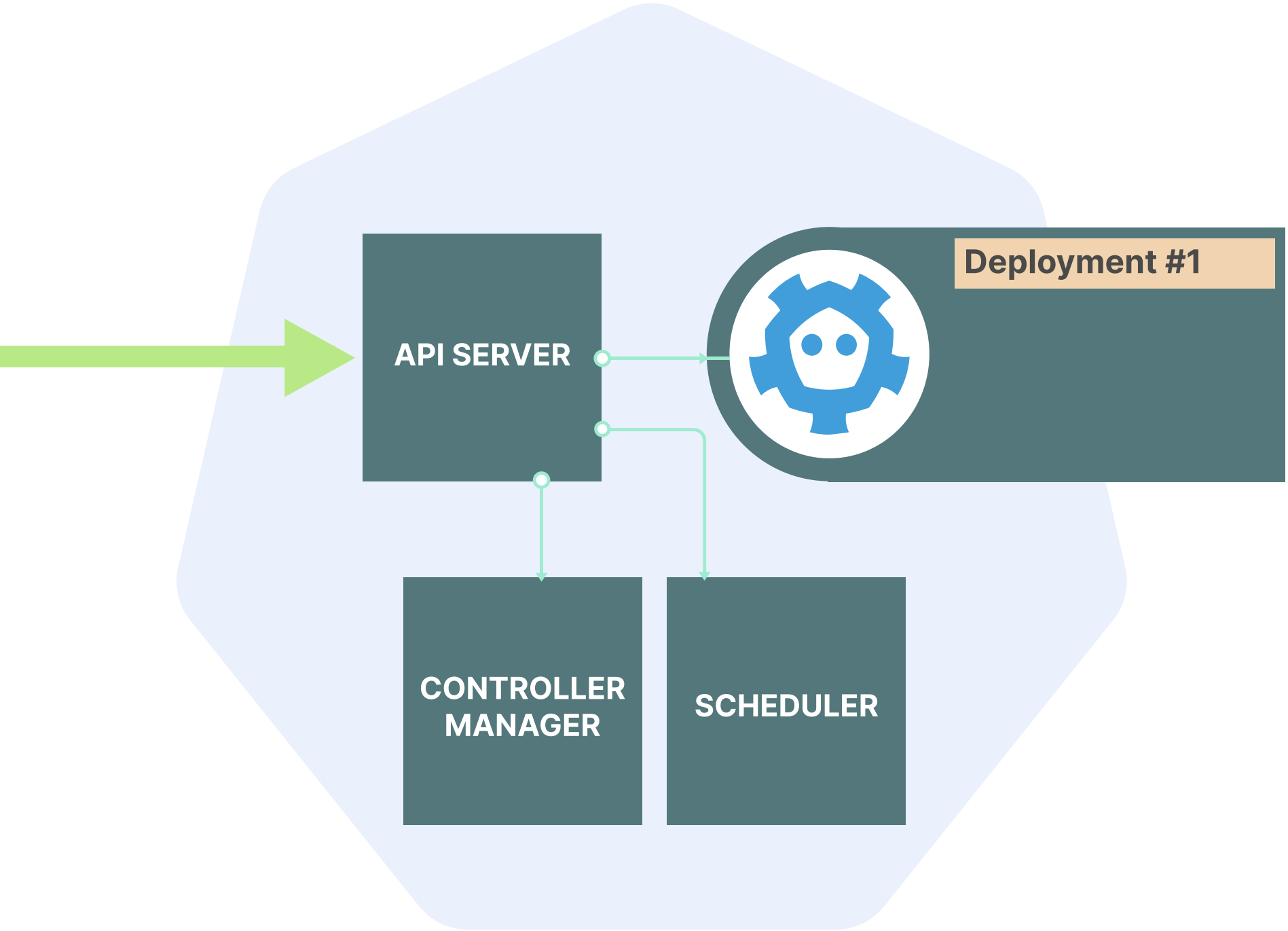
09

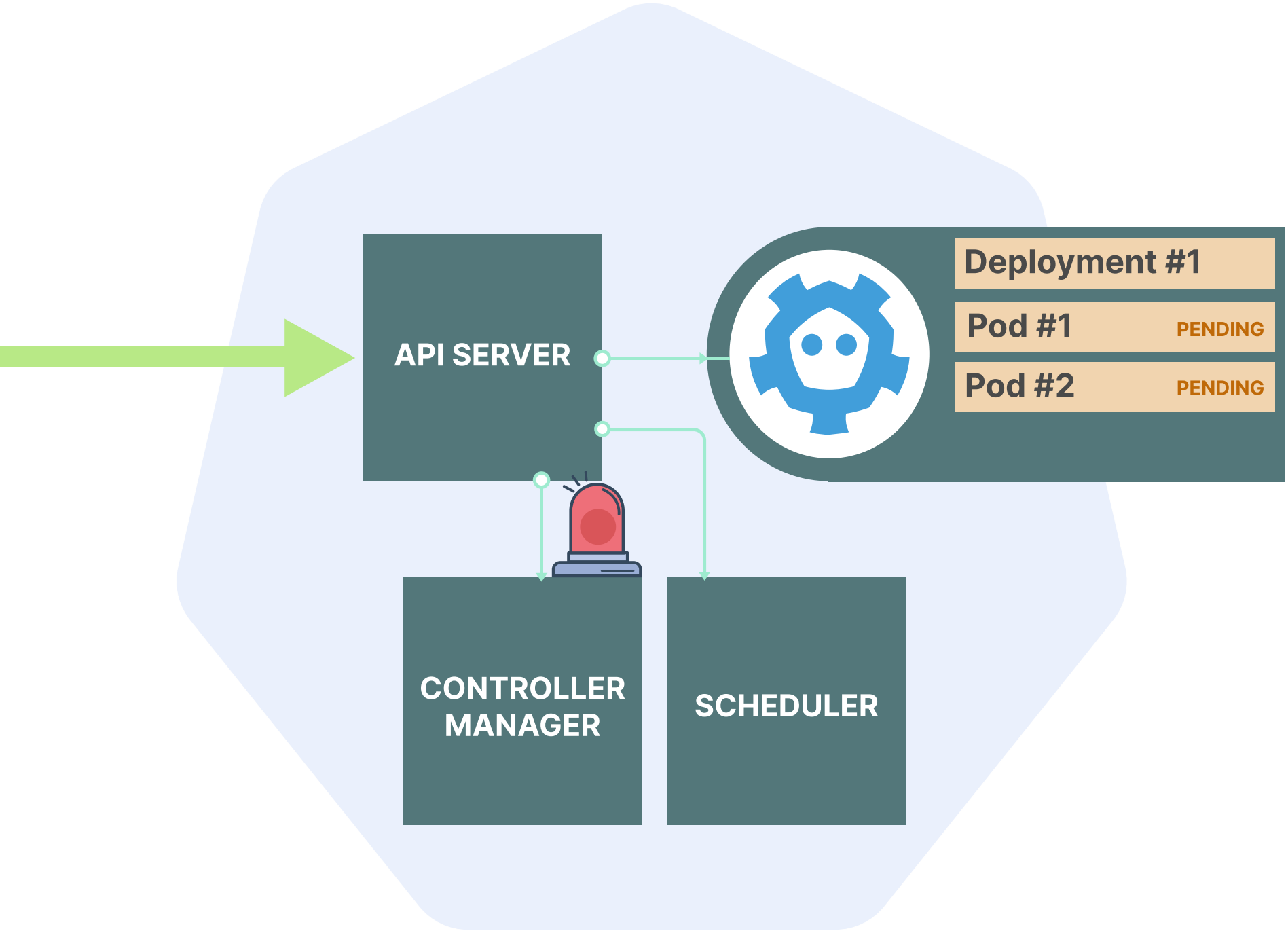
Kubernetes scheduler

Create a Deployment with 2 replicas





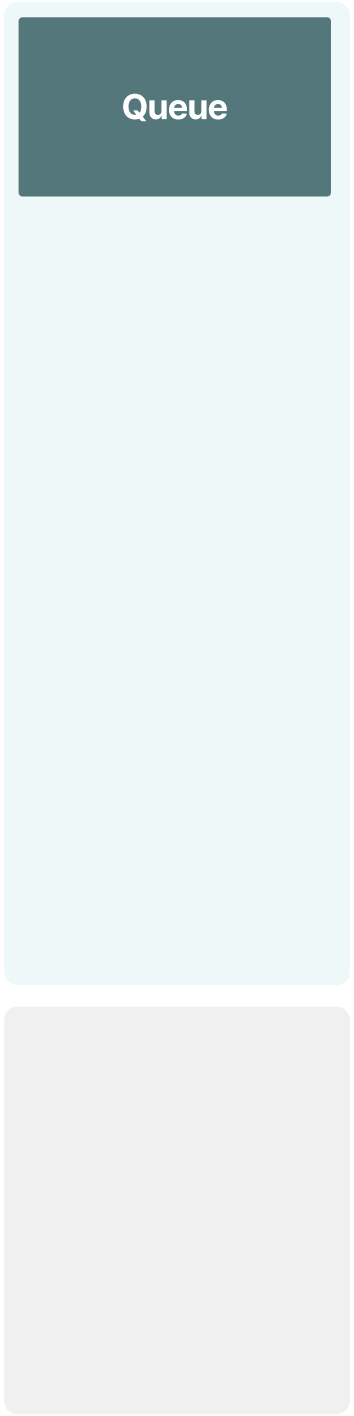

API SERVER
Deployment #1
CONTROLLER MANAGER
SCHEDULER


API SERVER
Deployment #1
Pod #1
PENDING
Pod #2
PENDING
CONTROLLER MANAGER
SCHEDULER

Scheduling & binding

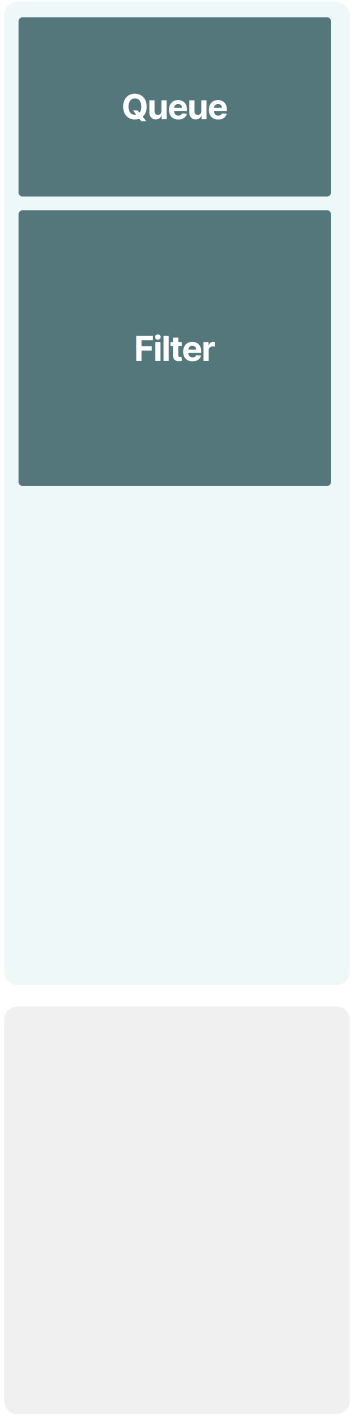
Scheduling

Binding



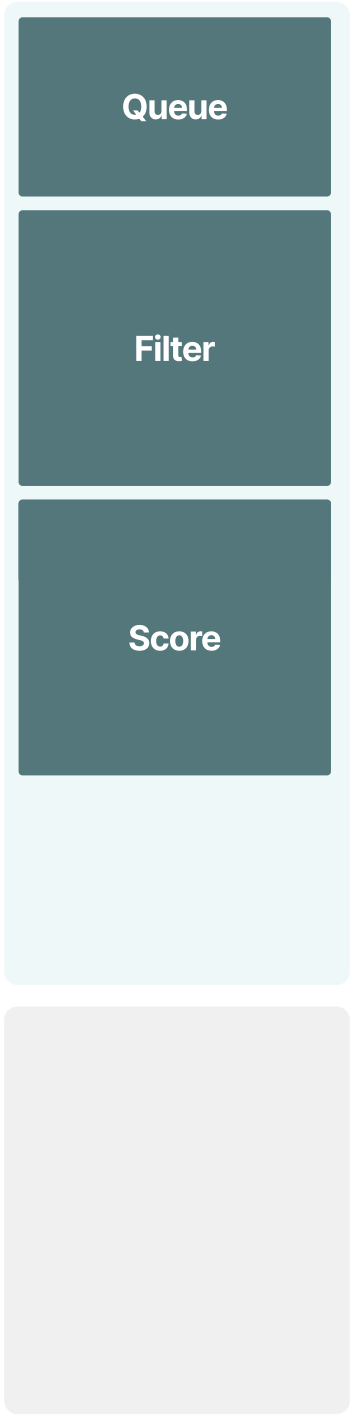
Scheduling

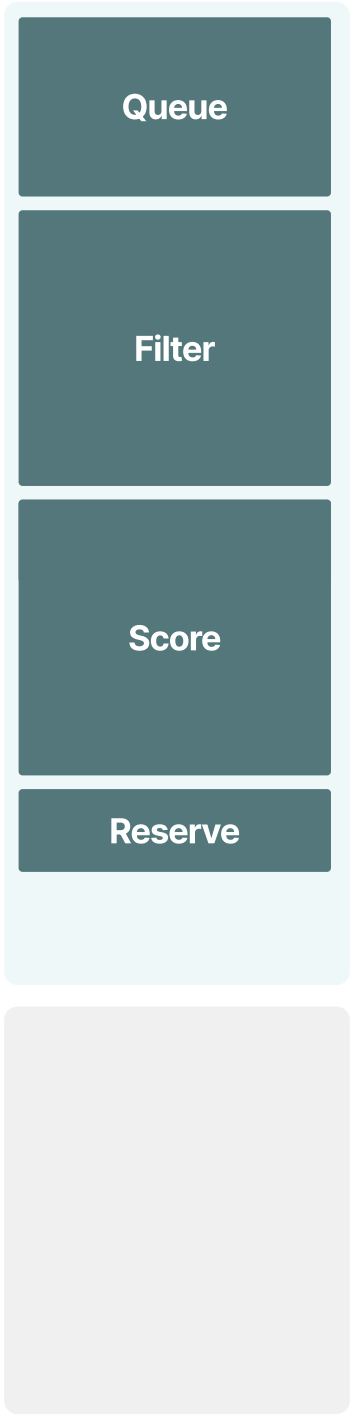
Binding



Scheduling

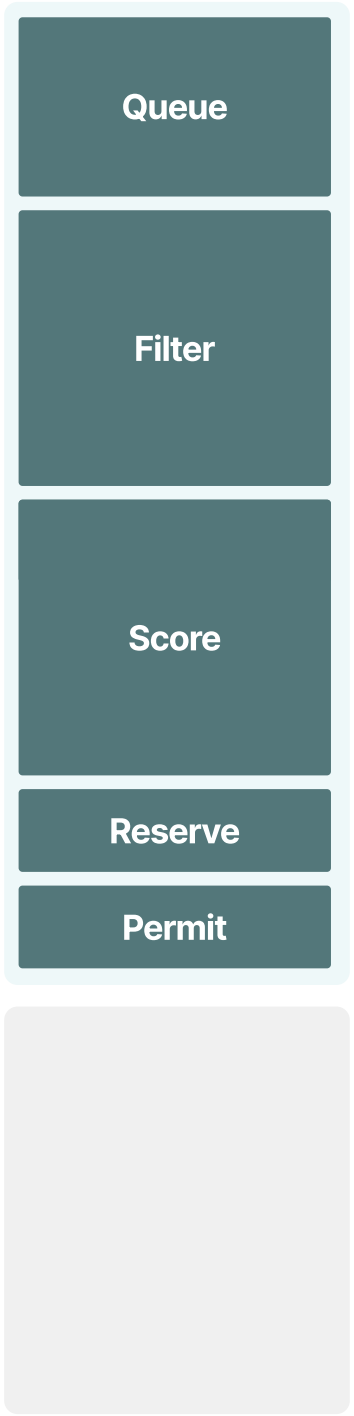
Binding


Queue
Filter
Score
Scheduling
Binding



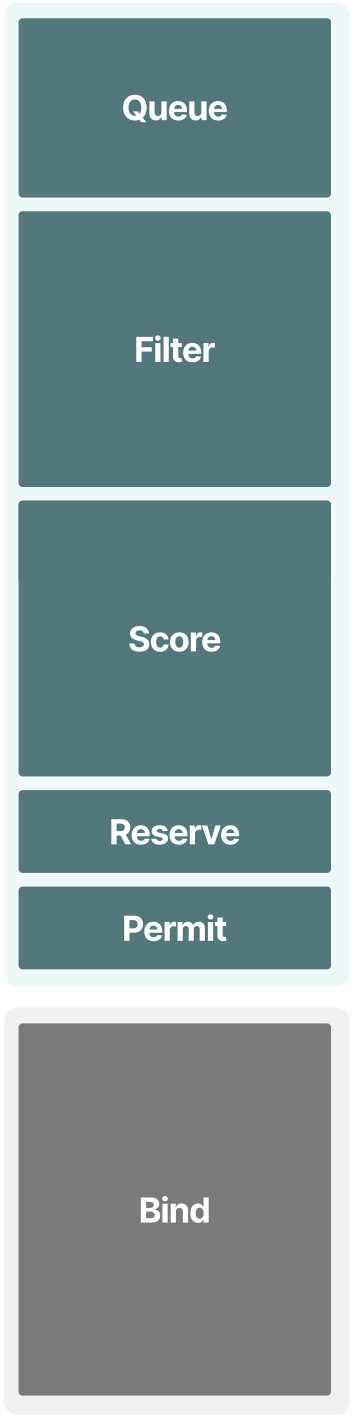
Scheduling

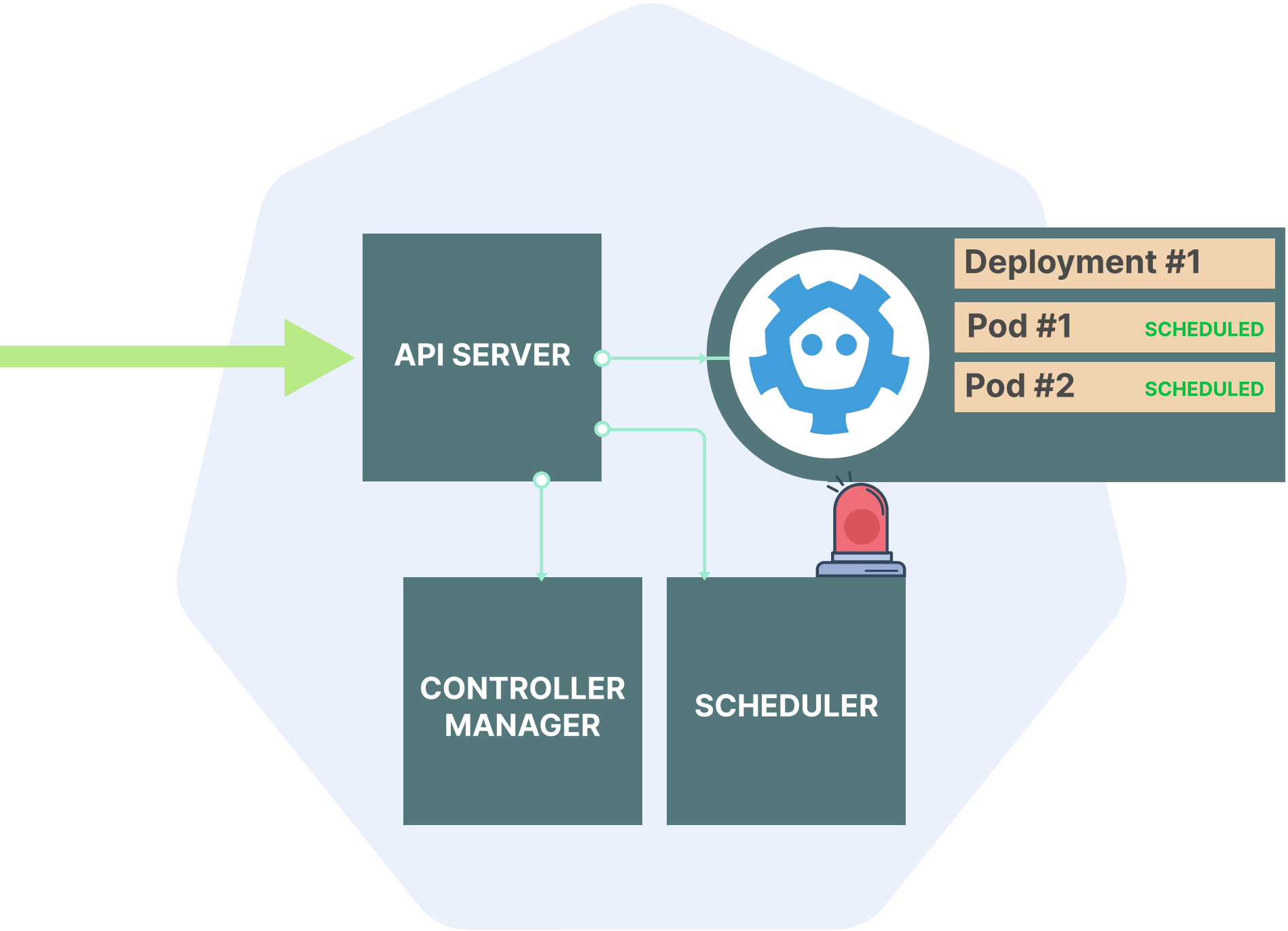
Binding



Scheduling

Binding

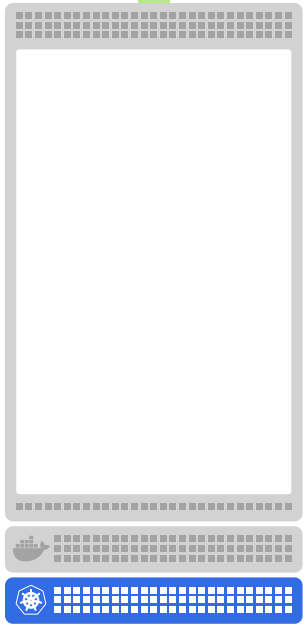

Queue
Filter
Score
Reserve
Permit
Bind
Scheduling
Binding


API SERVER
Deployment #1
Pod #1
SCHEDULED
Pod #2
SCHEDULED
CONTROLLER MANAGER
SCHEDULER

Deploy 3 replicas, please







Is there anything for me?

Scheduling & binding in practice

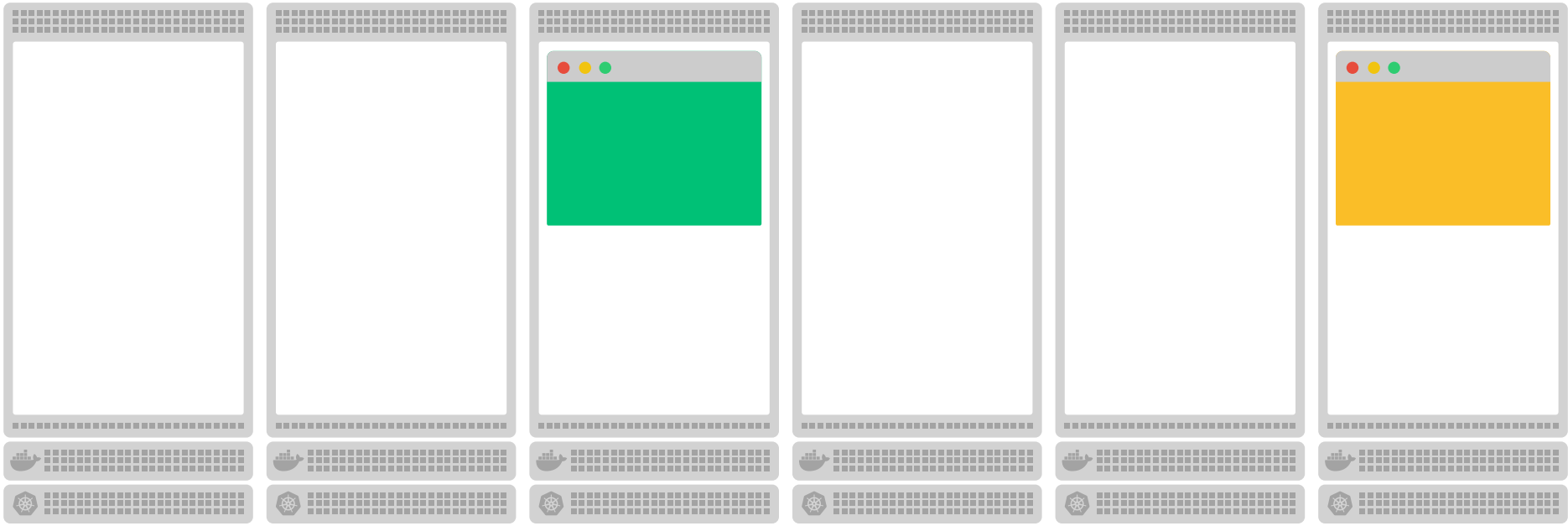
Create a Deployment with:

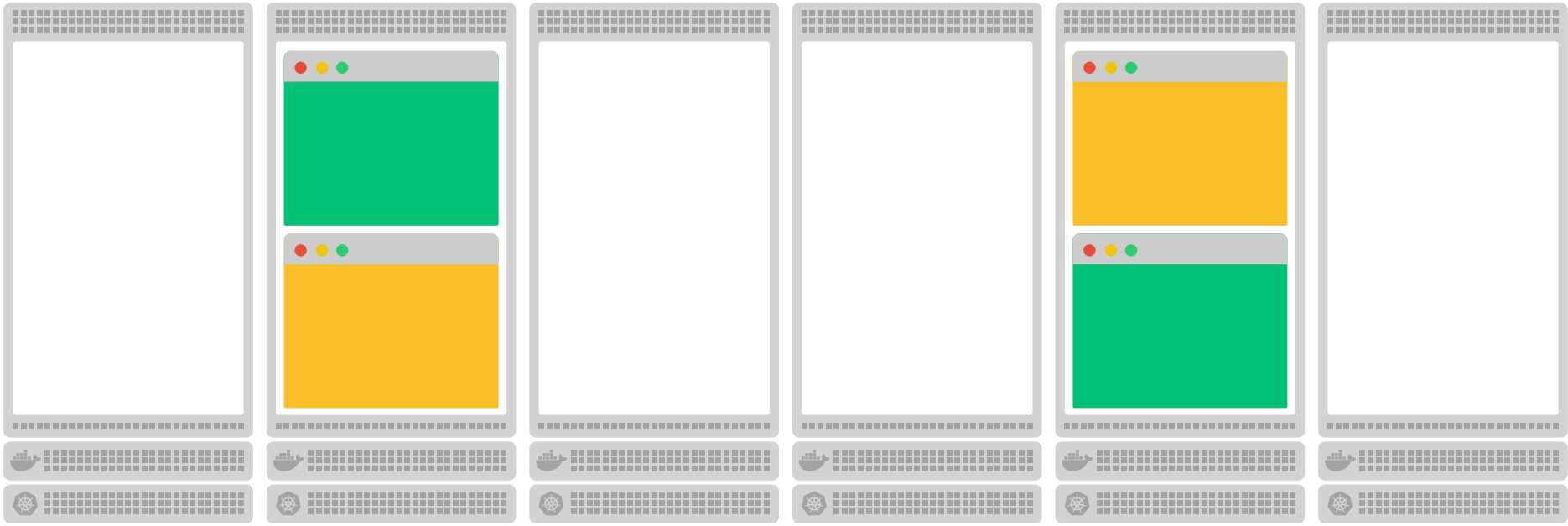


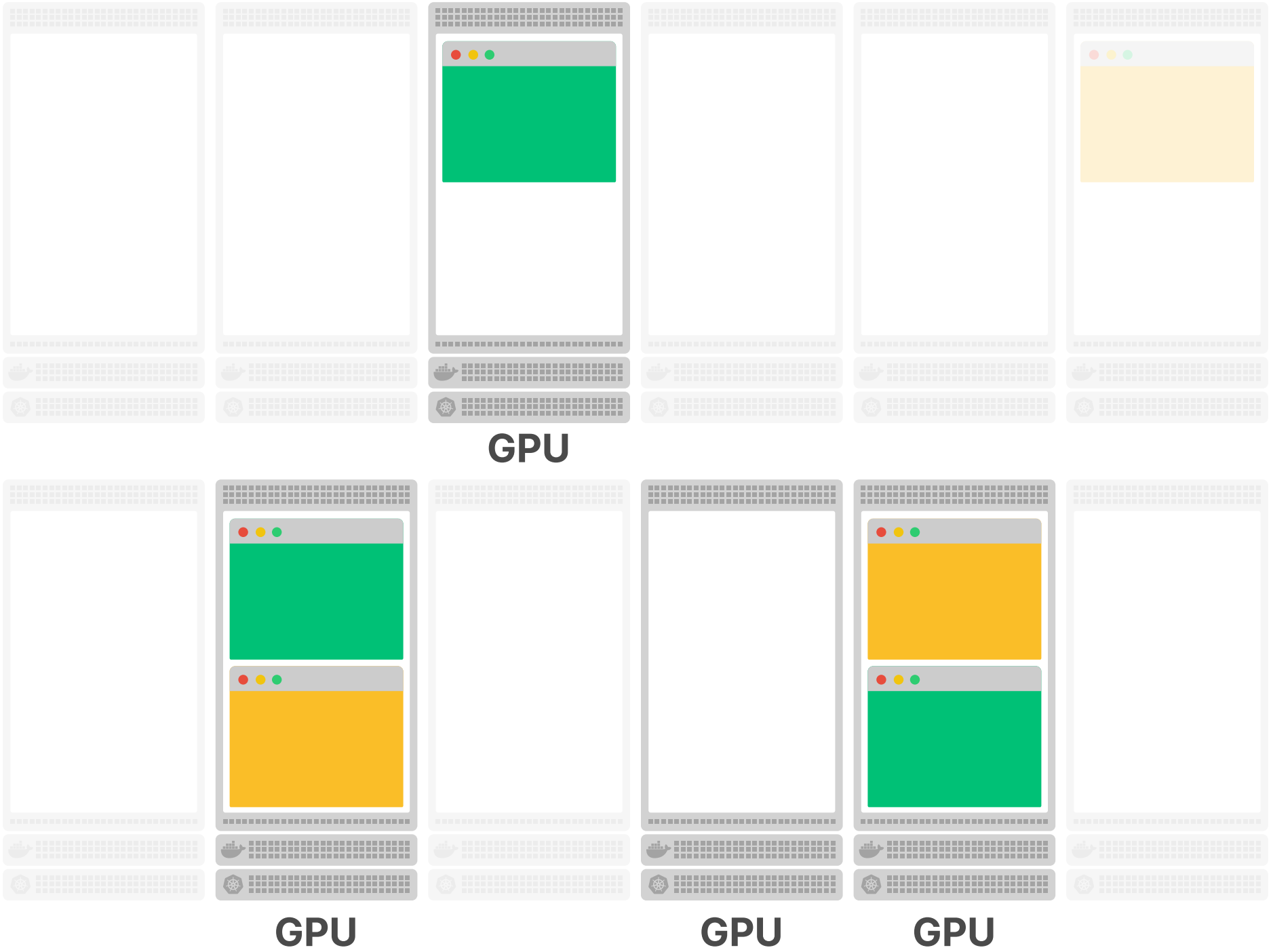
- 1 replica
- A GPU
- 1 CPU and 1 GB of memory

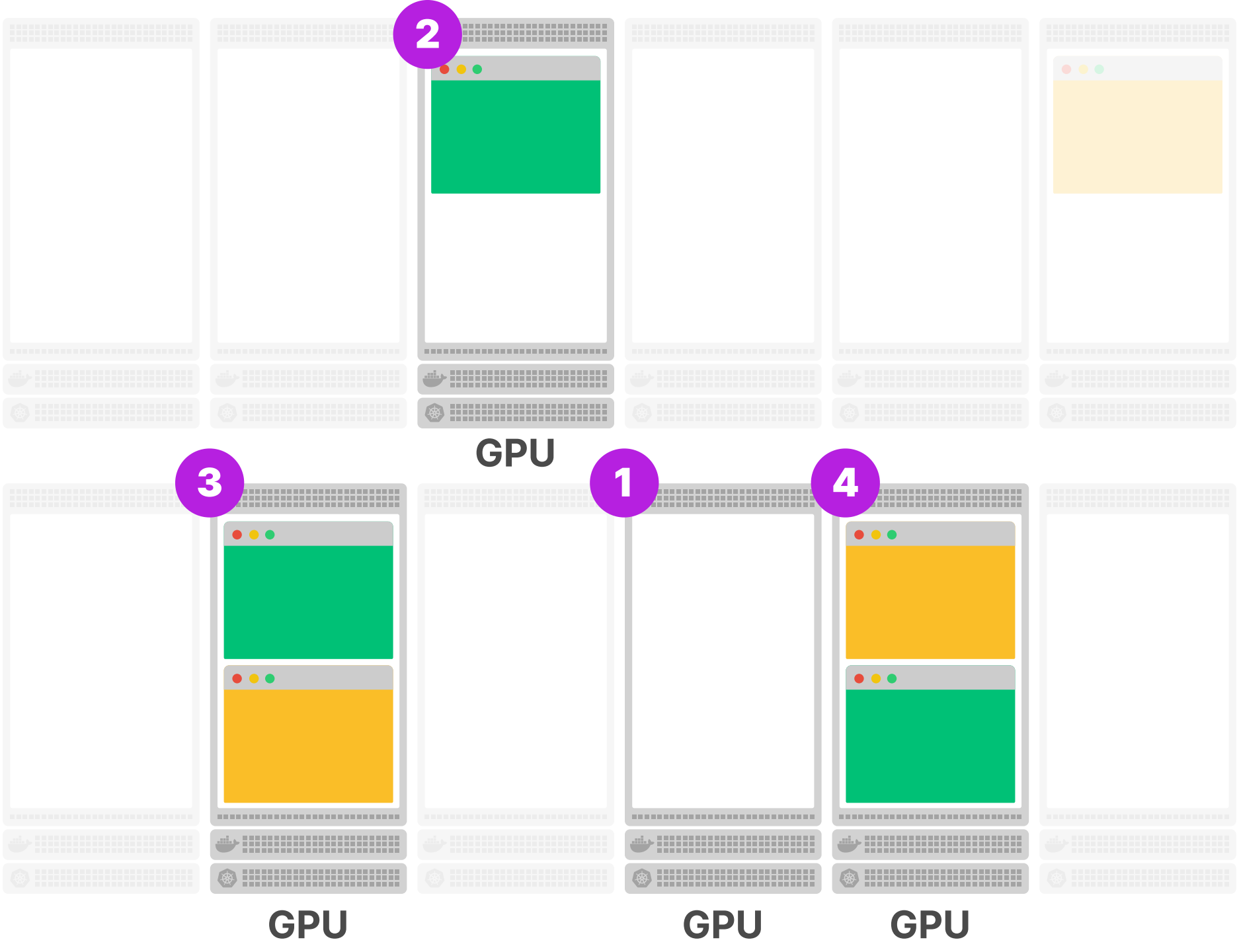


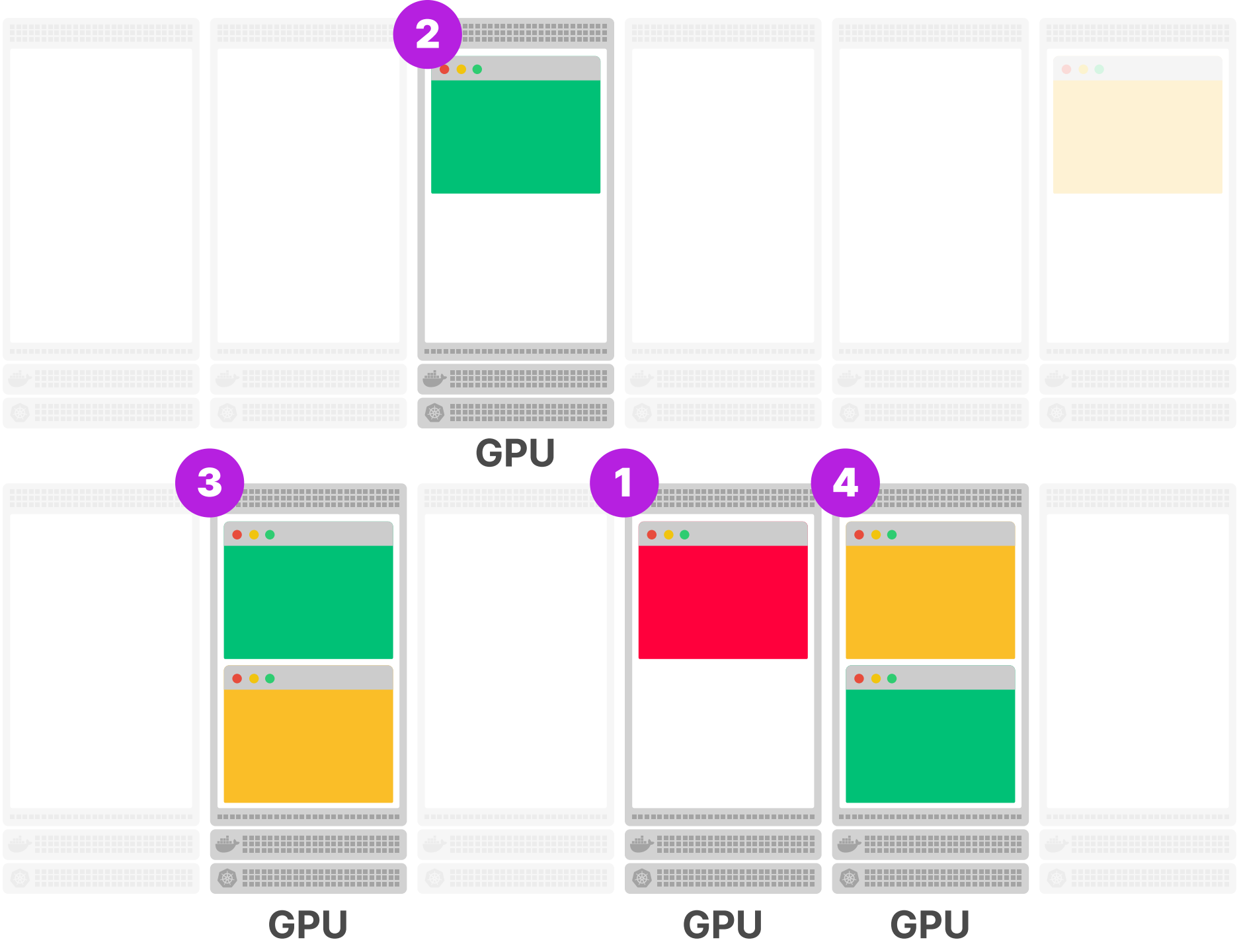



GPU

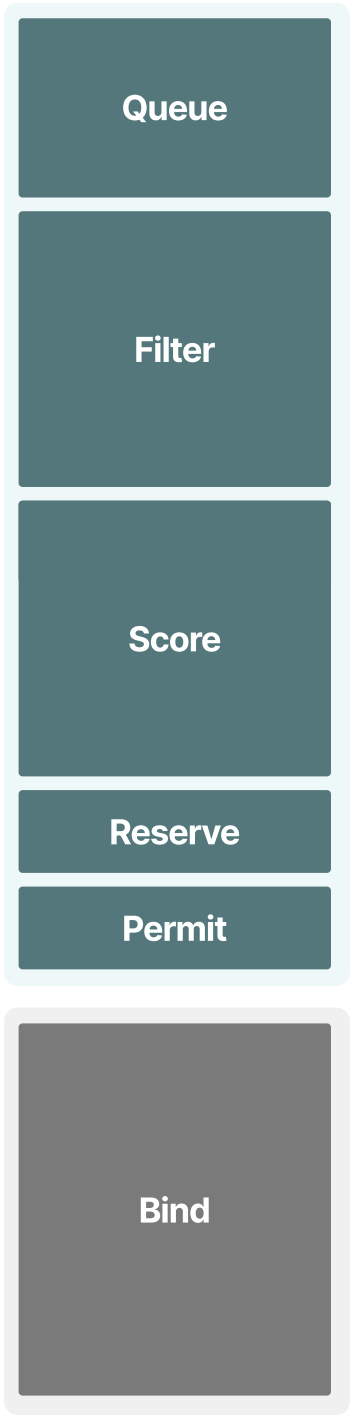

GPU
GPU
GPU


GPU
GPU
GPU
GPU


2
GPU
3
1
4
GPU
GPU
GPU

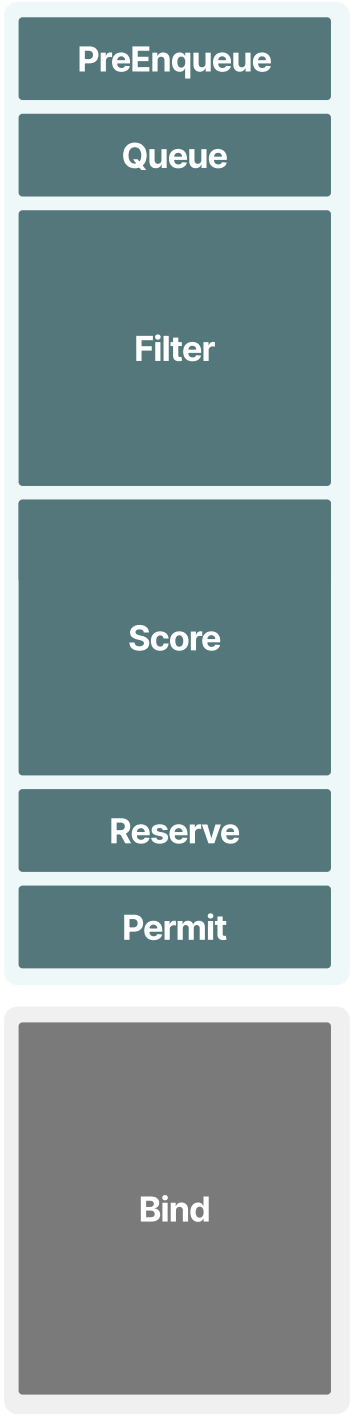

2
GPU
3
1
4
GPU
GPU
GPU

Pluggable and extensible



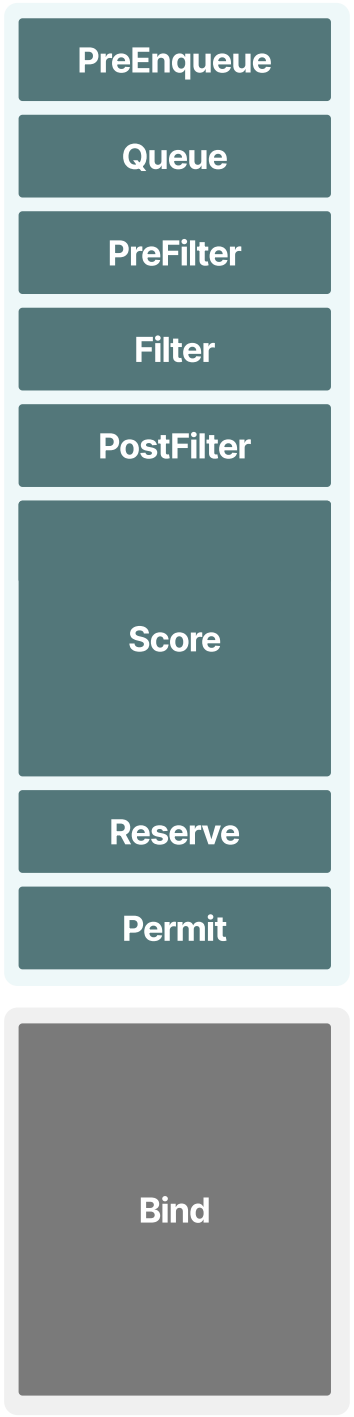
Scheduling

Binding



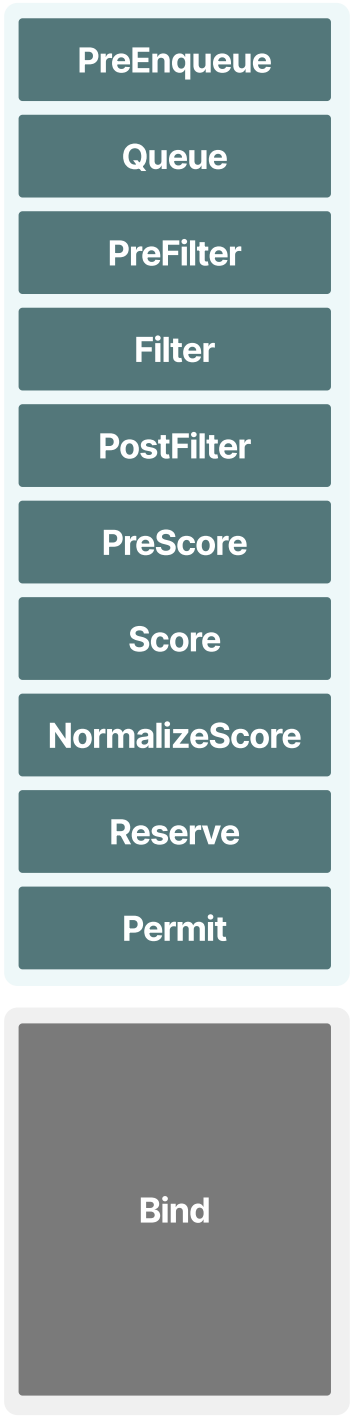
Scheduling

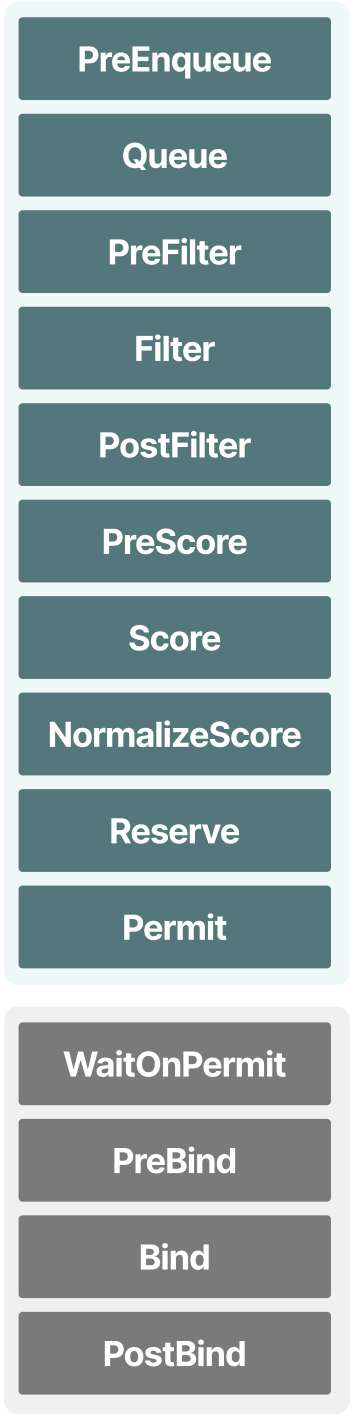
Binding



Scheduling

Binding

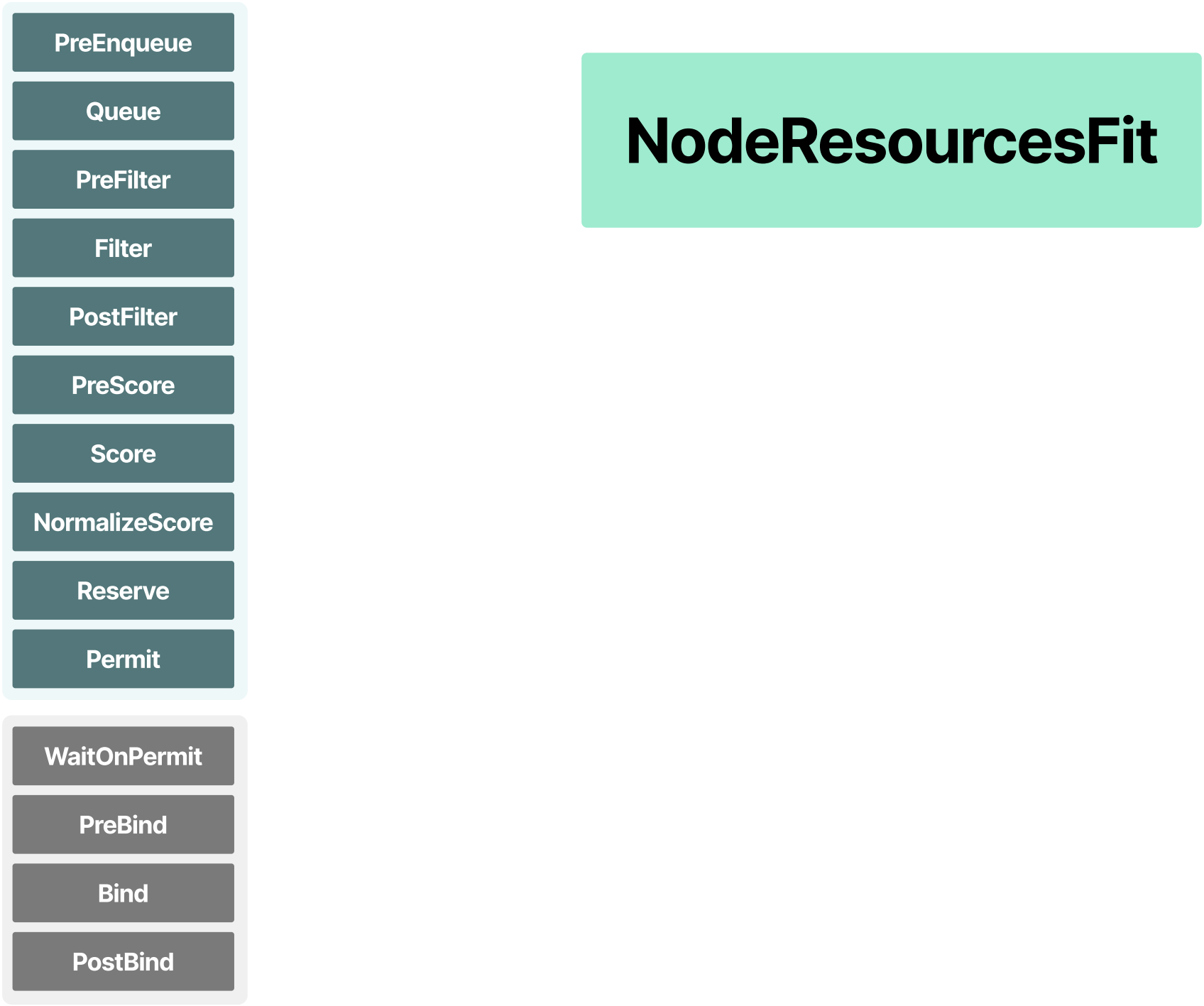

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
Bind
Scheduling
Binding

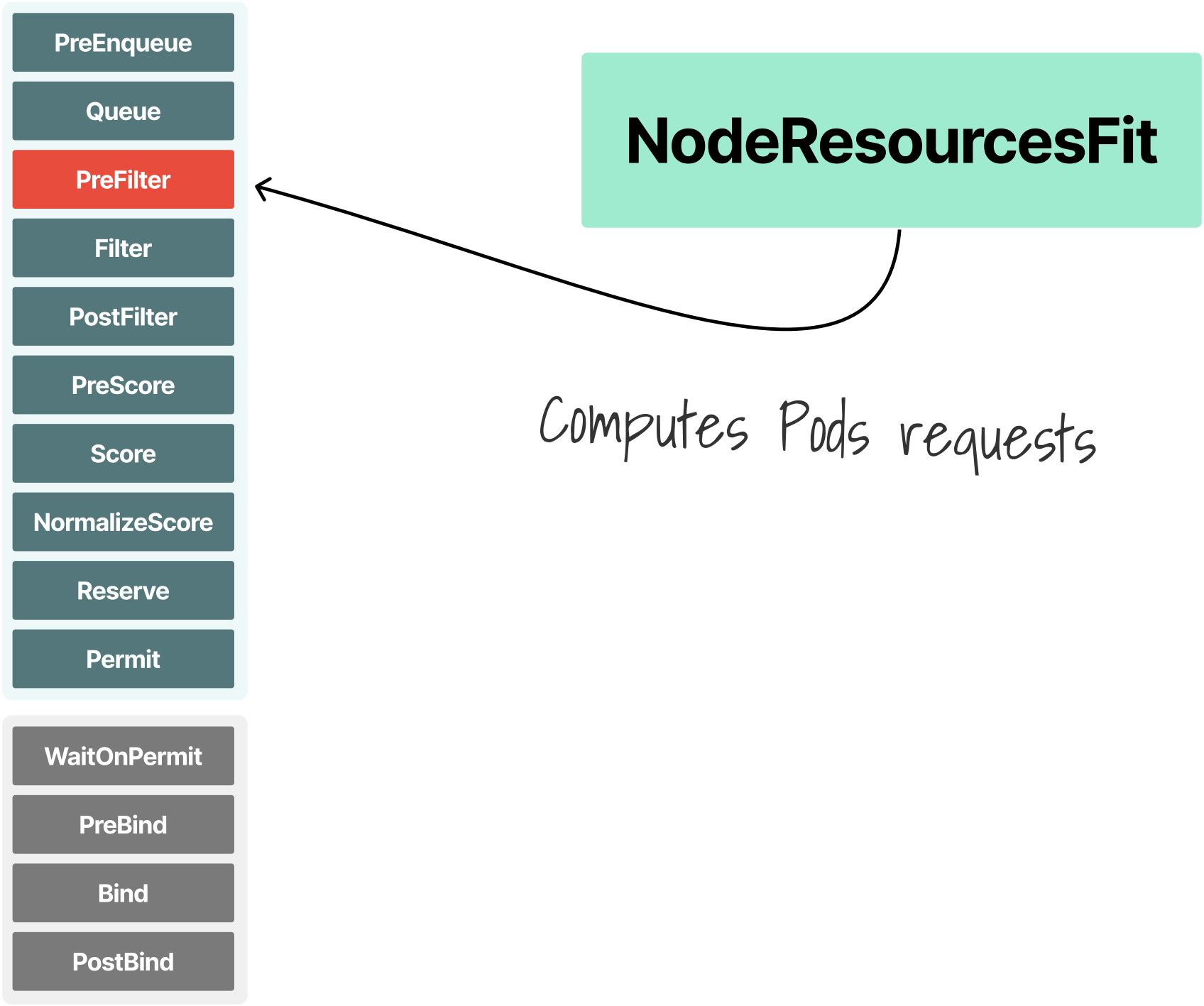


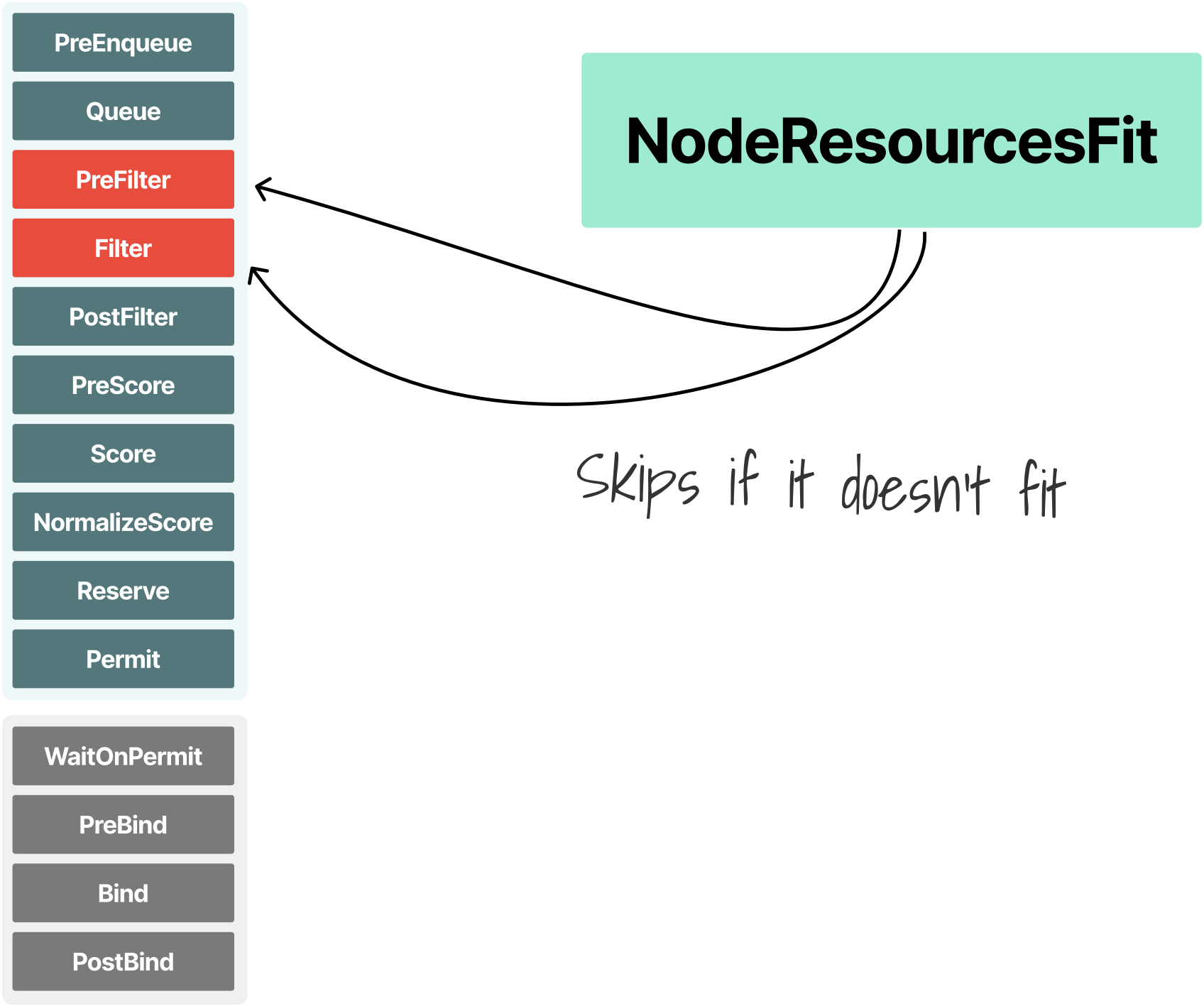
Scheduling

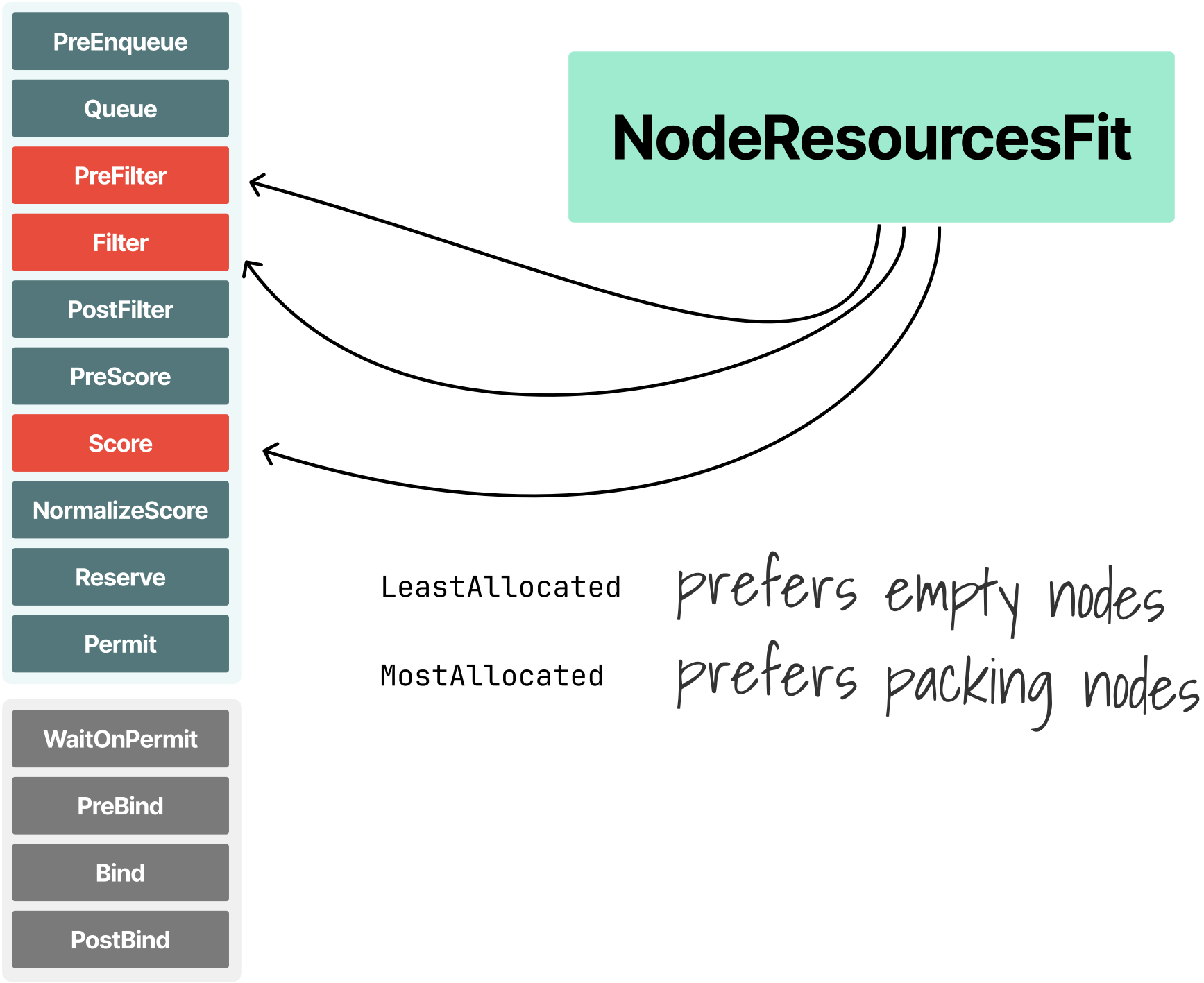
Binding

Plugins: NodeResourcesFit

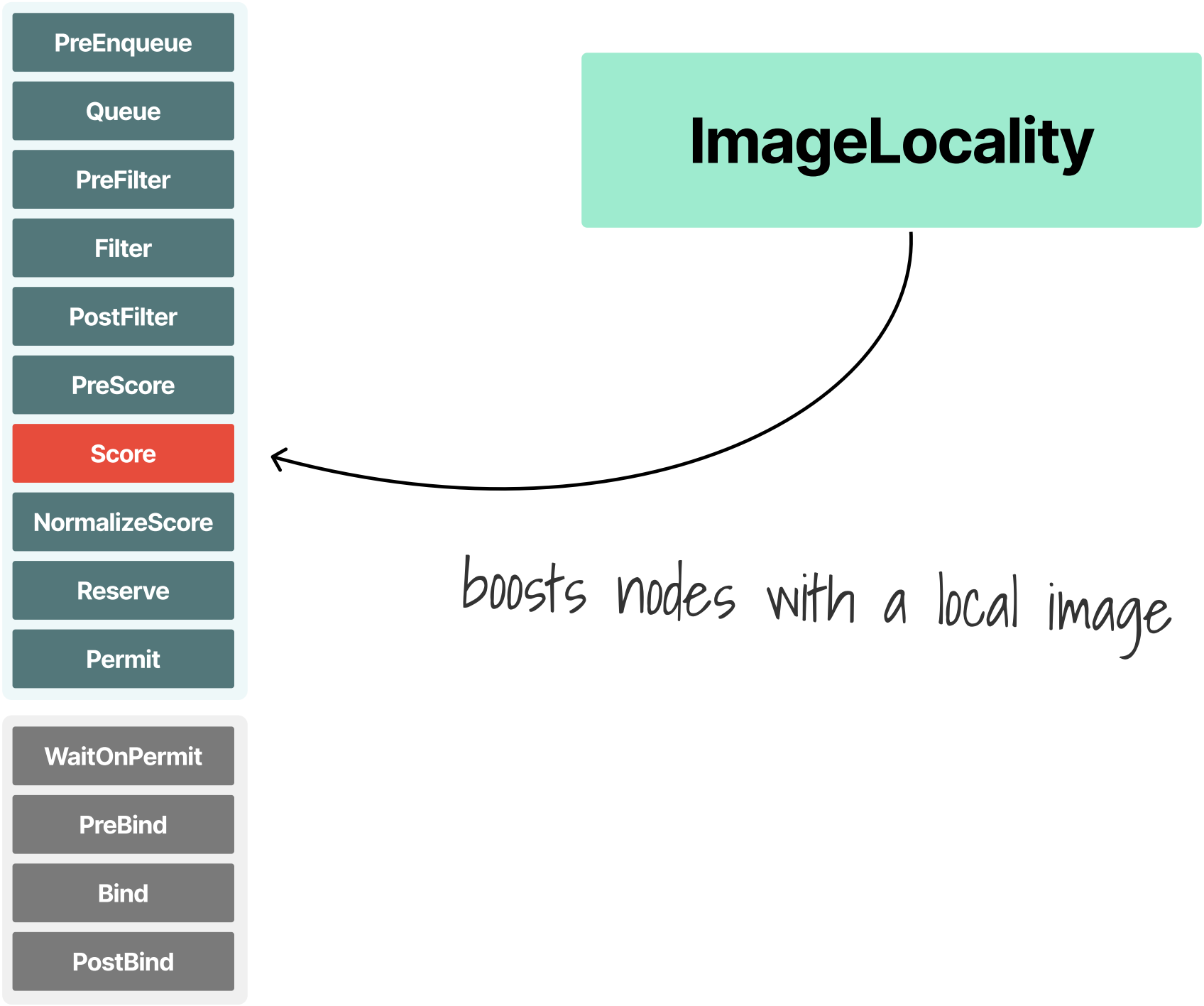

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeResourcesFit


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeResourcesFit
Computes Pods requests

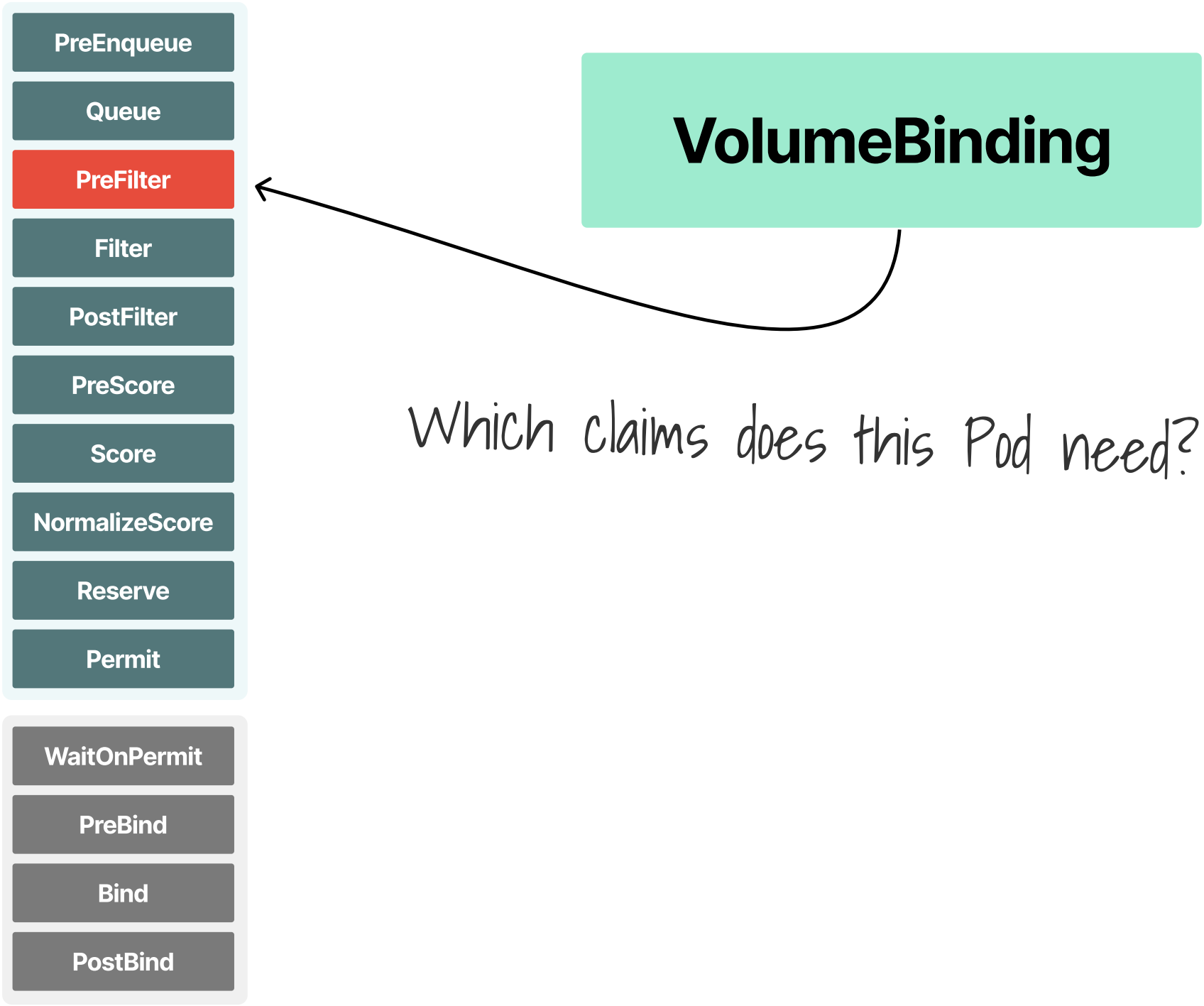

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeResourcesFit
Skips if it doesn't fit

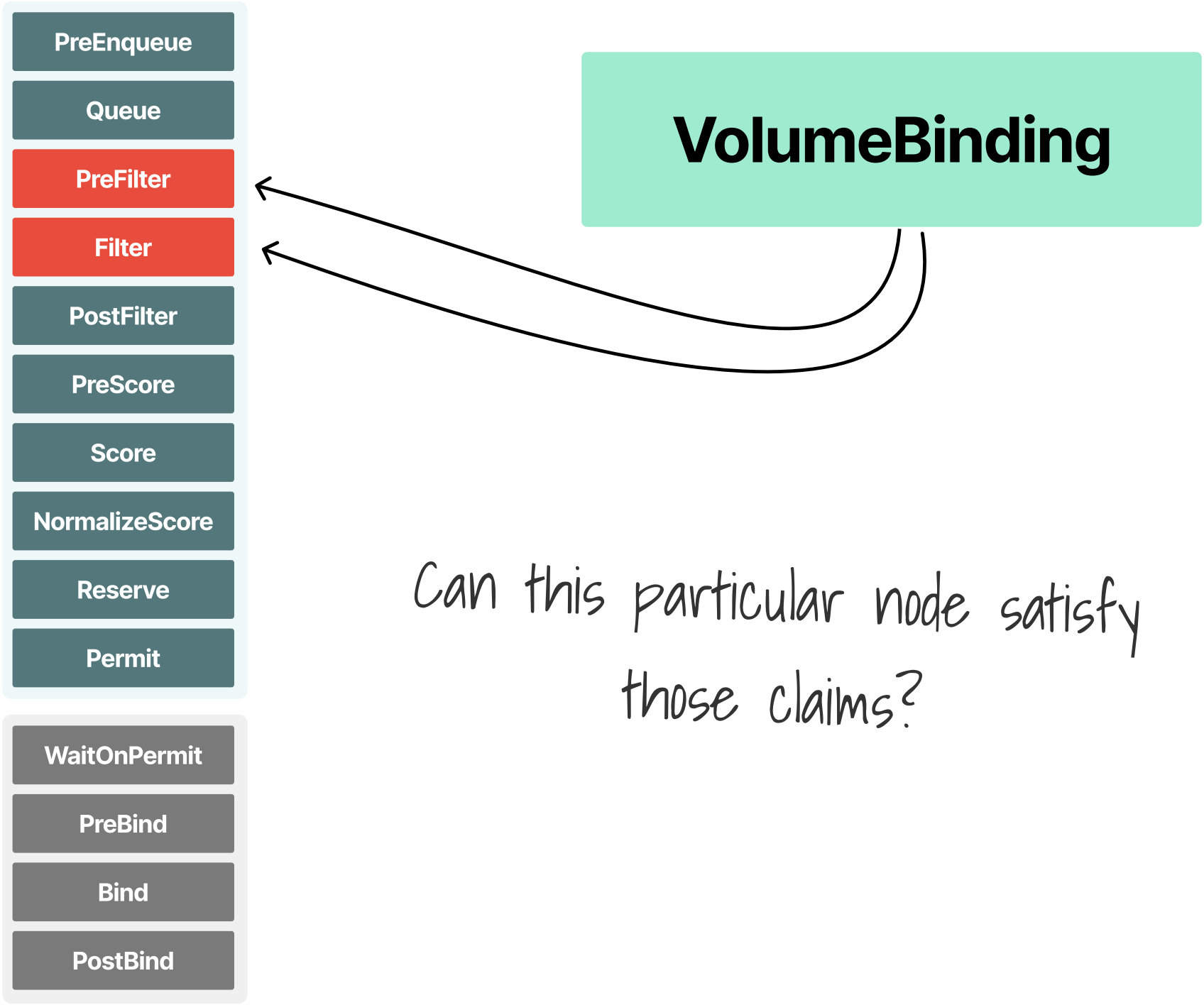

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeResourcesFit
LeastAllocated
prefers empty nodes
MostAllocated
prefers packing nodes

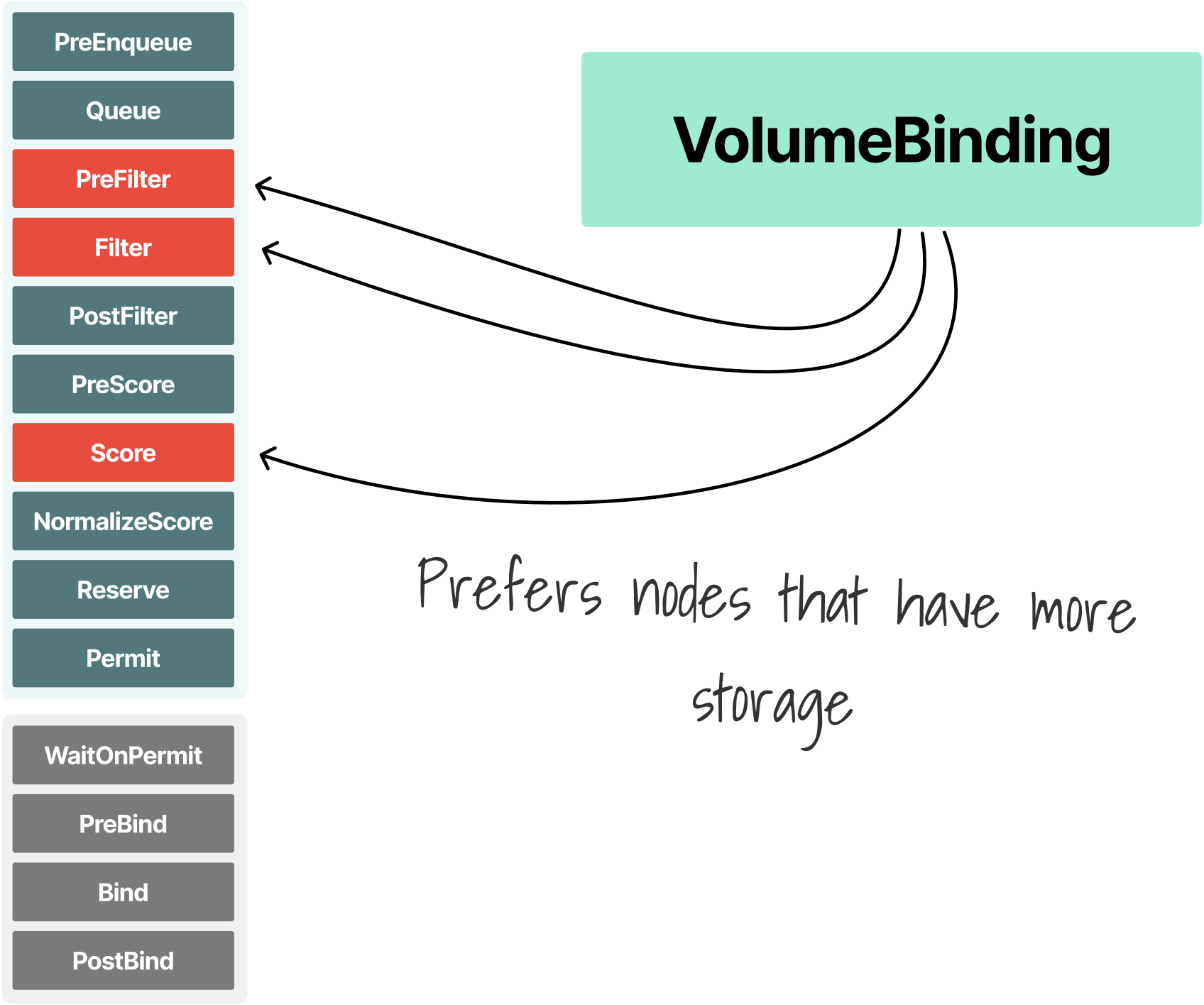
Plugins: ImageLocality

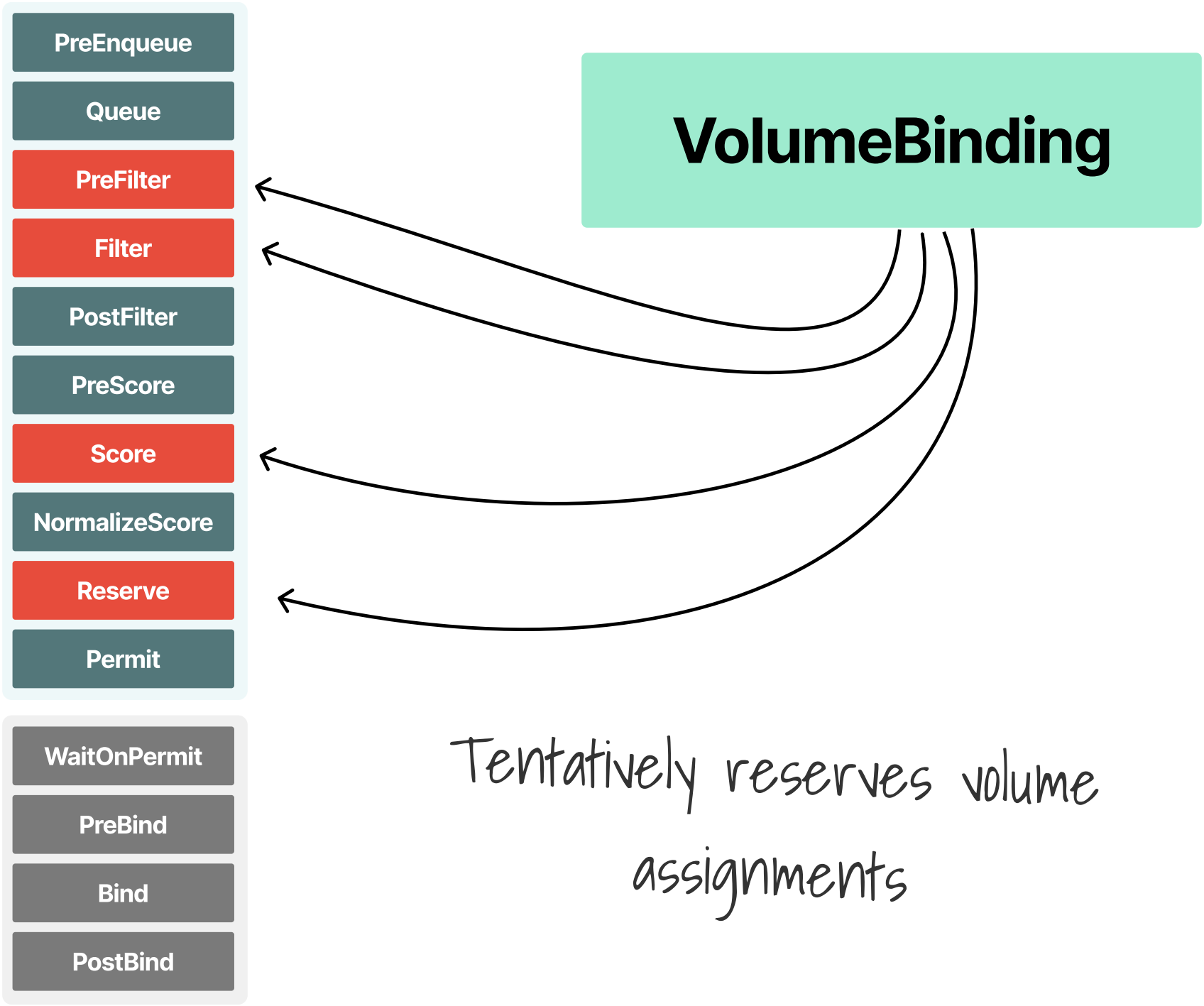

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
ImageLocality
boosts nodes with a local image

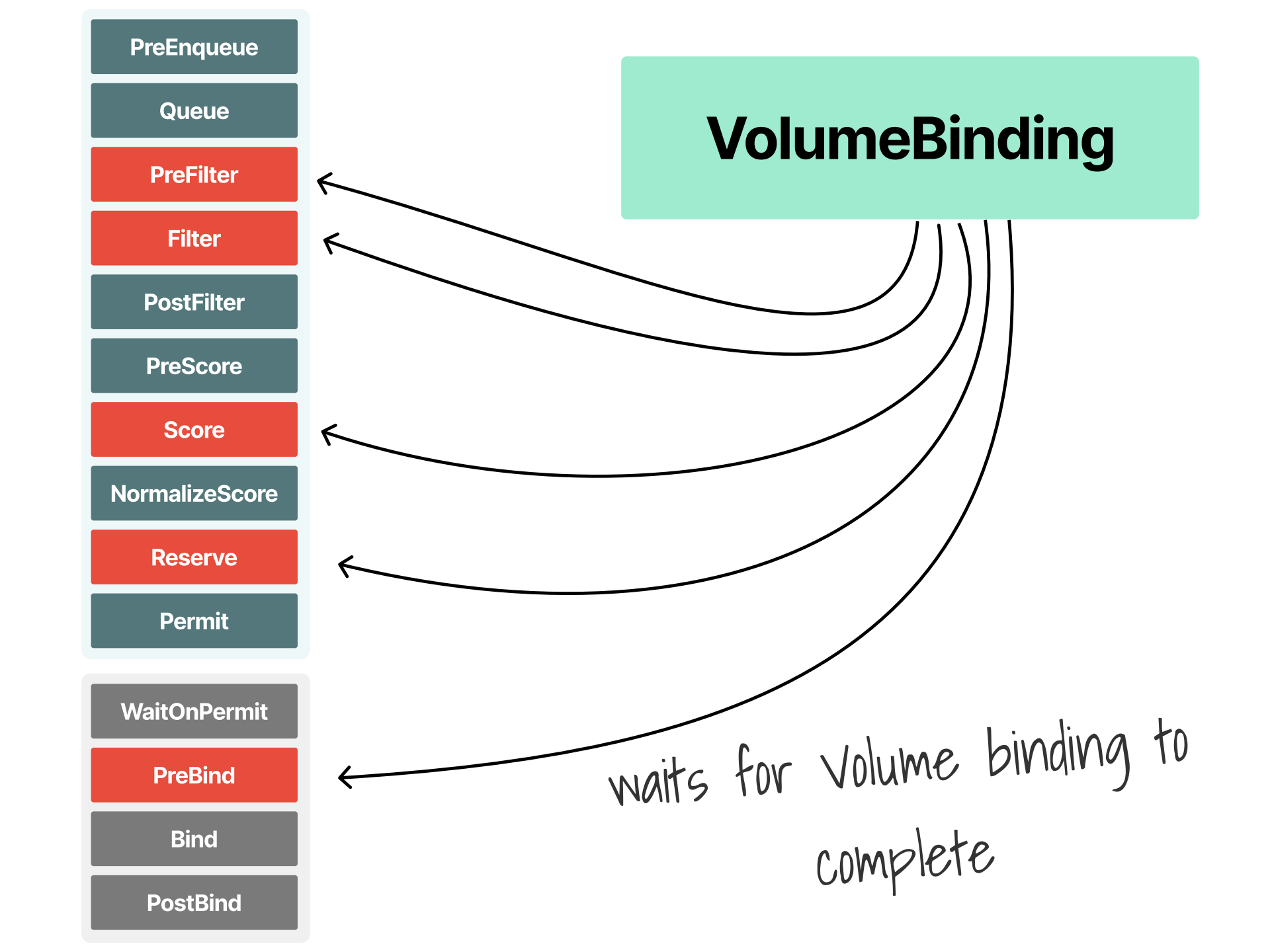
Plugins: VolumeBinding


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
VolumeBinding
Which claims does this Pod need?

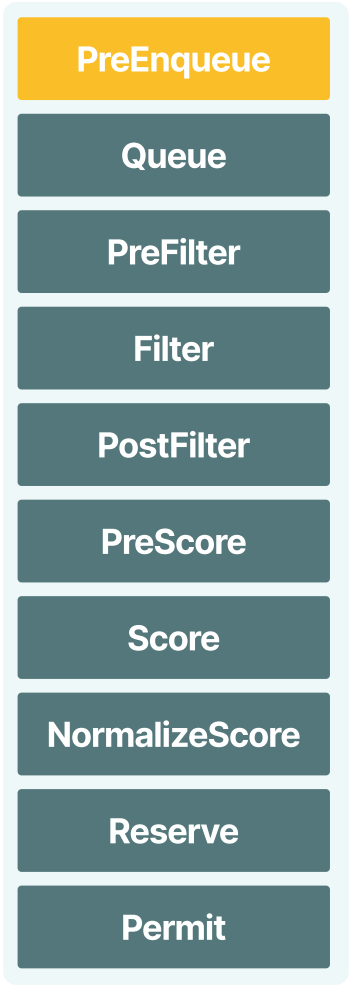

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
VolumeBinding
Can this particular node satisfy those claims?

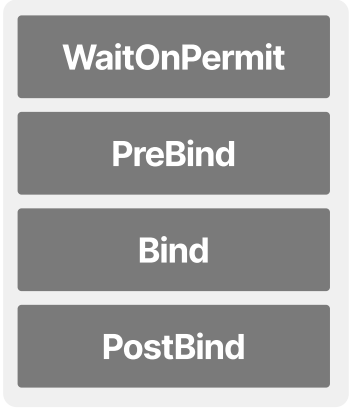

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
VolumeBinding
Prefers nodes that have more storage

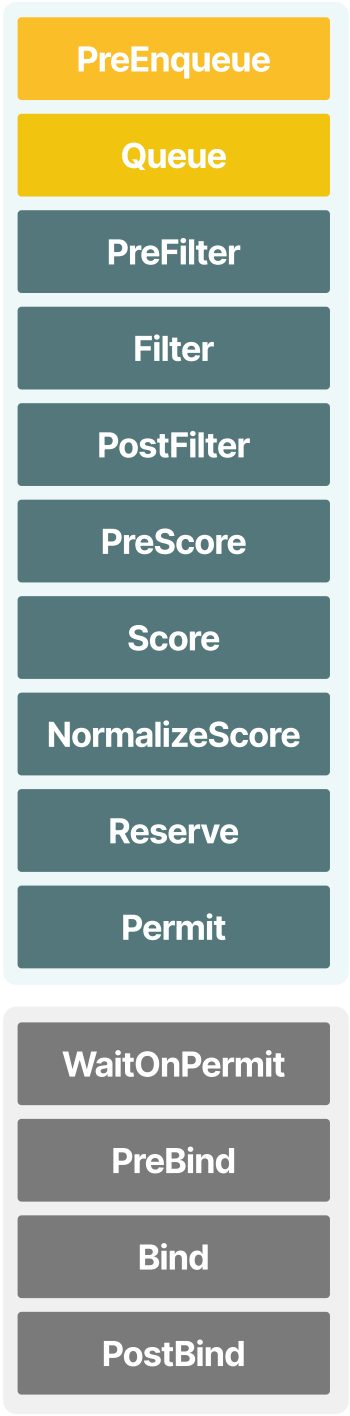

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
VolumeBinding
Tentatively reserves volume assignments

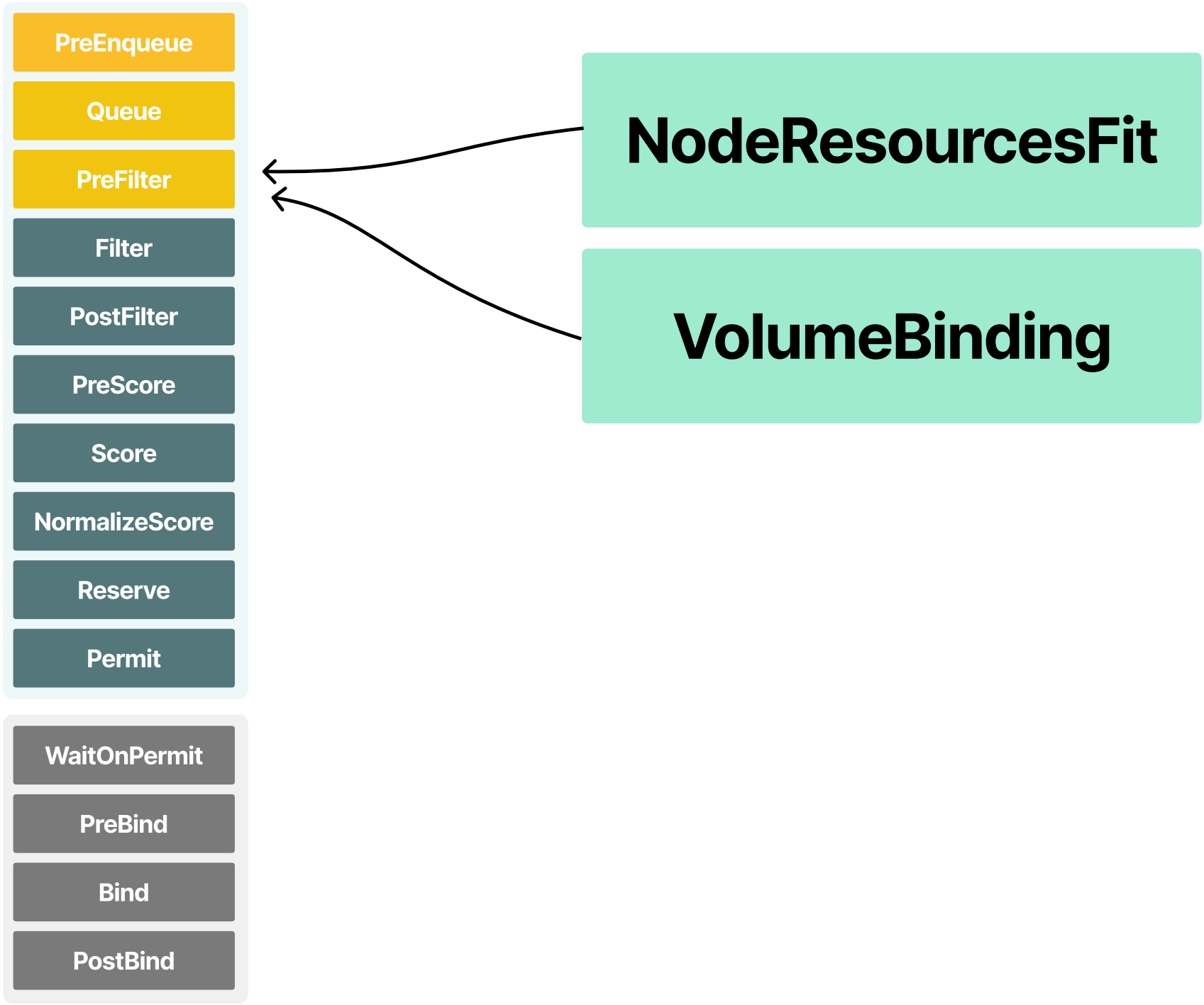

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
VolumeBinding
waits for Volume binding to complete

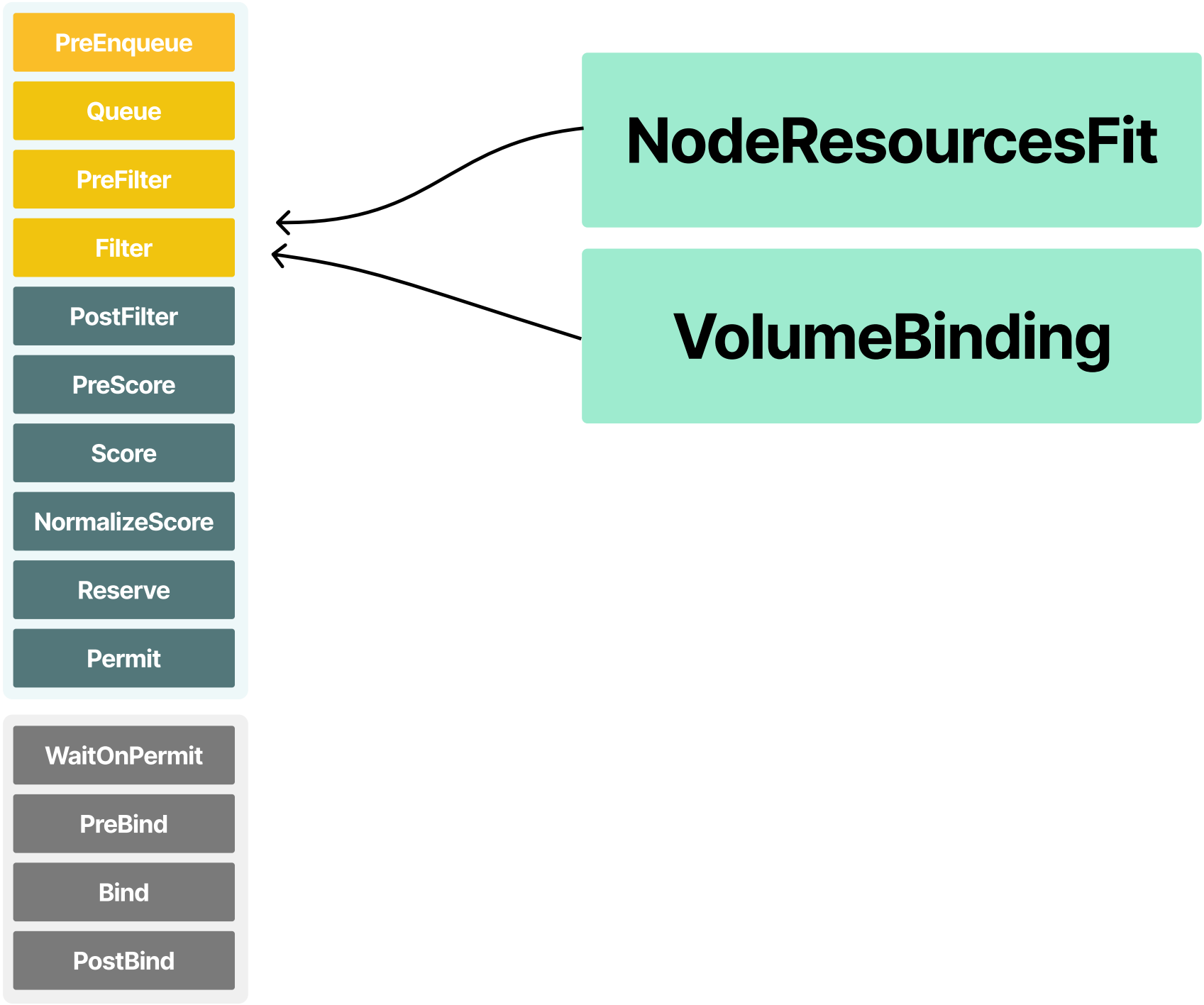
All together

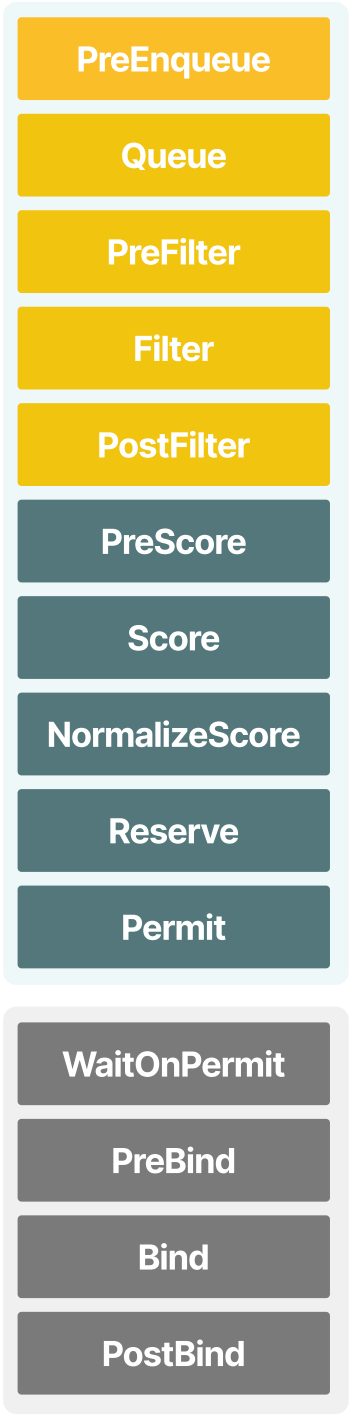

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit

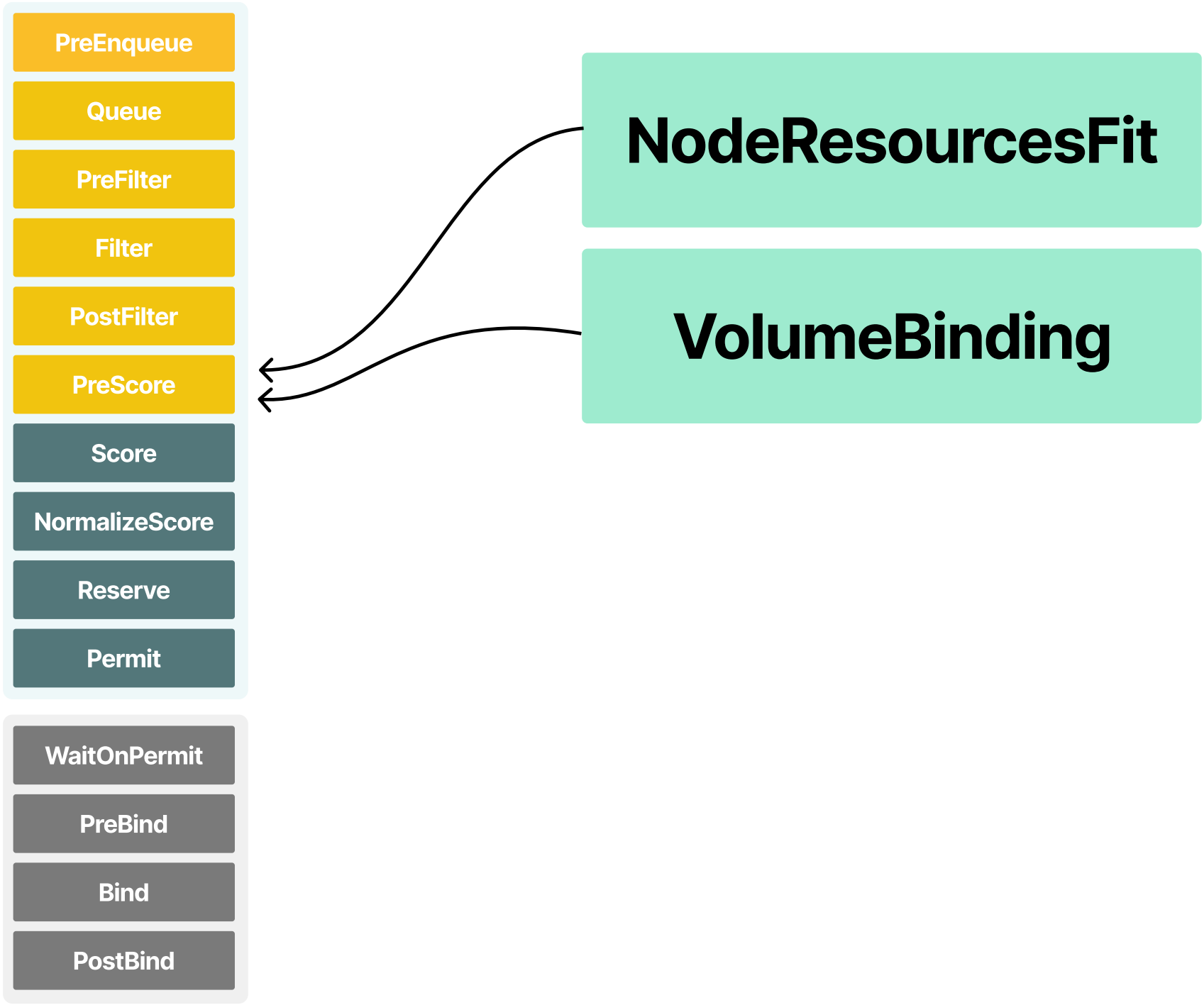

WaitOnPermit
PreBind
Bind
PostBind

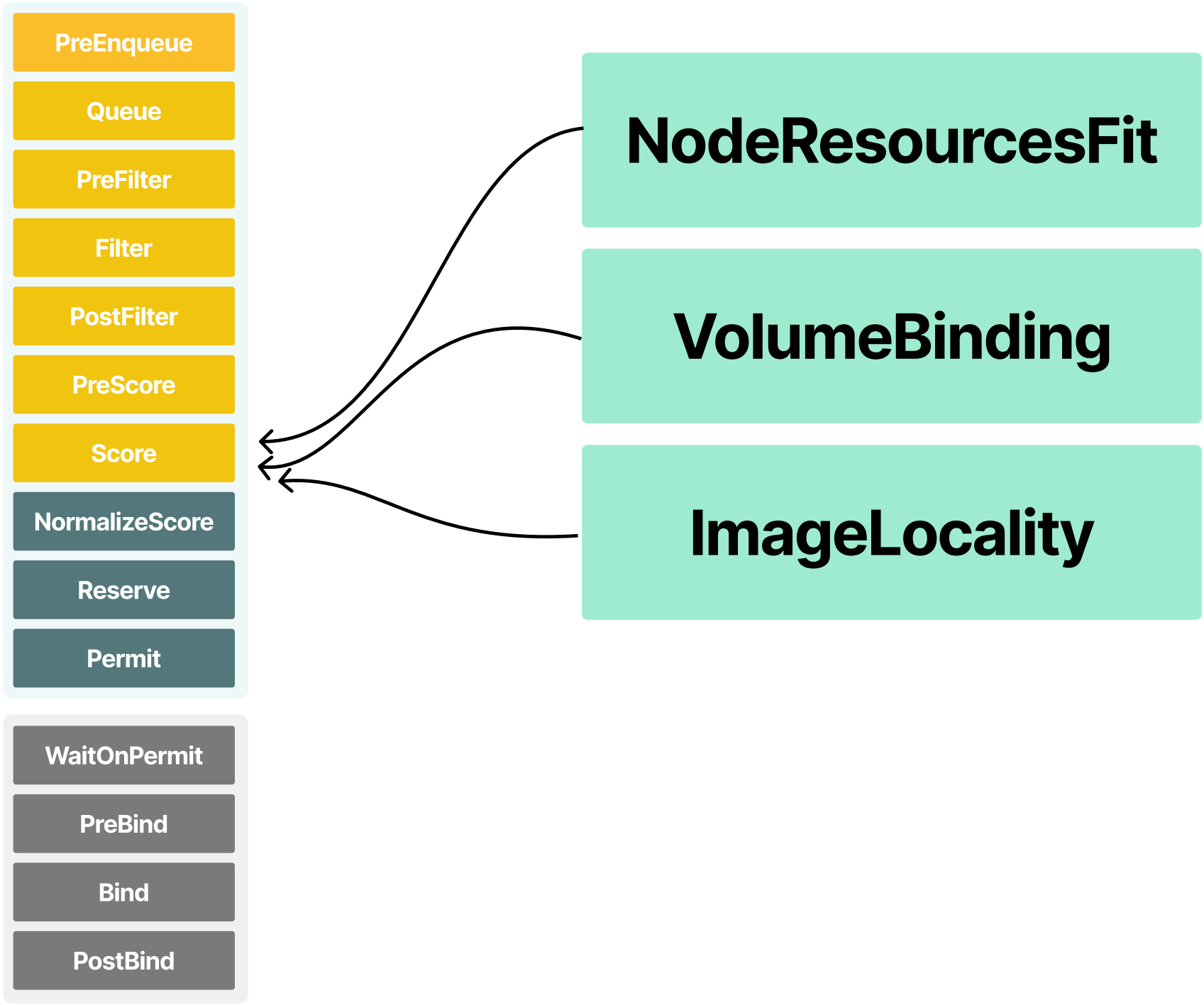

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind

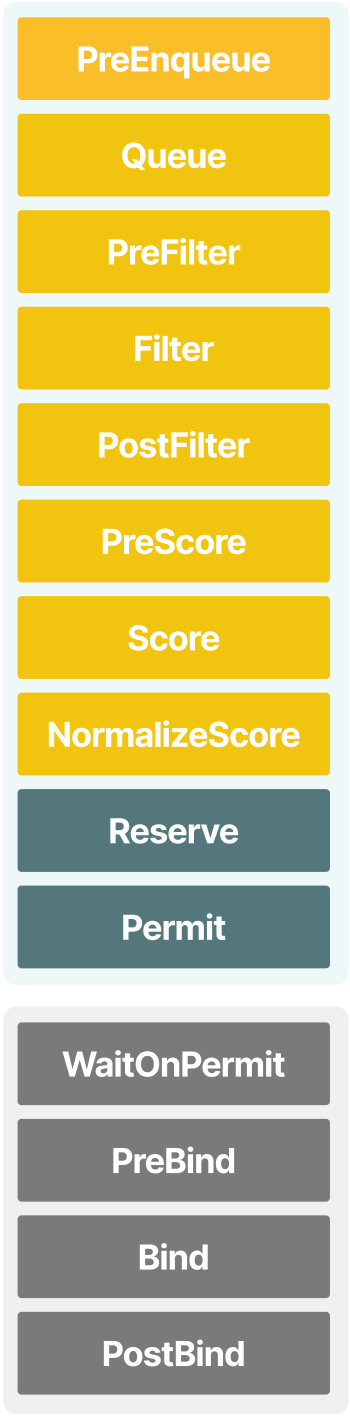

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeResourcesFit
VolumeBinding

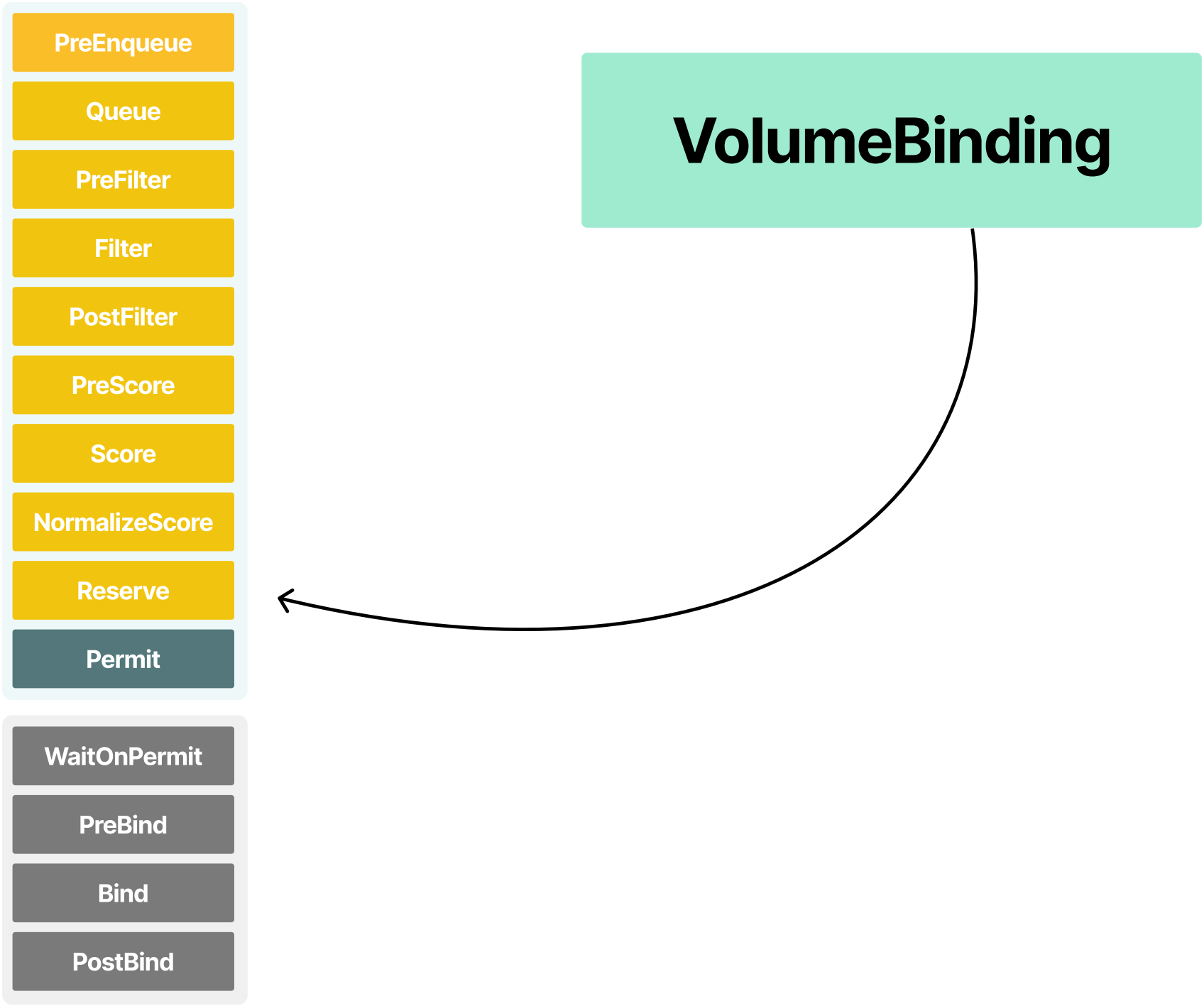

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeResourcesFit
VolumeBinding

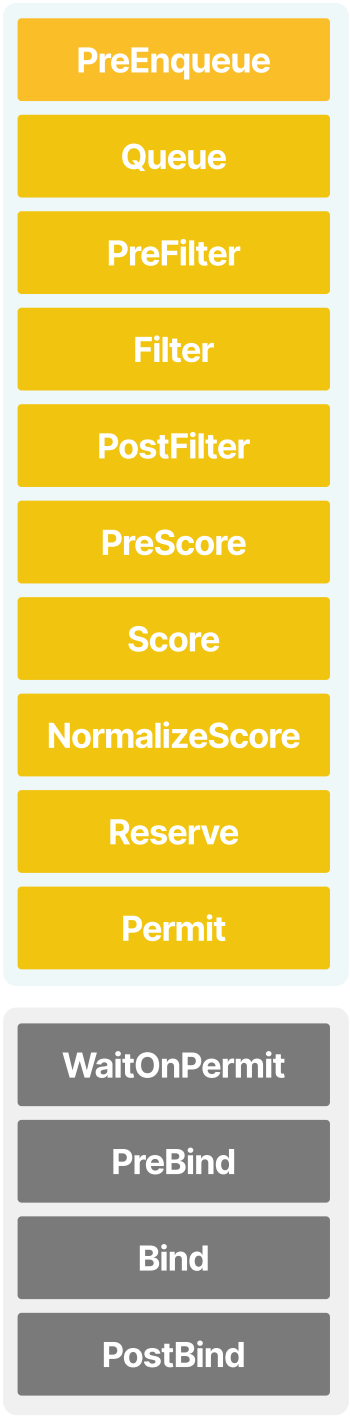

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind

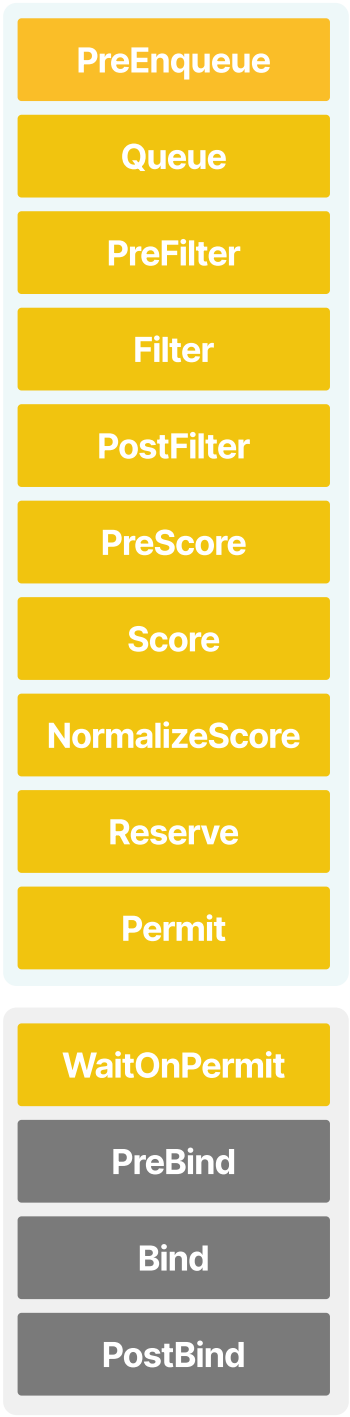

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeResourcesFit
VolumeBinding

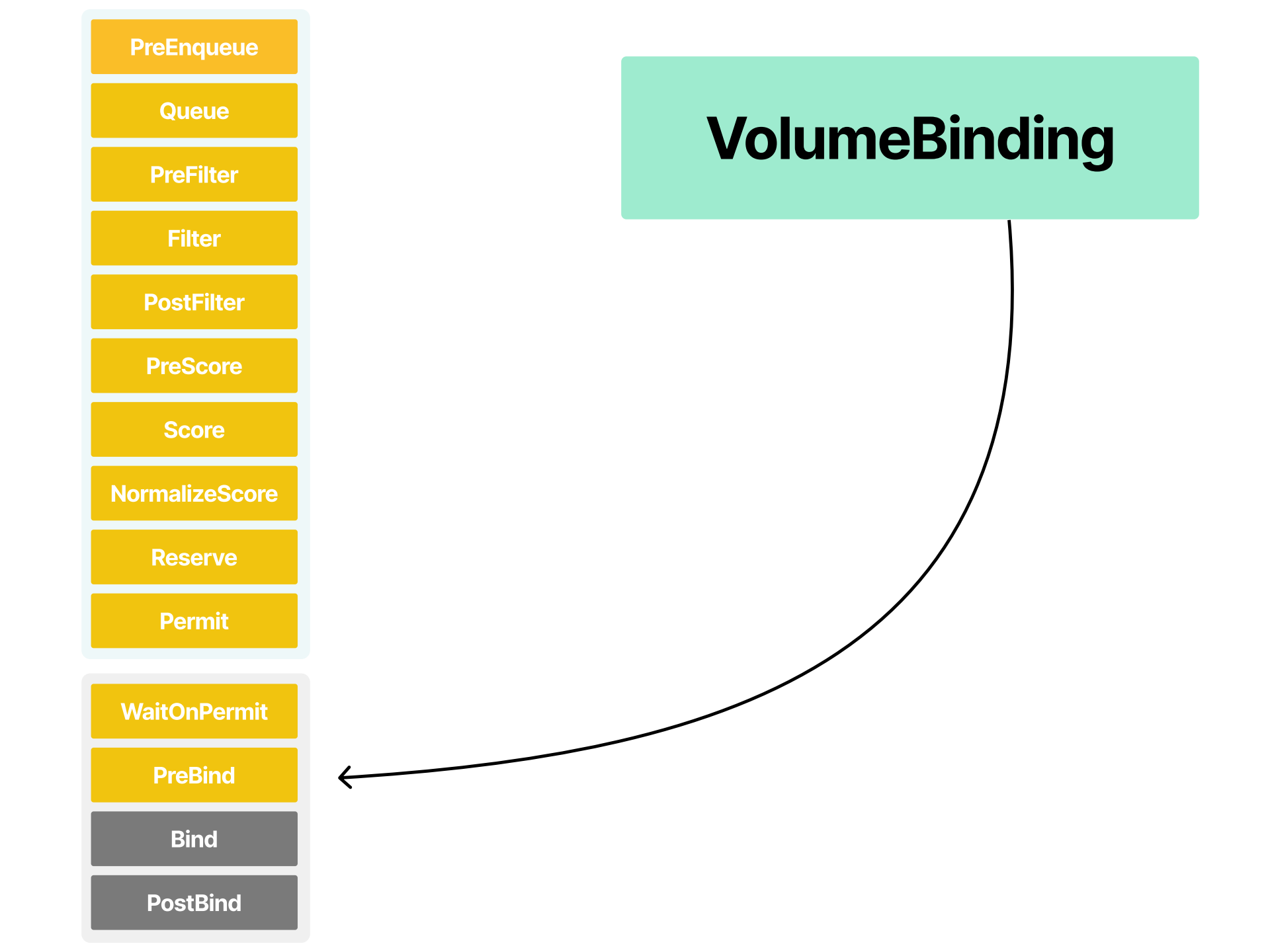

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeResourcesFit
VolumeBinding
ImageLocality

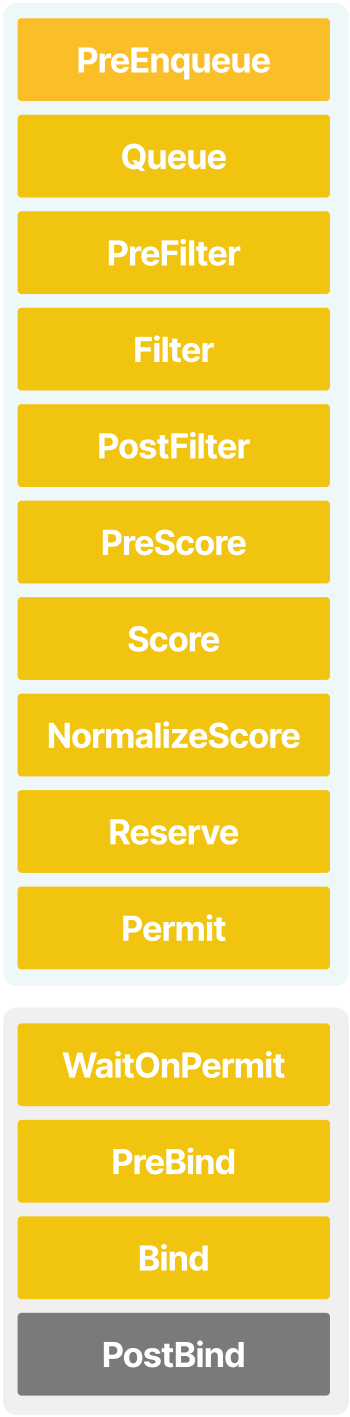

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind

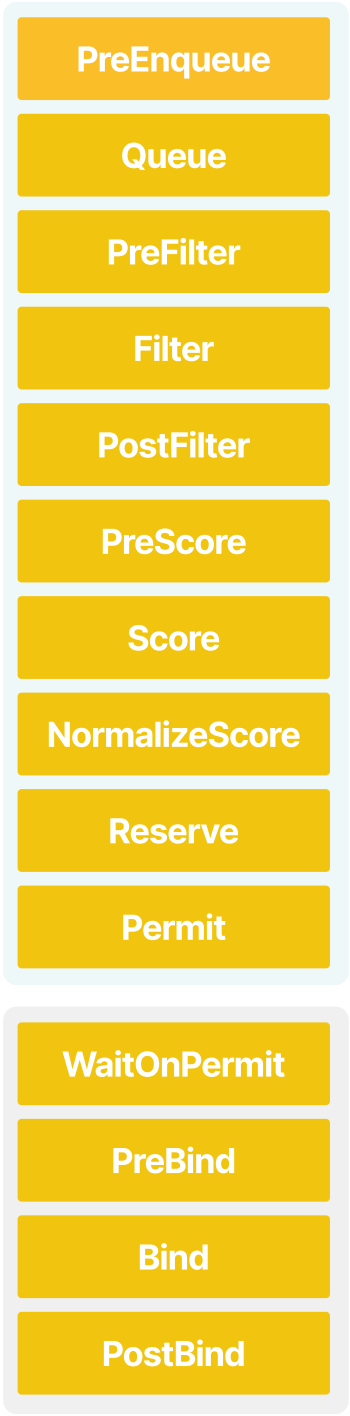

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
VolumeBinding


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
VolumeBinding


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind

Scheduling hints

1. nodeSelector

Scheduling hints

1. nodeSelector

2. Node Affinity

Scheduling hints

1. nodeSelector

2. Node Affinity

3. **Pod affinity/anti-affinity**

Scheduling hints

1. nodeSelector
2. Node Affinity
3. Pod affinity/anti-affinity
4. **Taints & tolerations**

nodeSelector

01

nodeSelector

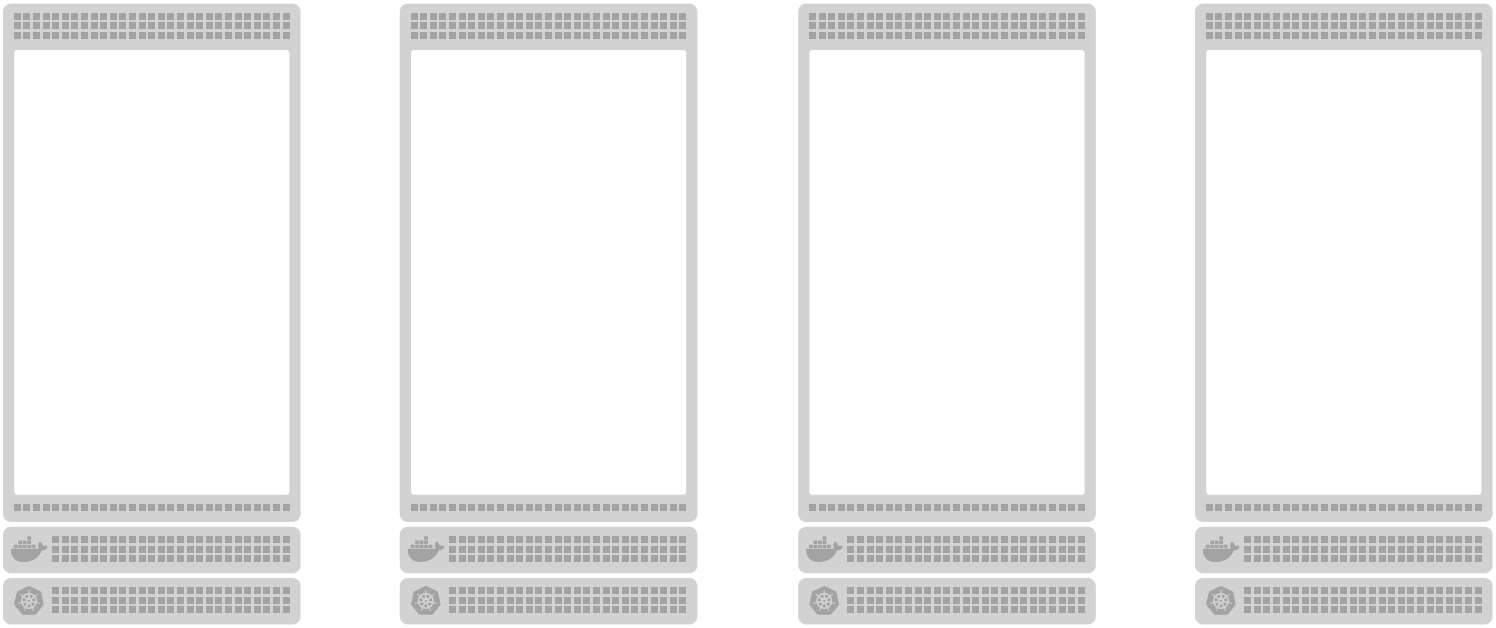
Nodes have labels

Pods have selectors

nodeSelector

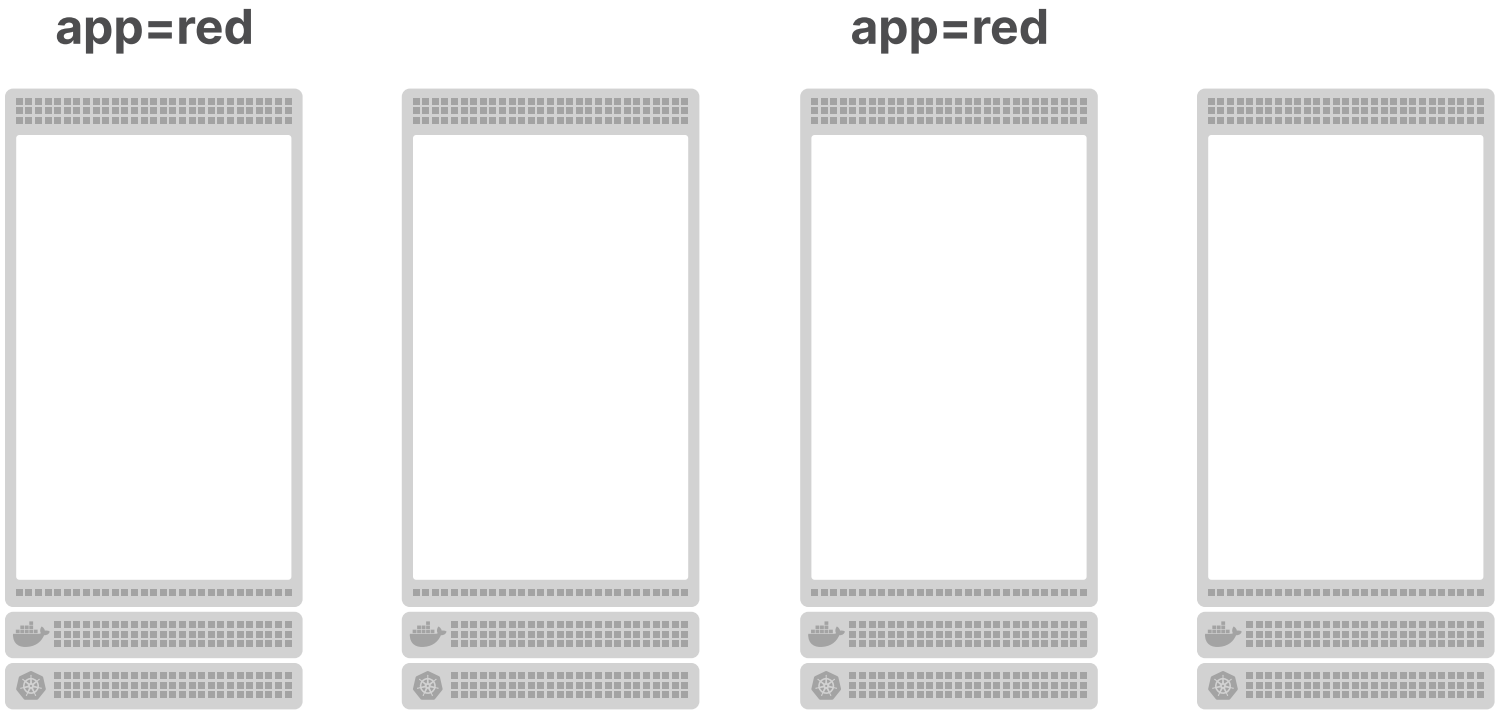
Nodes have labels

Pods have selectors

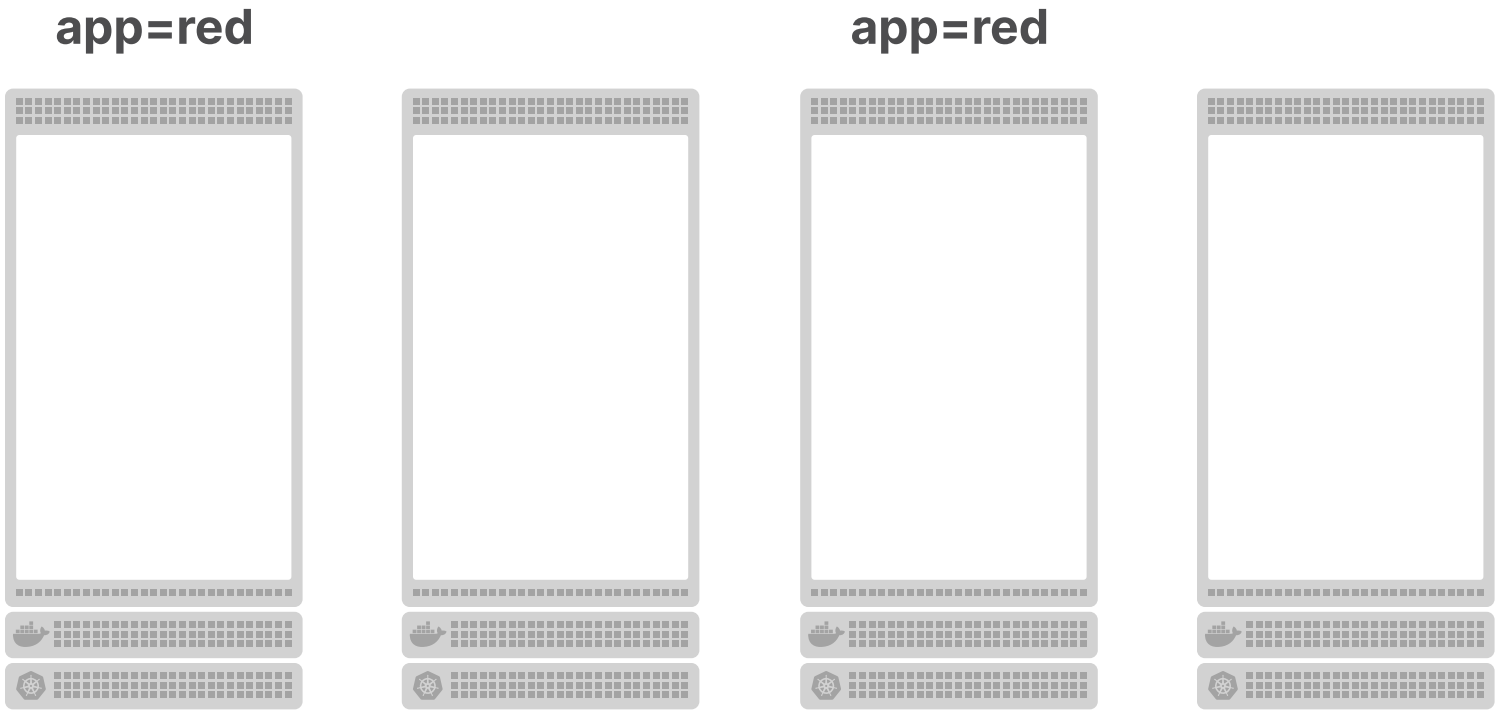


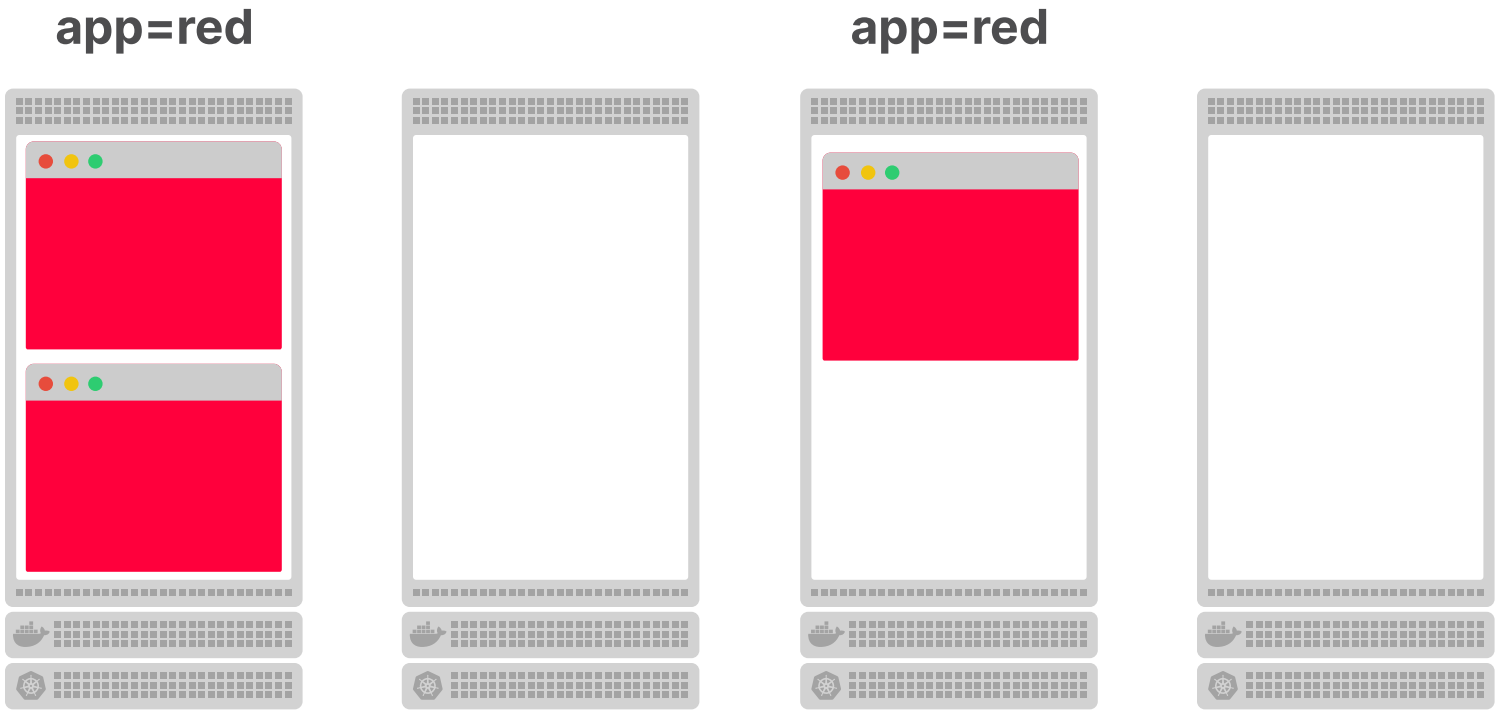


```
~$ kubectl label nodes \
 worker1 app=red
```


app=red
app=red

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ns-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.7.9
      nodeSelector:
        app: red
```


app=red
app=red


app=red
app=red

nodeSelector

Exact match

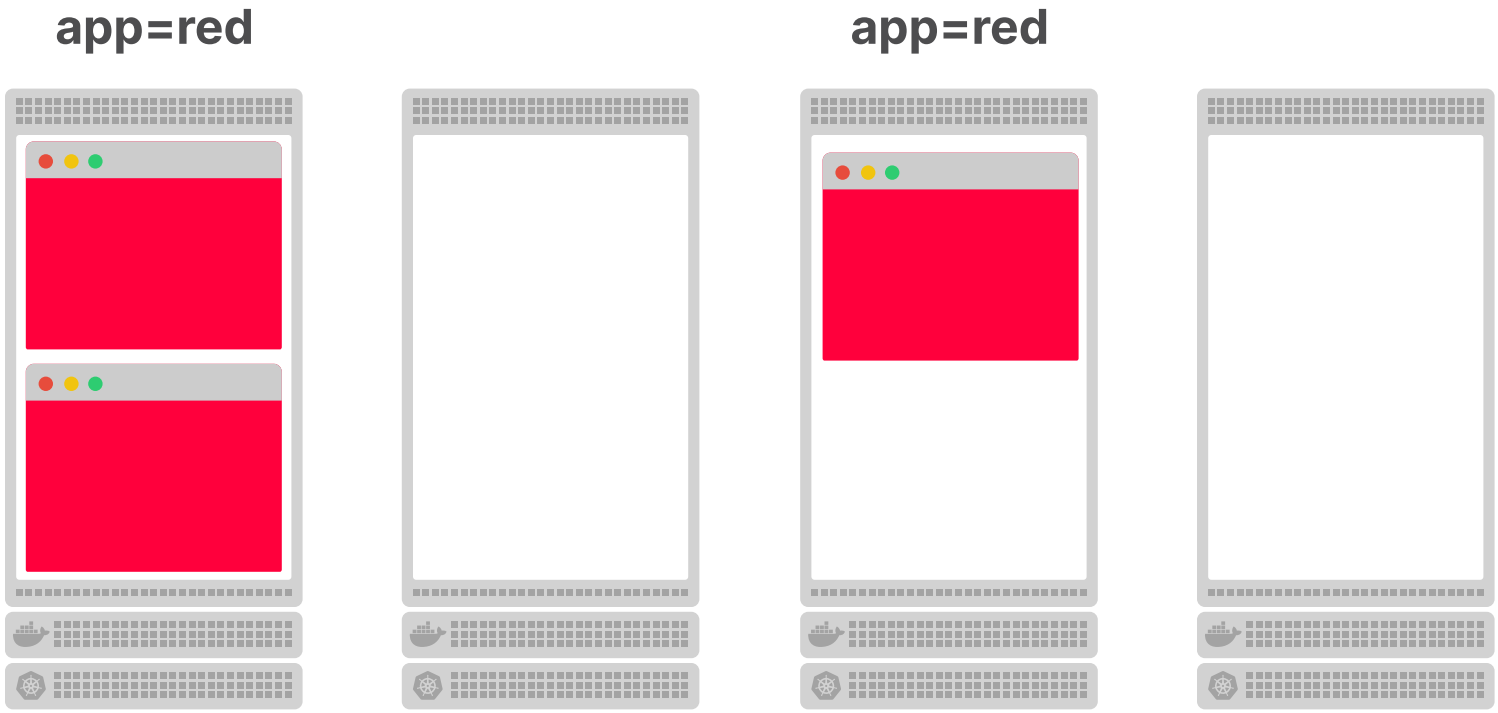


Hard requirements

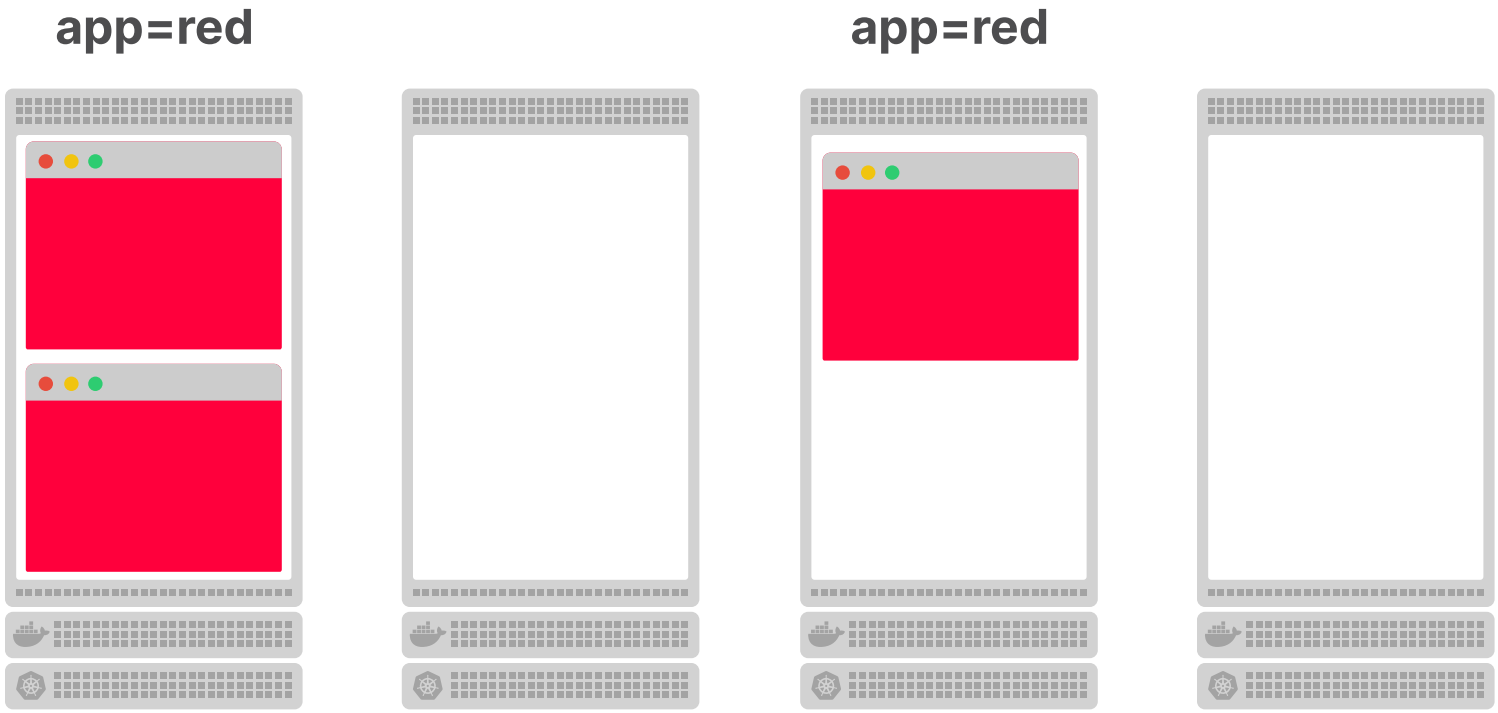
nodeSelector

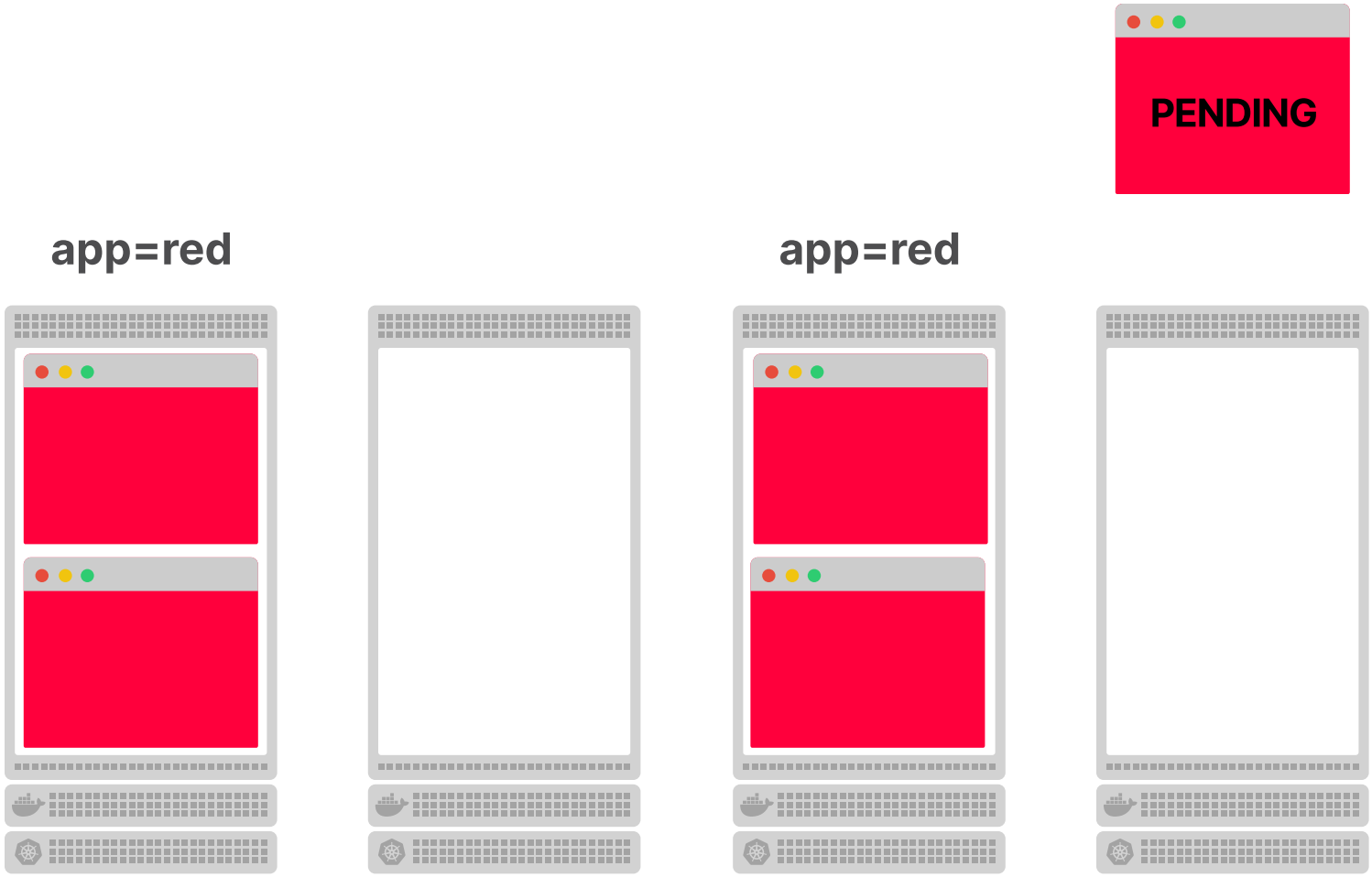
Exact match

Hard requirements 


app=red
app=red

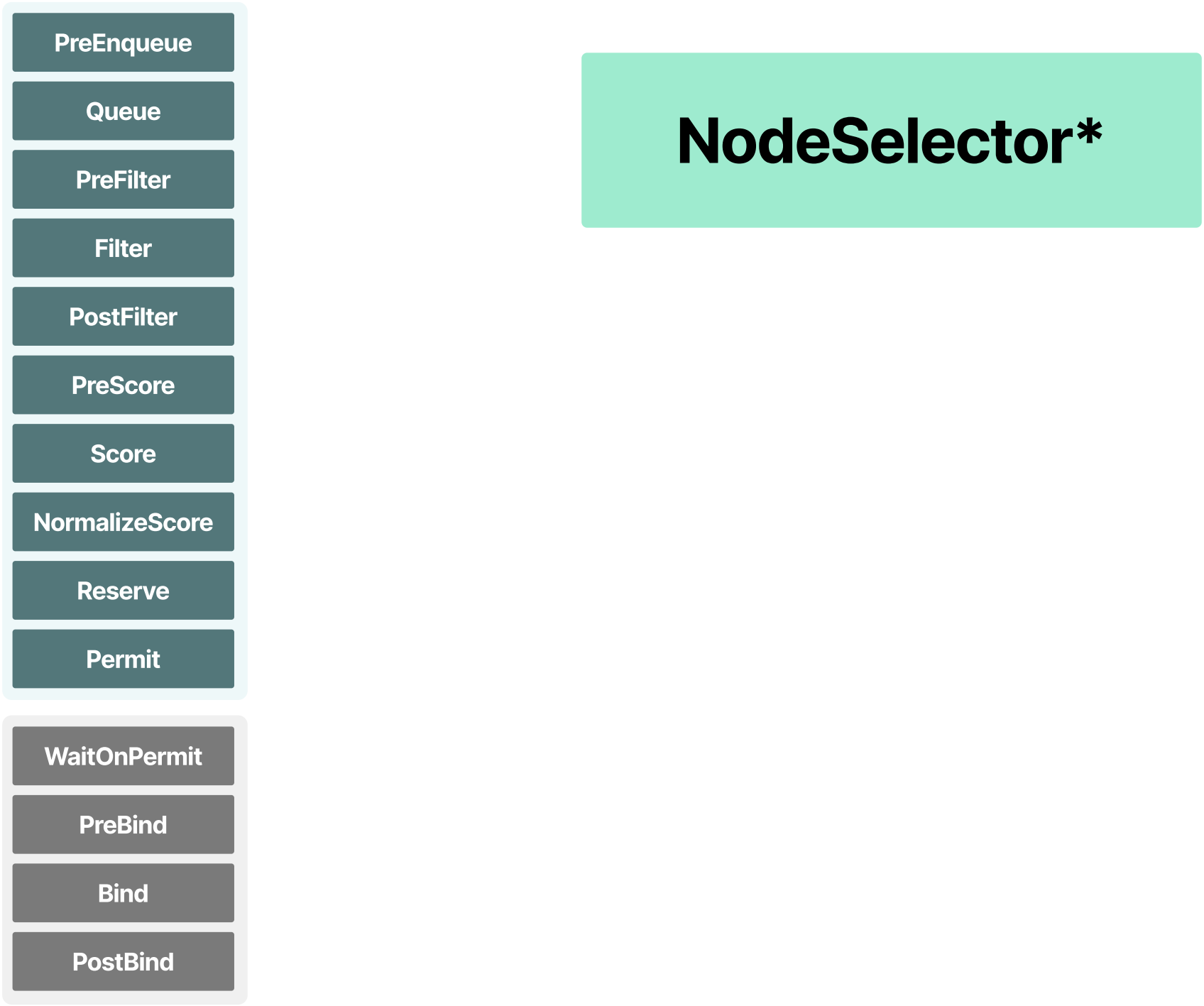
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ns-deployment
spec:
  replicas: 5
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.7.9
      nodeSelector:
        app: red
```

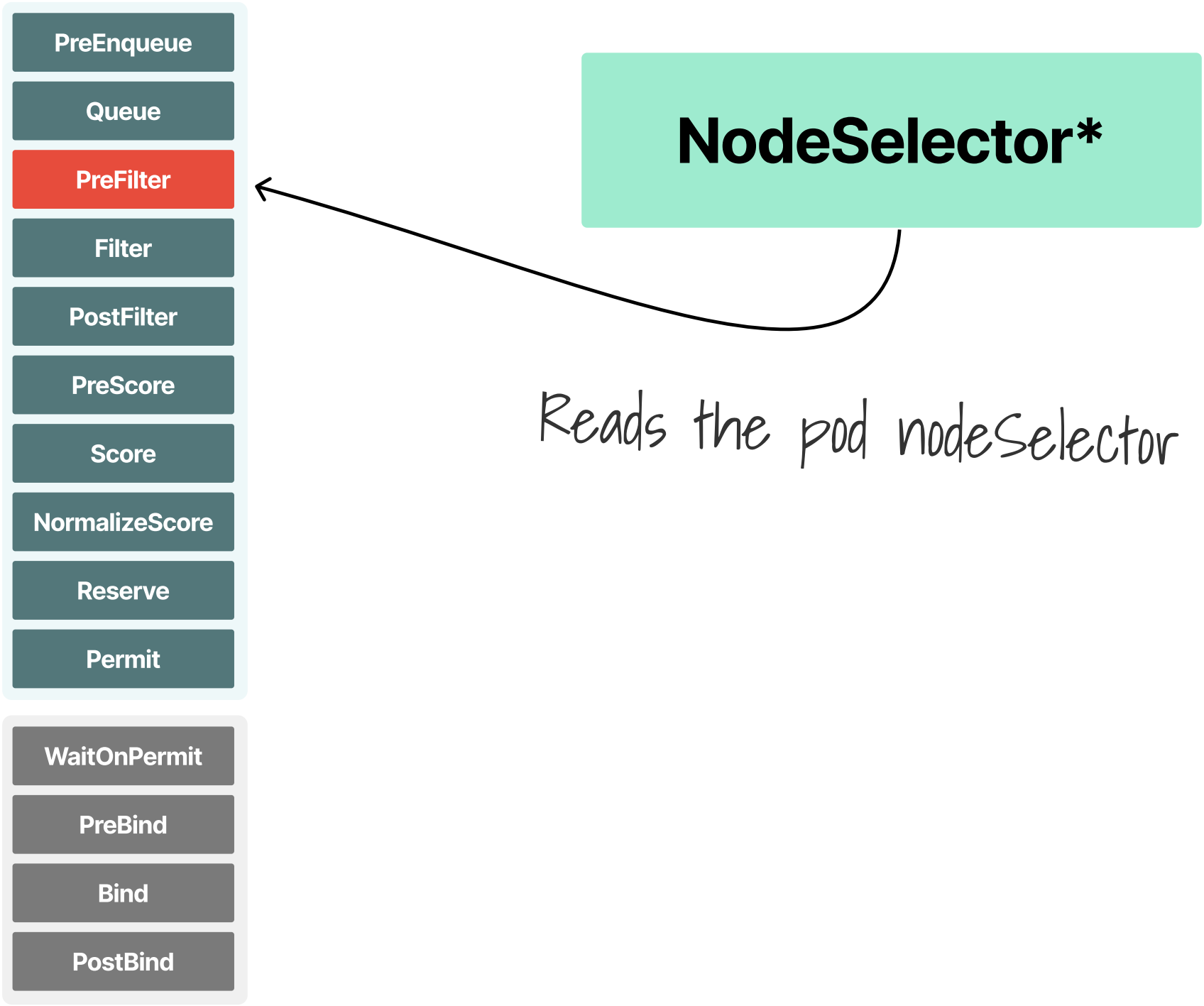

app=red
app=red

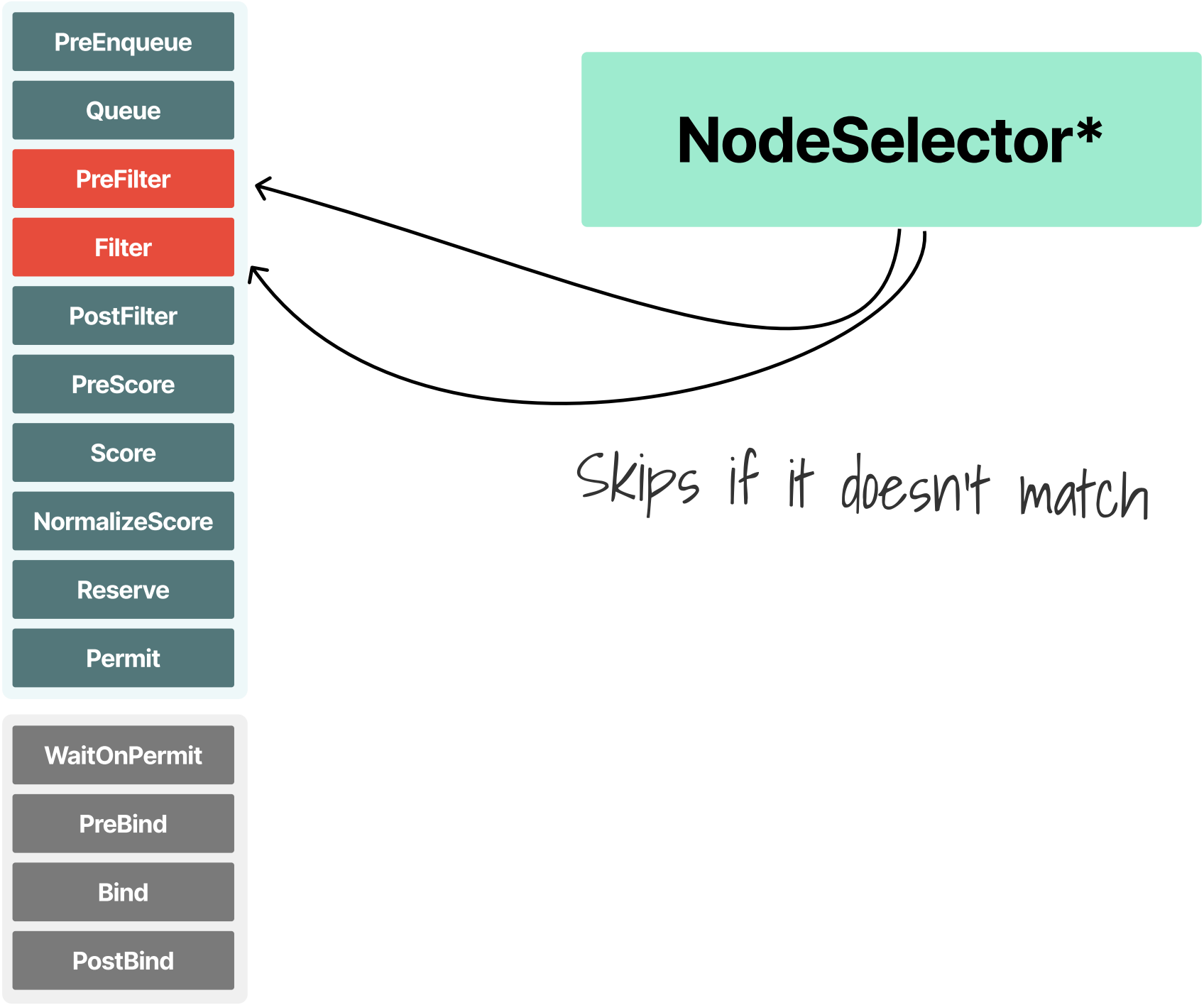

PENDING
app=red
app=red

nodeSelector plugin

01


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeSelector*


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeSelector*
Reads the pod nodeSelector


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeSelector*
Skips if it doesn't match

Node affinity

02

Node affinity

requiredDuringSchedulingIgnoredDuringExecution

preferredDuringSchedulingIgnoredDuringExecution

Node affinity

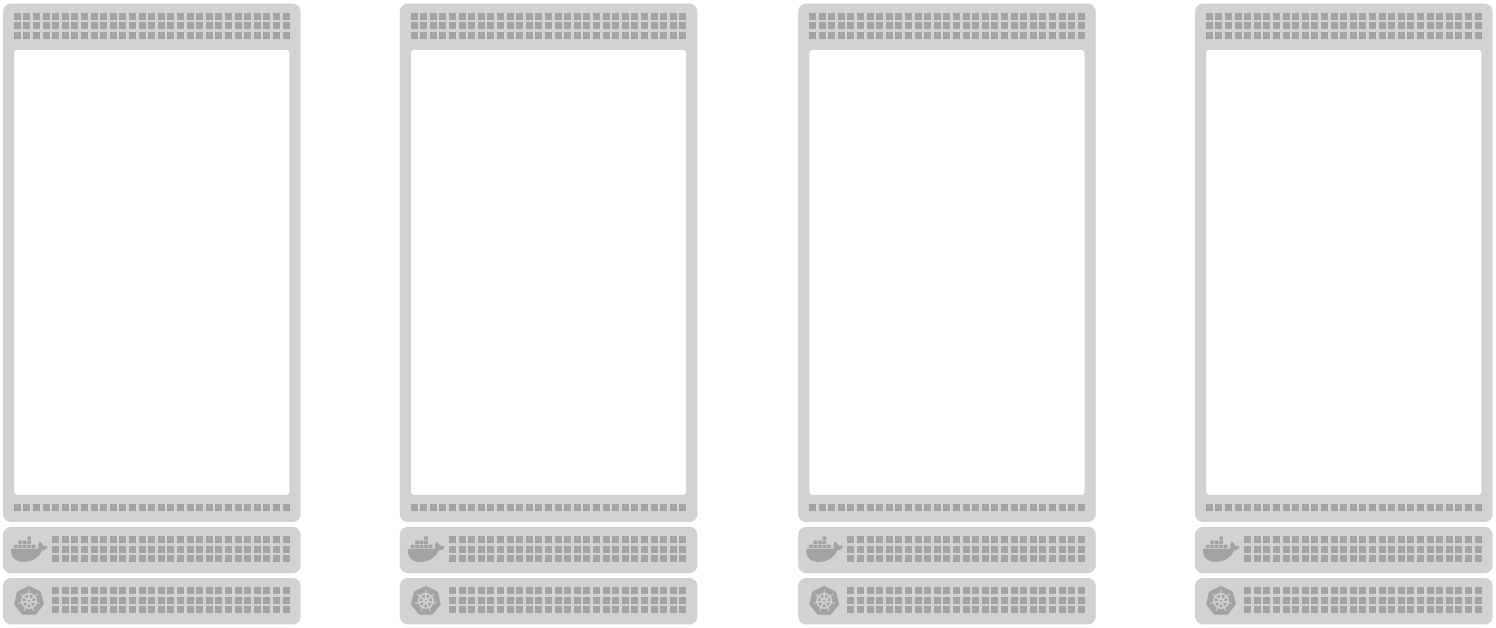
Required
Preferred

This database must run on nodes with SSDs

Node affinity hard

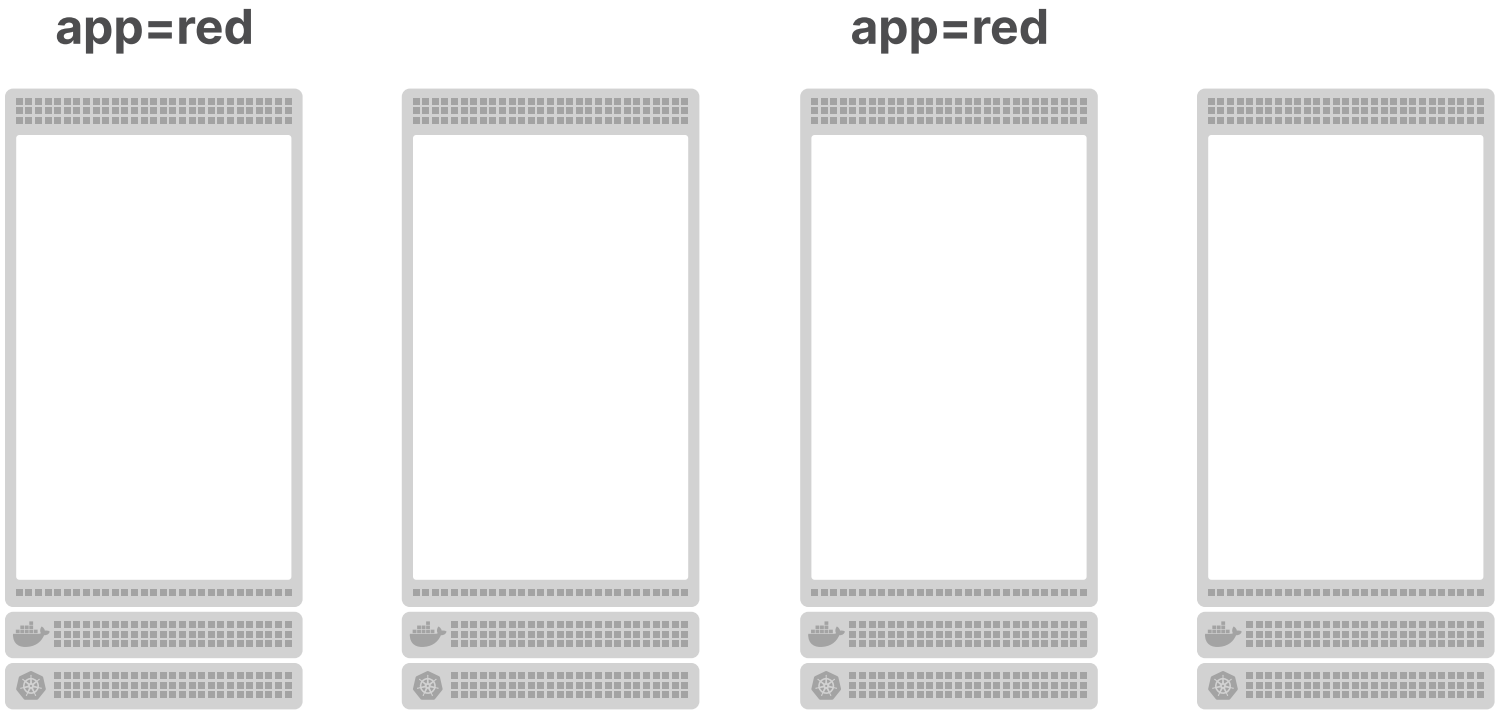
Prefer SSD nodes, but allow other nodes

Node affinity soft

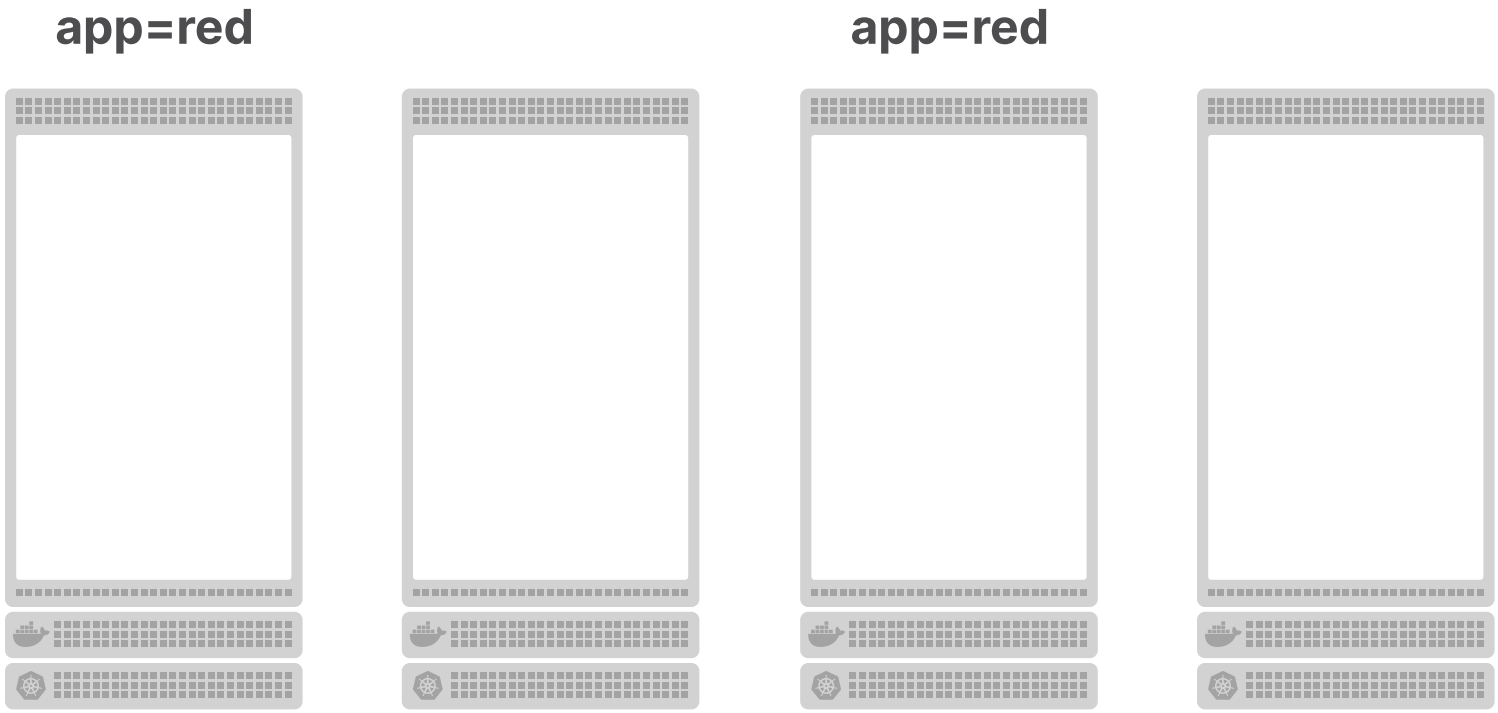


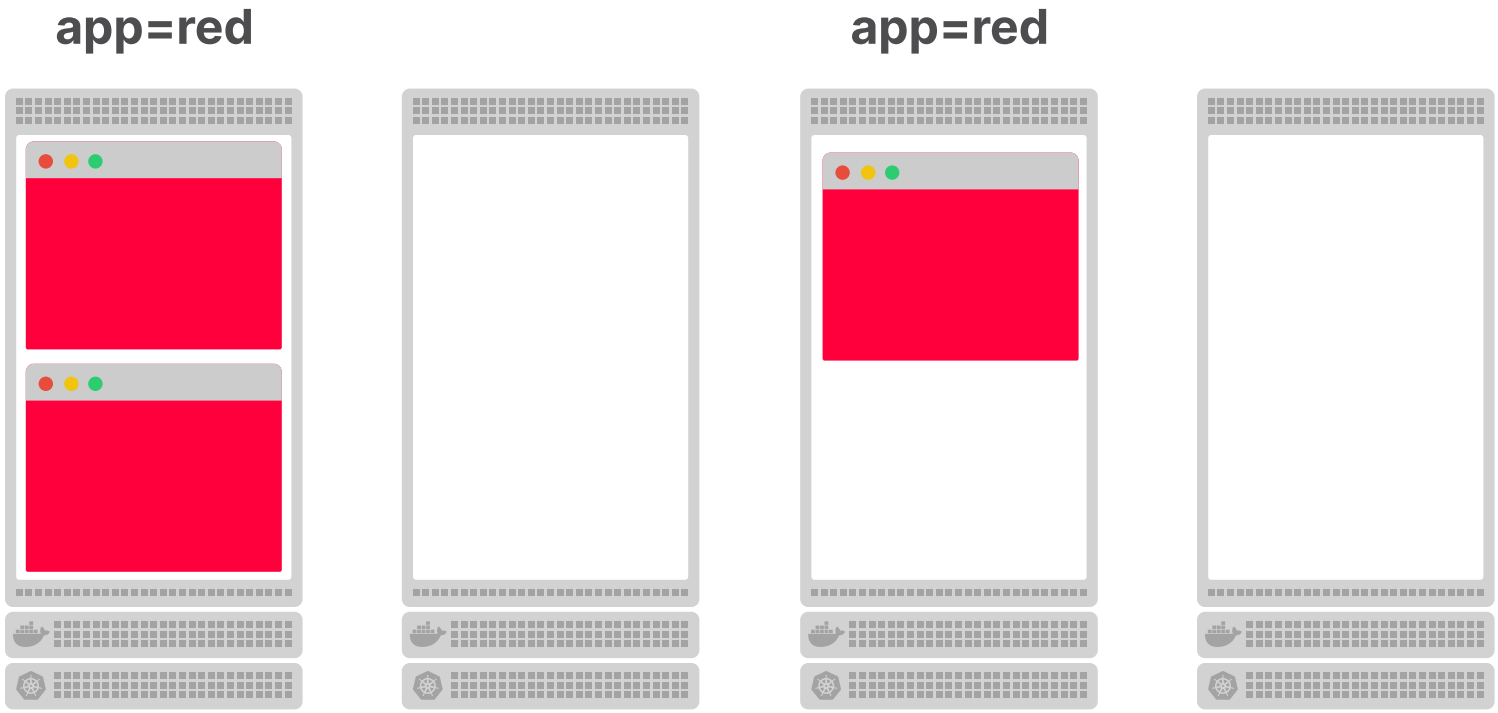


```
~$ kubectl label nodes \
 worker1 app=red
```

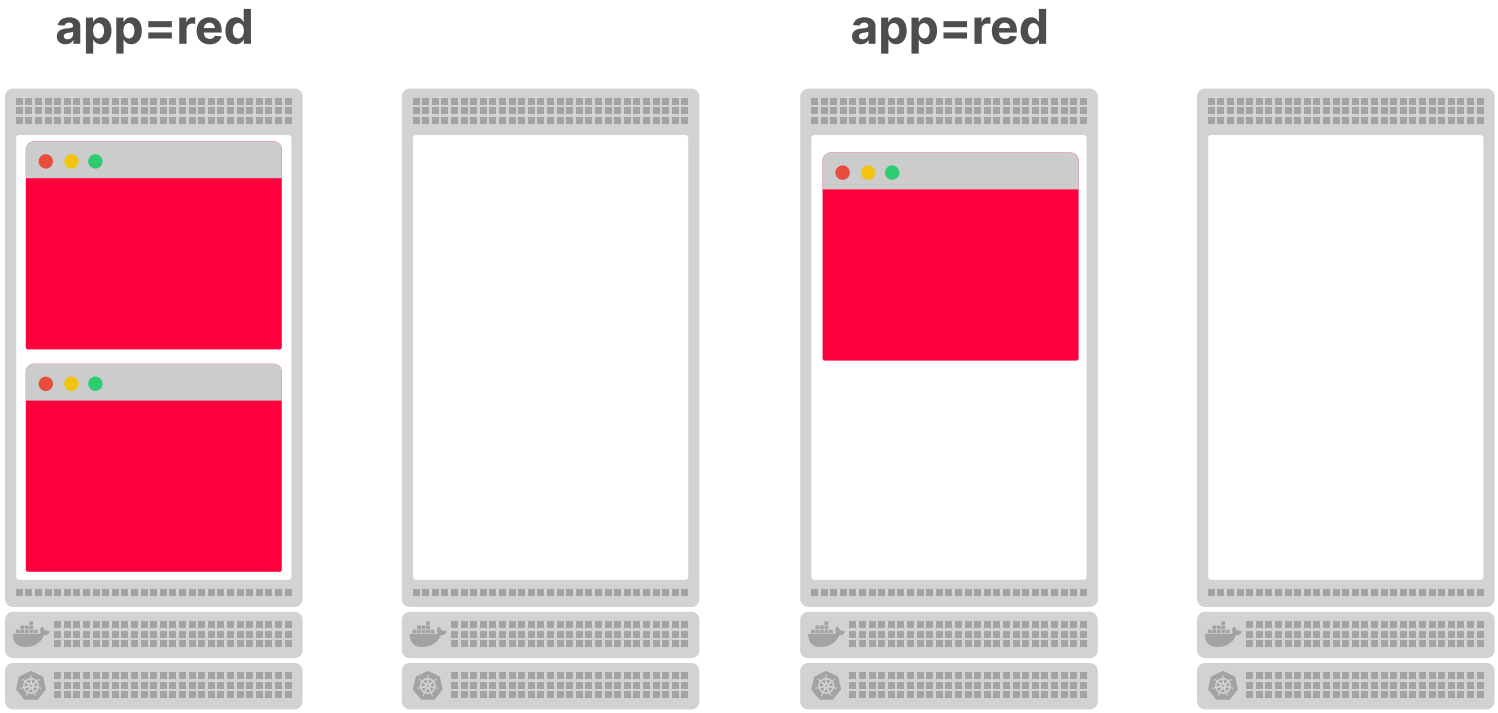

app=red
app=red

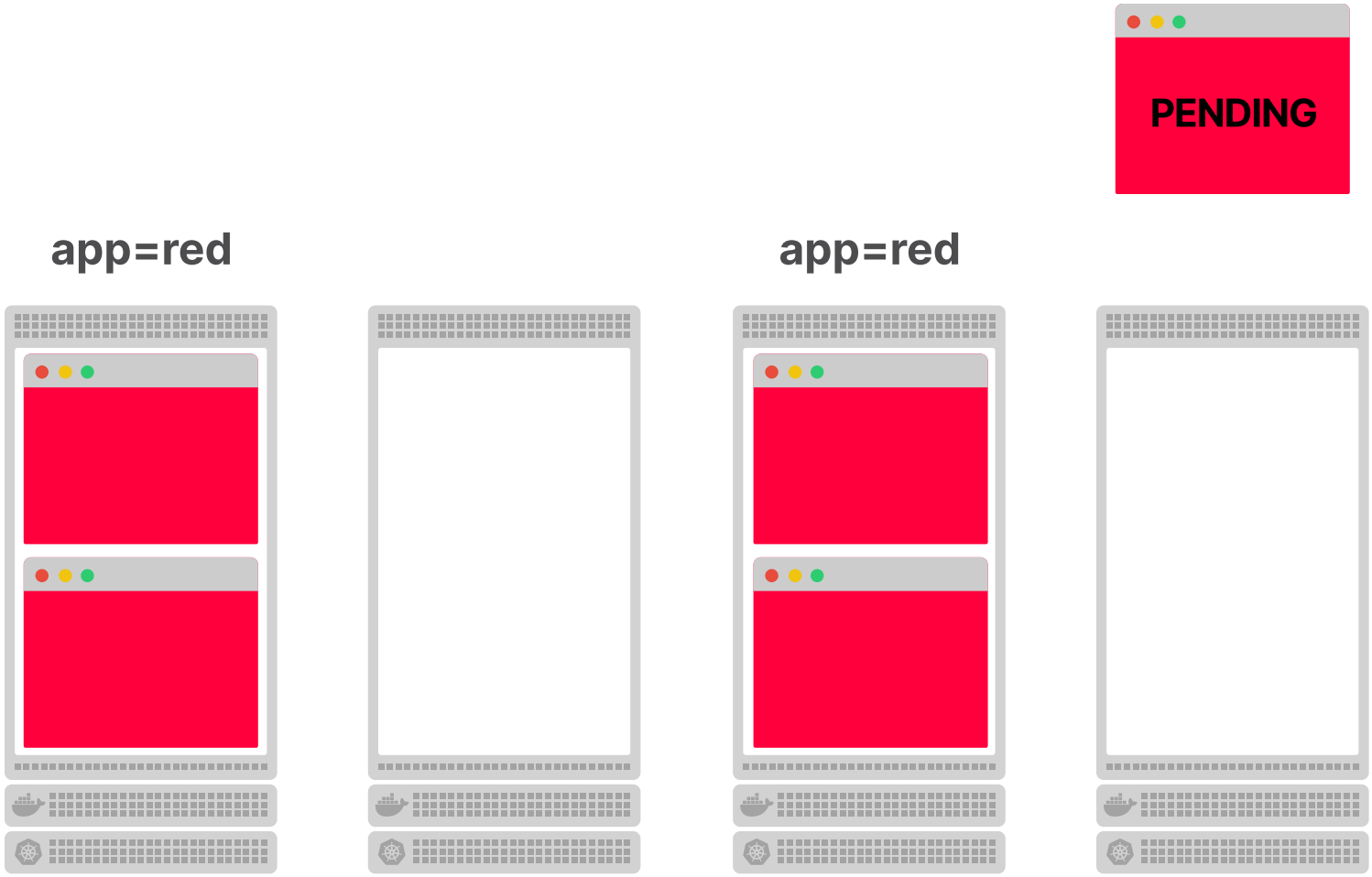
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ns-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      affinity:
        nodeAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            nodeSelectorTerms:
            - matchExpressions:
              - key: app
                operator: In
                values:
                - red
      containers:
      - name: nginx
        image: nginx:1.7.9
```


app=red
app=red

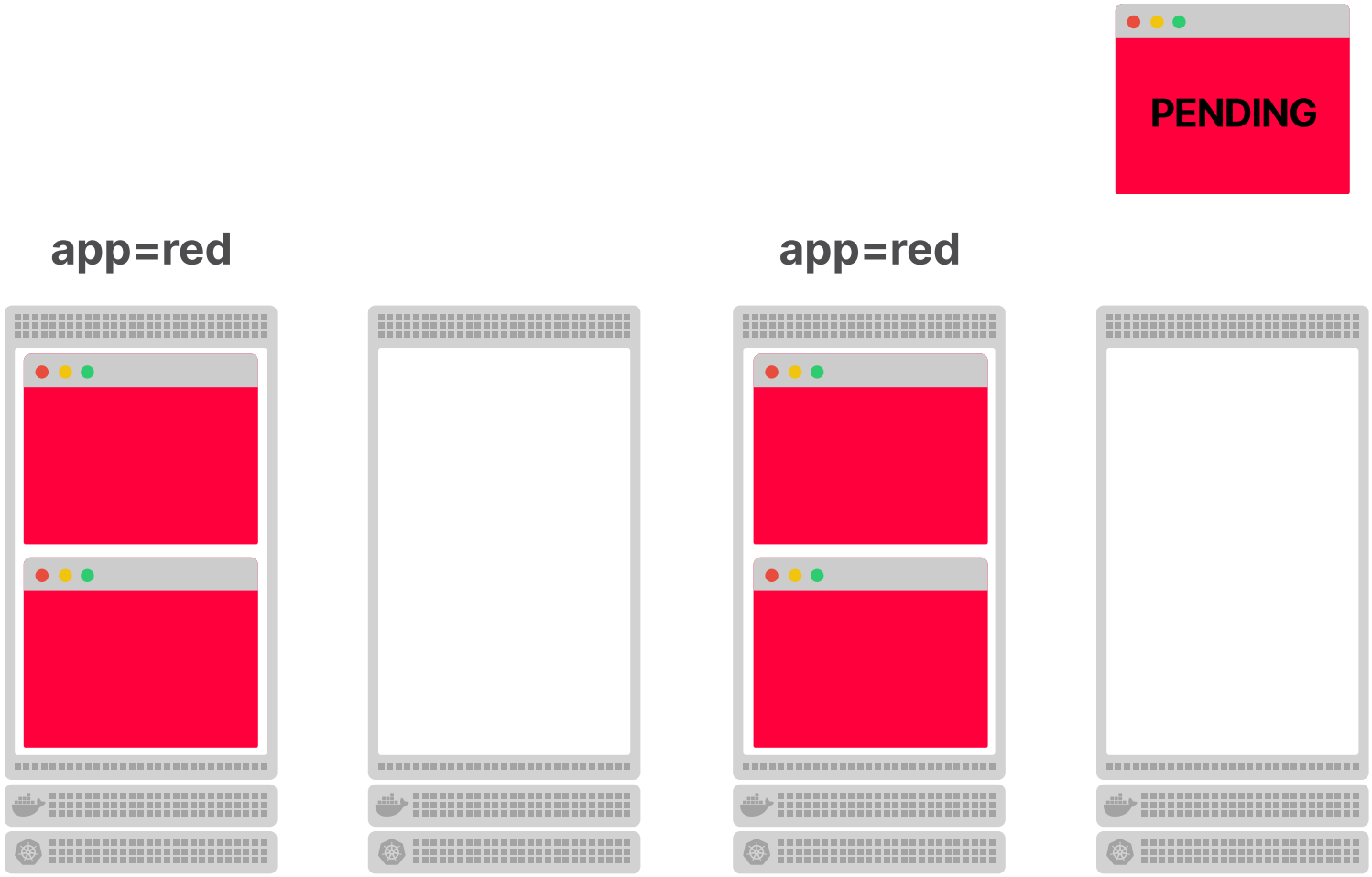

app=red
app=red

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ns-deployment
spec:
  replicas: 5
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      affinity:
        nodeAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            nodeSelectorTerms:
            - matchExpressions:
              - key: app
                operator: In
                values:
                - red
      containers:
      - name: nginx
        image: nginx:1.7.9
```


app=red
app=red

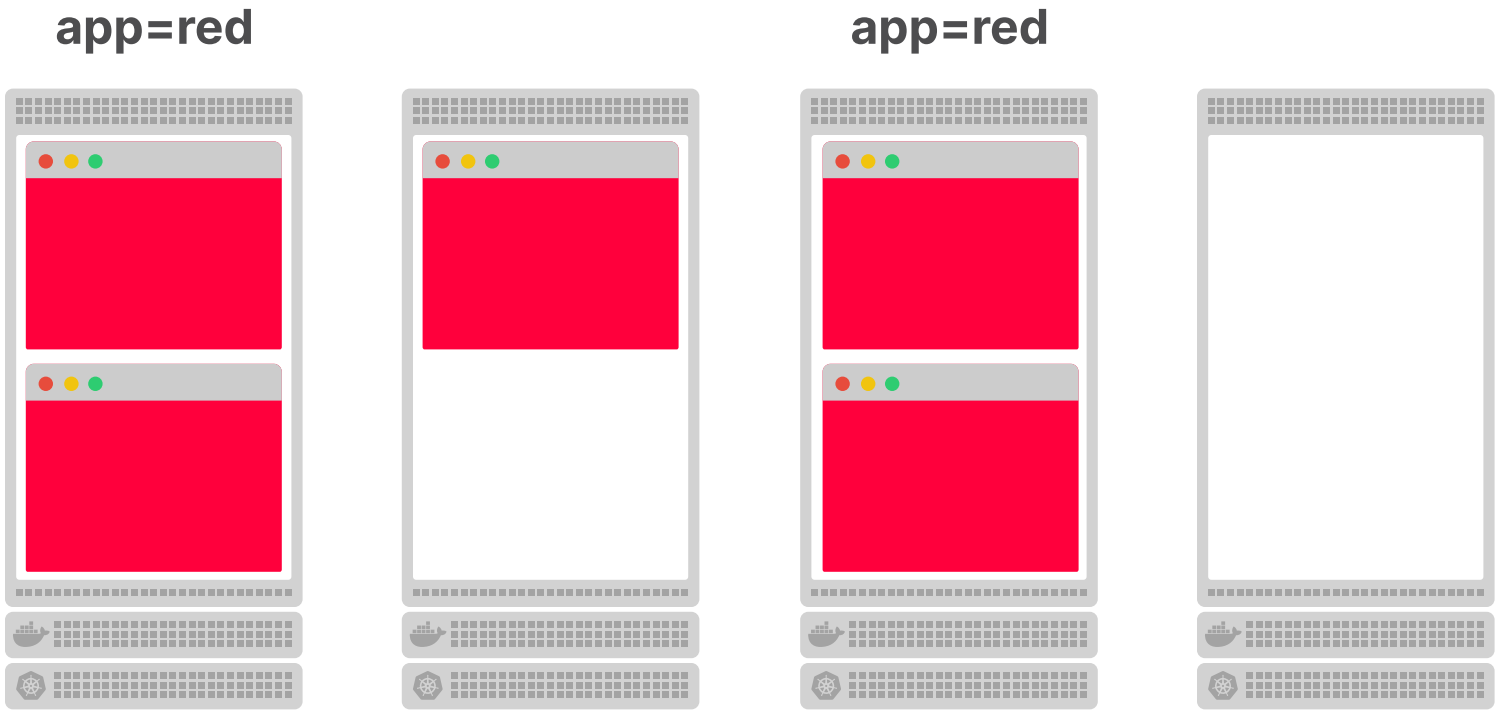

PENDING
app=red
app=red

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ns-deployment
spec:
  replicas: 5
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      affinity:
        nodeAffinity:
          preferredDuringSchedulingIgnoredDuringExecution:
          - weight: 100
            preference:
              matchExpressions:
              - key: app
                operator: In
                values:
                - red
      containers:
      - name: nginx
        image: nginx:1.7.9
```


PENDING
app=red
app=red

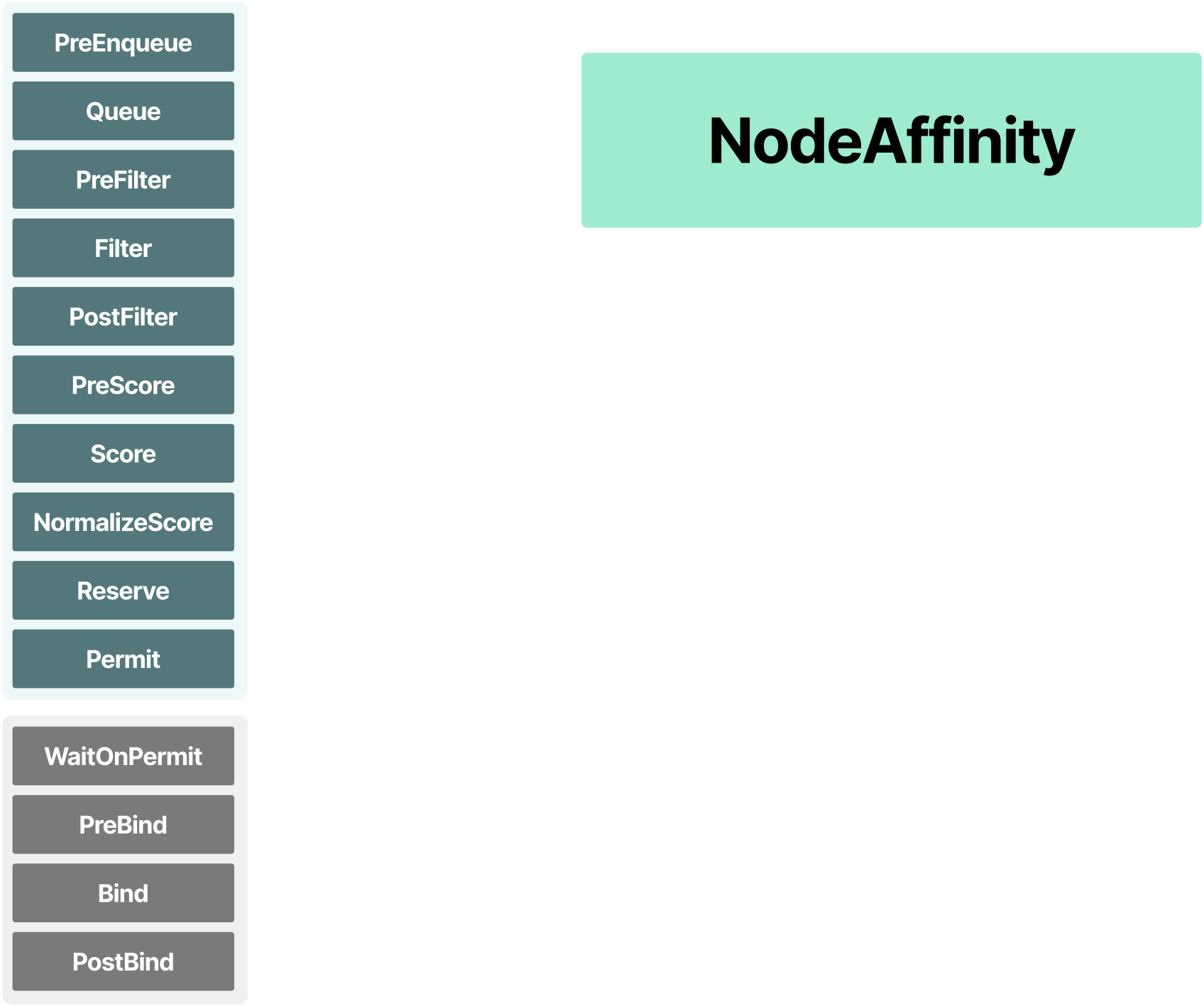


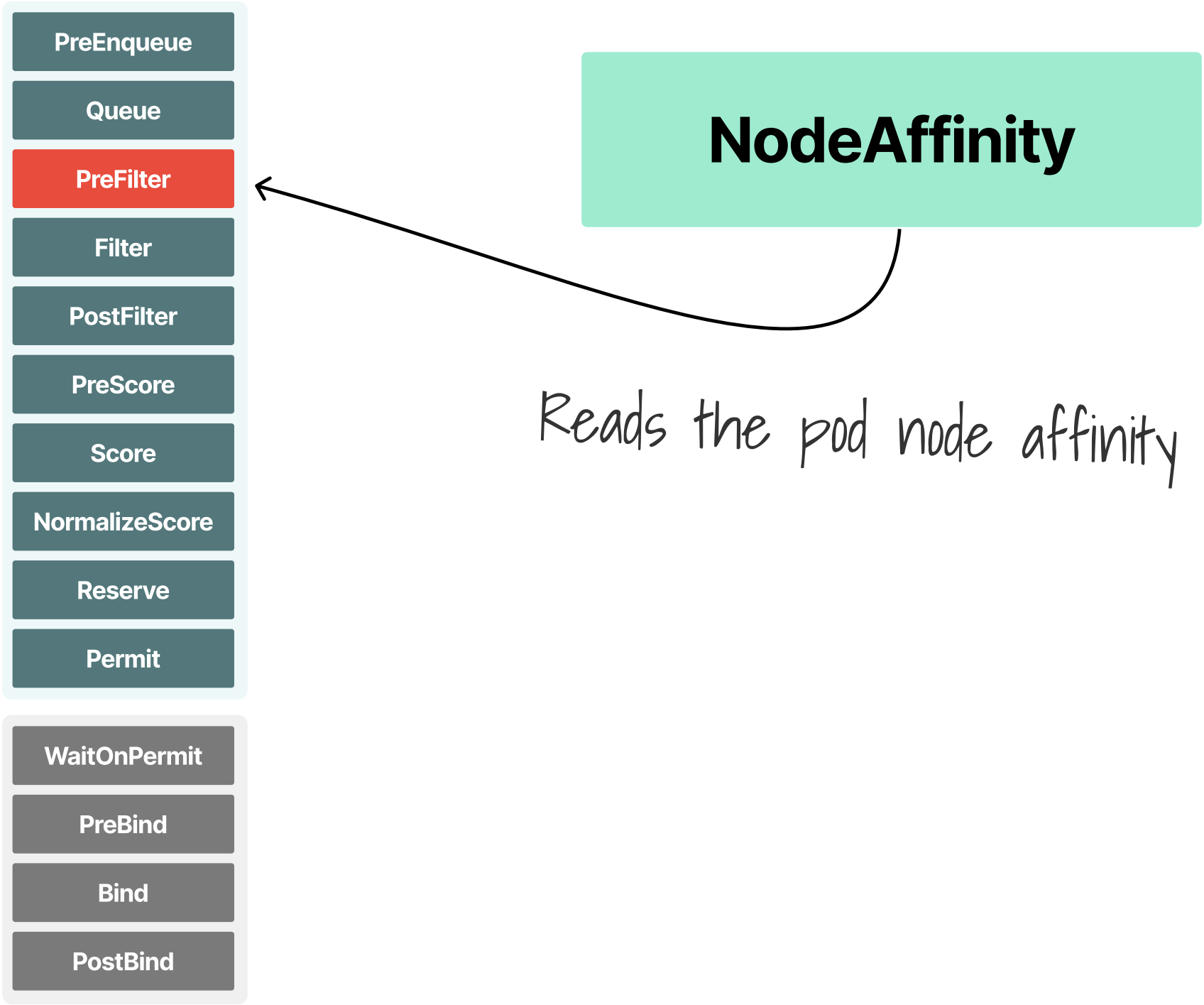
```
~$ kubectl apply -f deploy.yaml
```

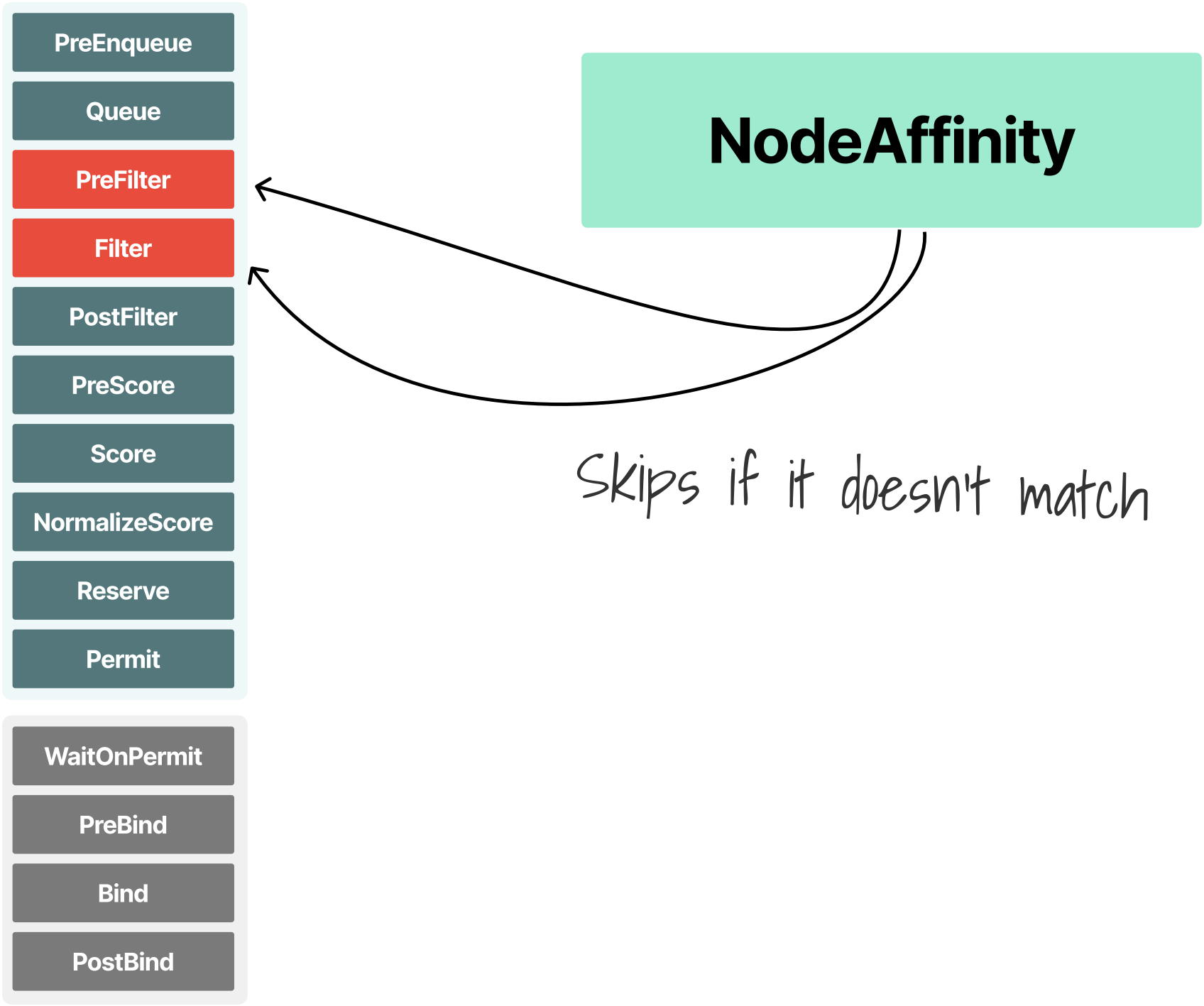

app=red
app=red

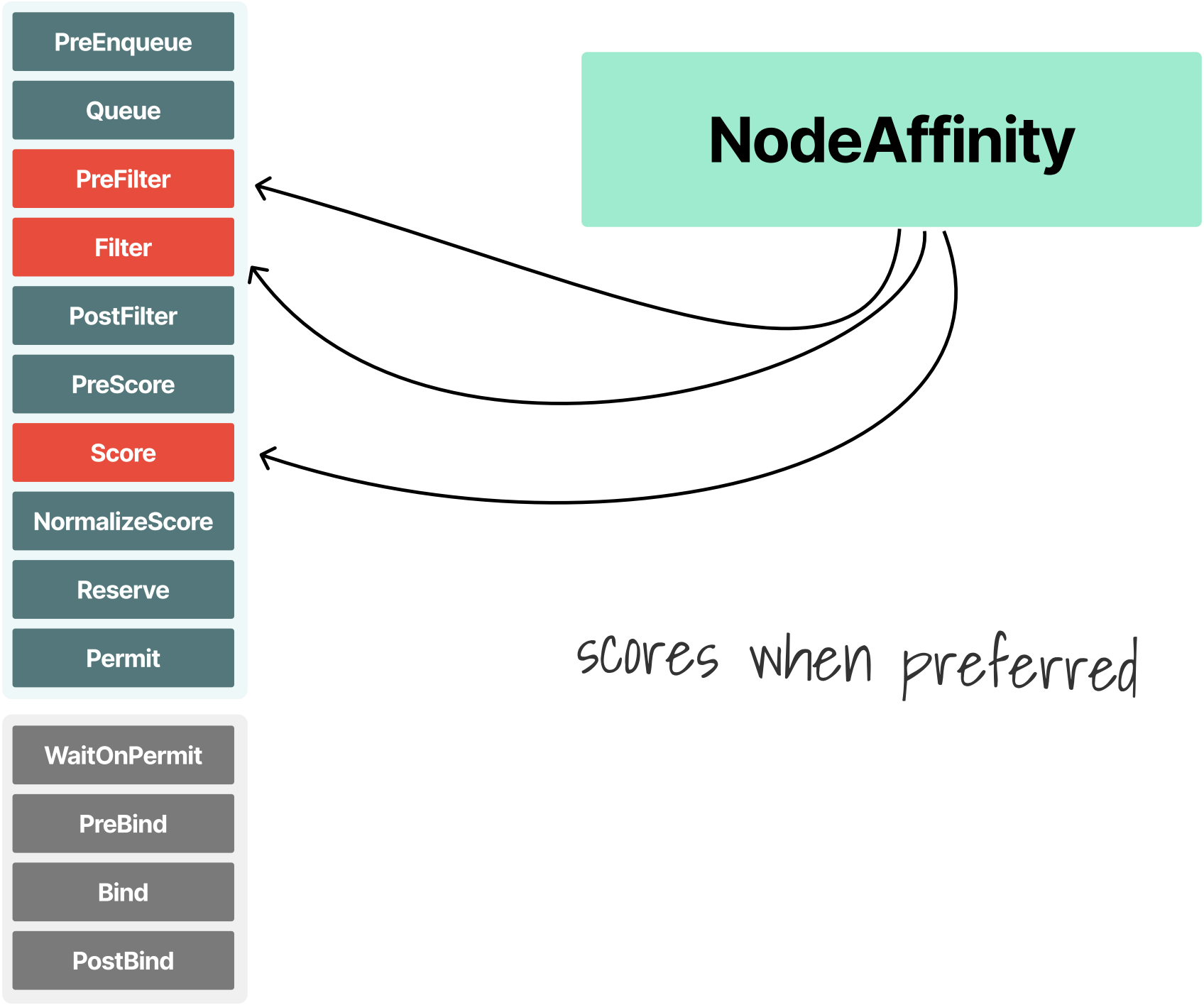
Node Affinity plugin

02

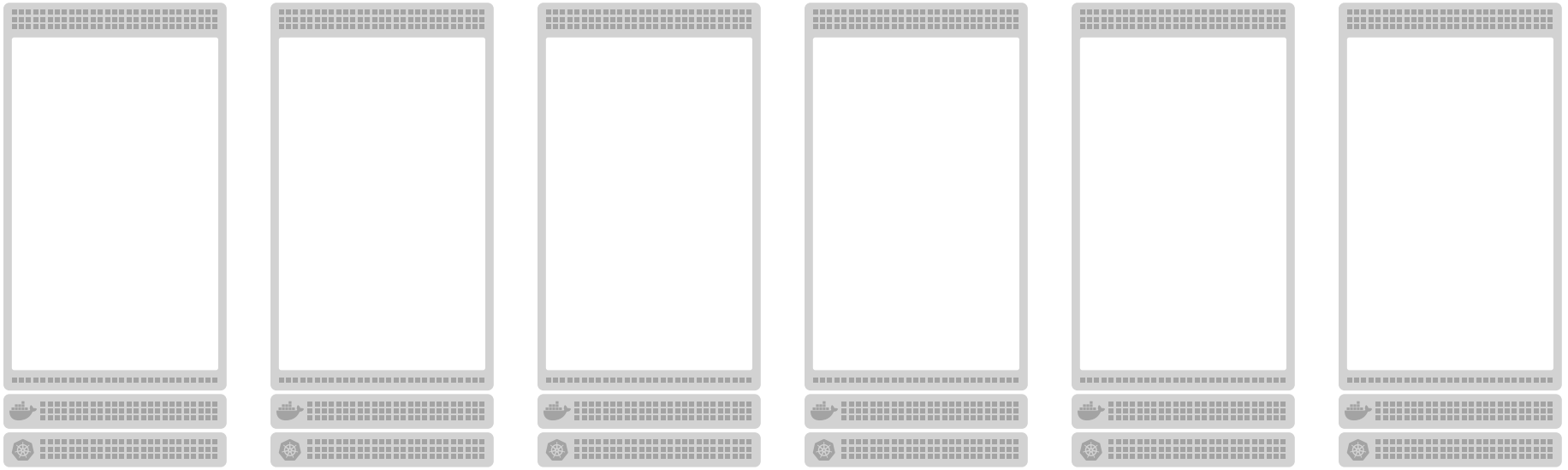

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeAffinity

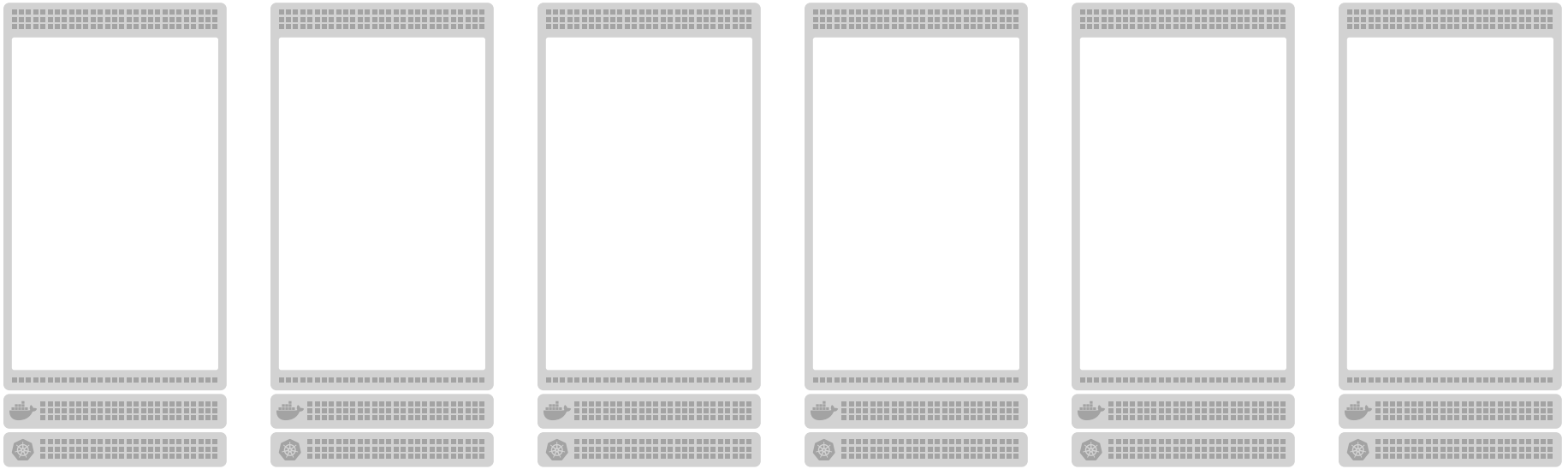

PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeAffinity
Reads the pod node affinity


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeAffinity
Skips if it doesn't match


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
NodeAffinity
scores when preferred

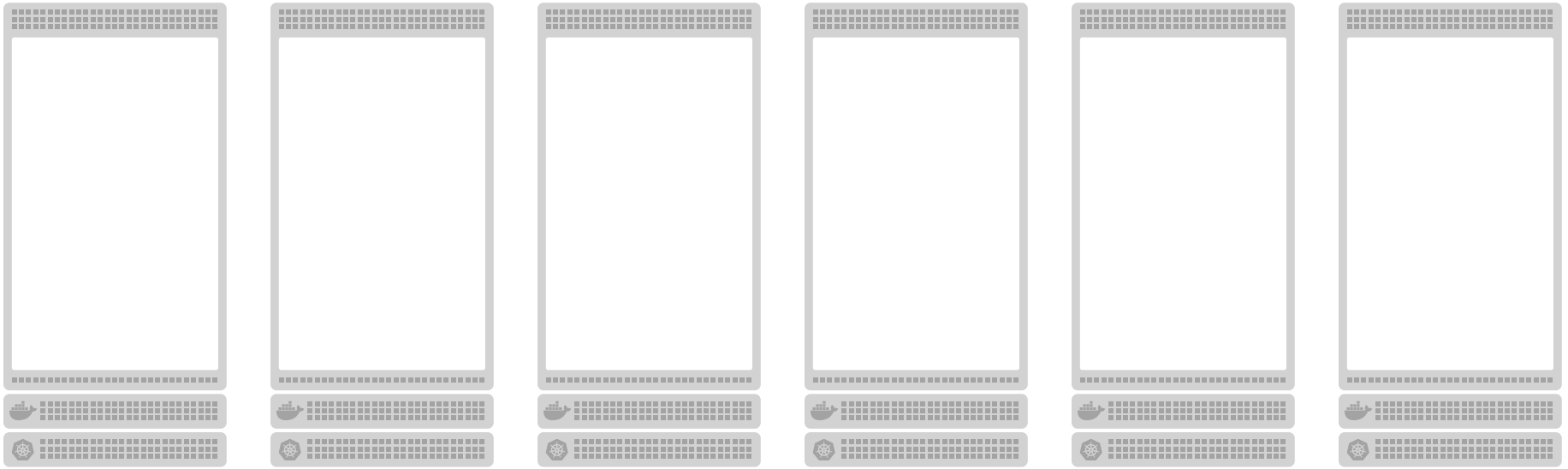
Labels and topology

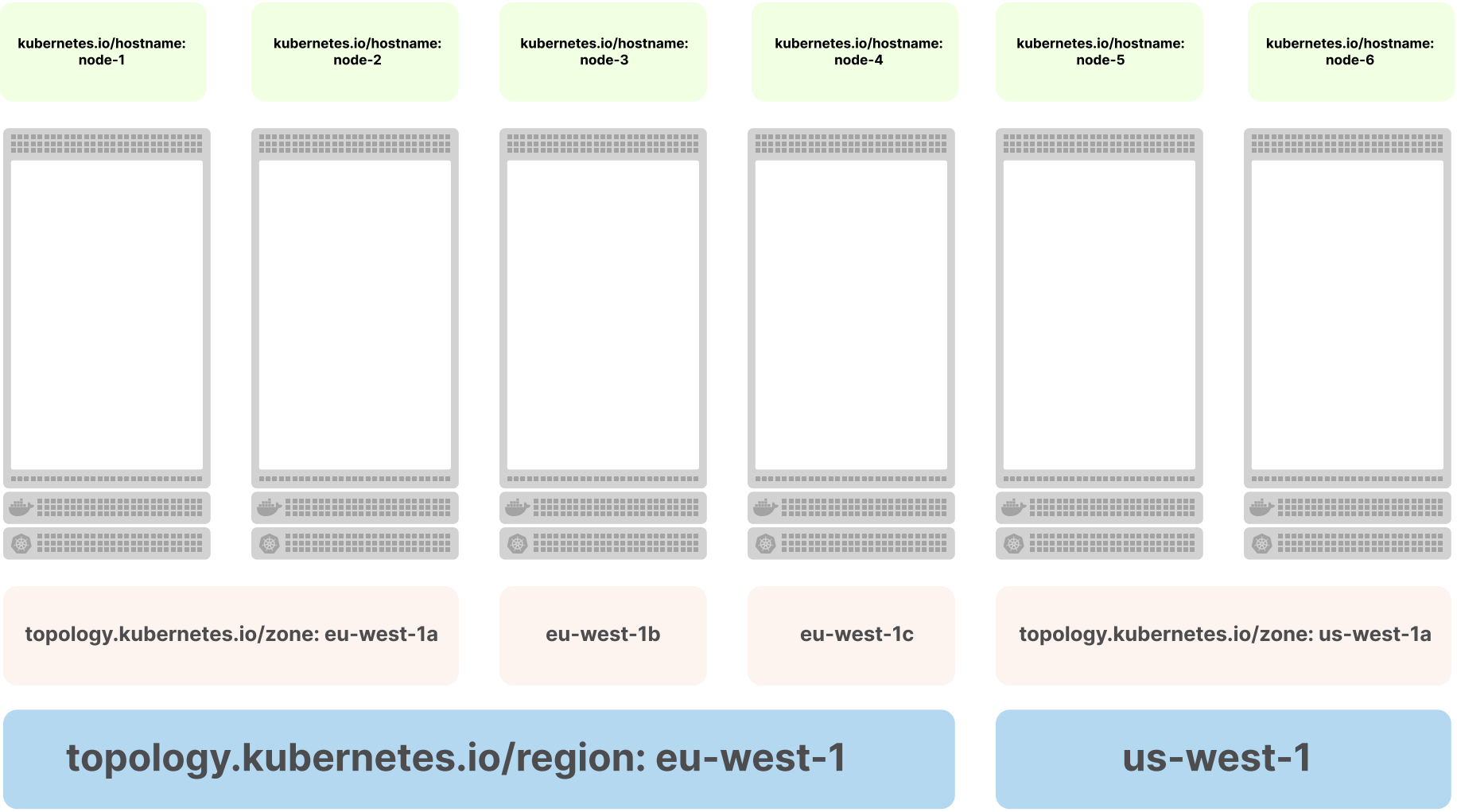




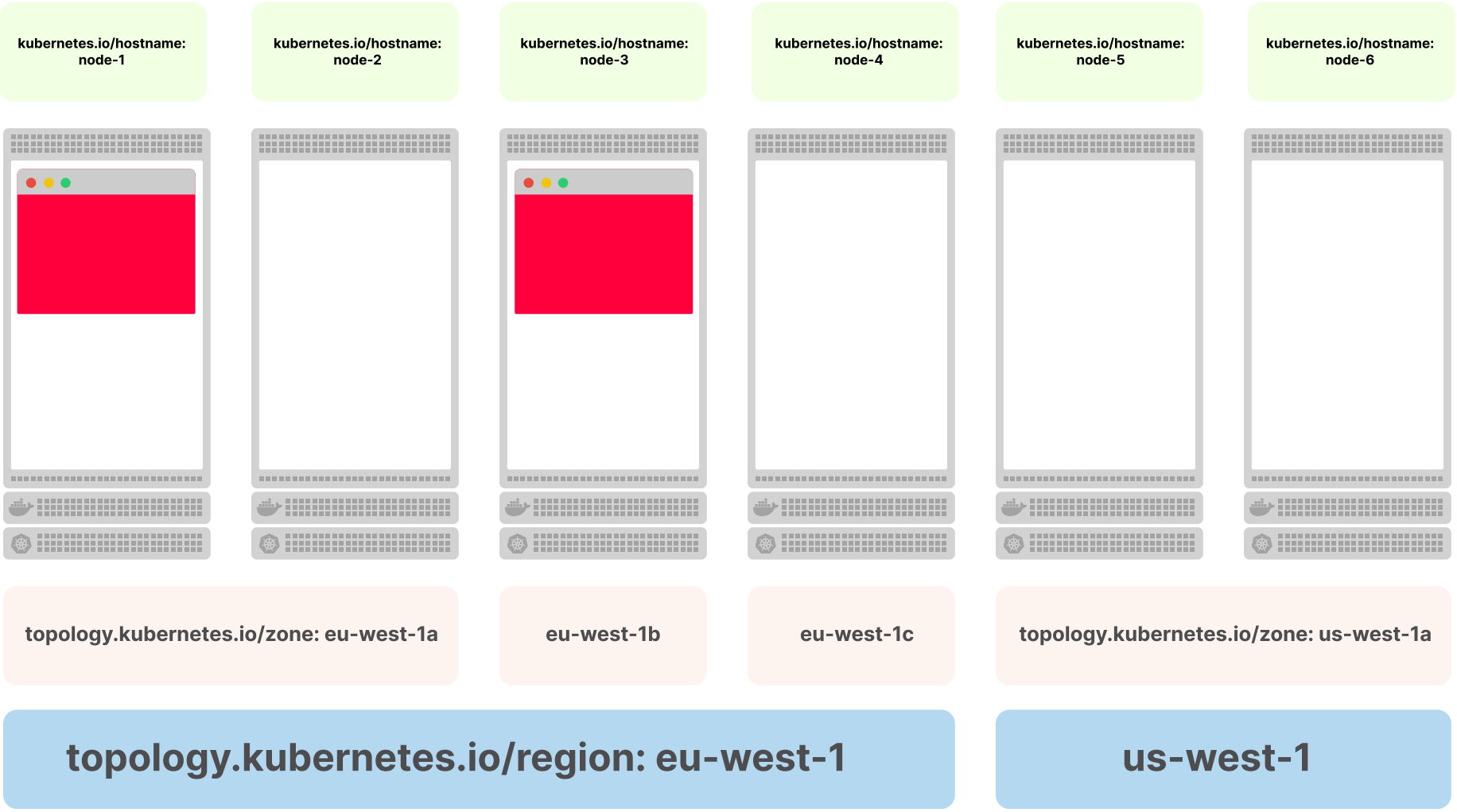
topology.kubernetes.io/region: eu-west-1

us-west-1

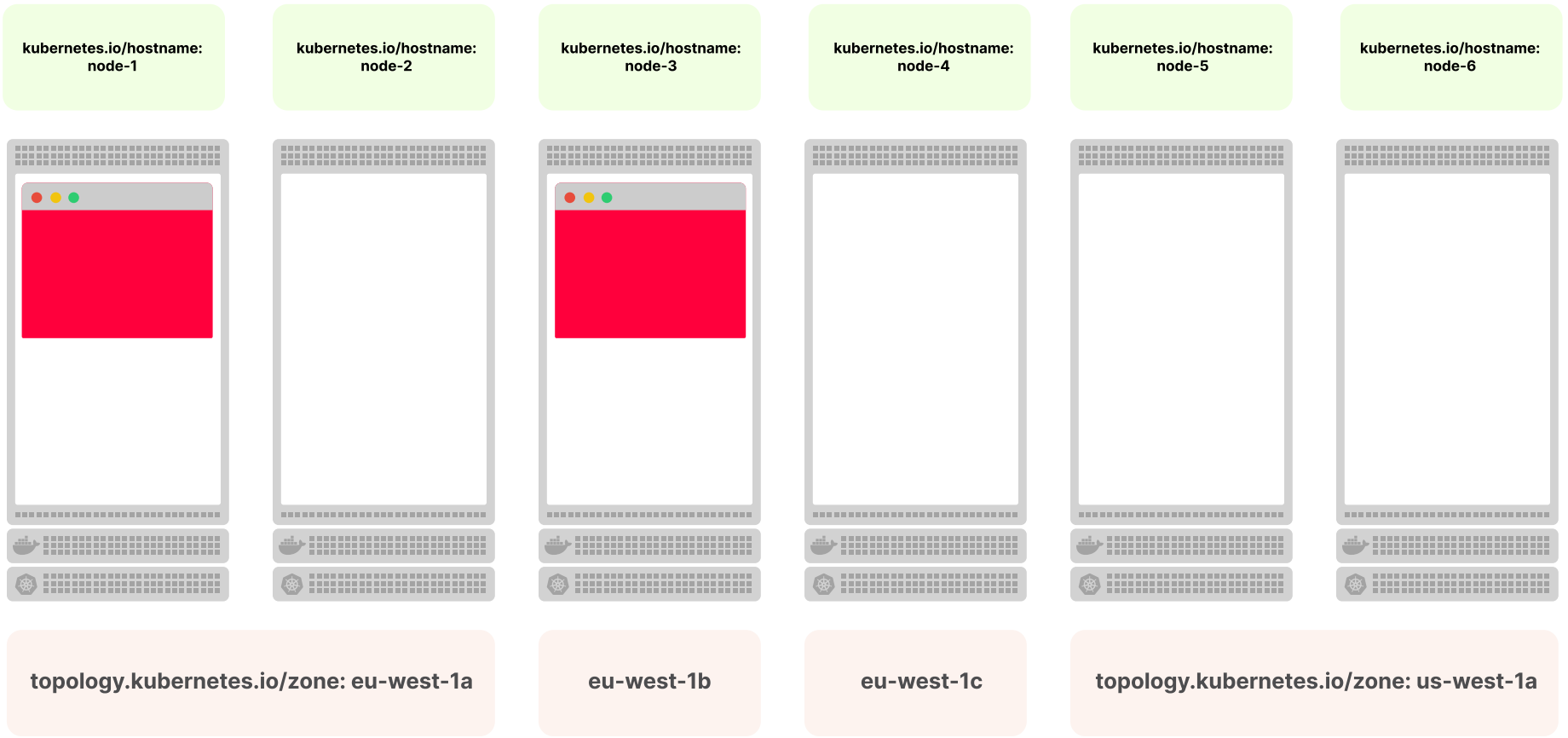

topology.kubernetes.io/zone: eu-west-1a
eu-west-1b
eu-west-1c
topology.kubernetes.io/zone: us-west-1a
topology.kubernetes.io/region: eu-west-1
us-west-1


kubernetes.io/hostname: node-1
kubernetes.io/hostname: node-2
kubernetes.io/hostname: node-3
kubernetes.io/hostname: node-4
kubernetes.io/hostname: node-5
kubernetes.io/hostname: node-6
topology.kubernetes.io/zone: eu-west-1a
eu-west-1b
eu-west-1c
topology.kubernetes.io/zone: us-west-1a
topology.kubernetes.io/region: eu-west-1
us-west-1

are they "together"?

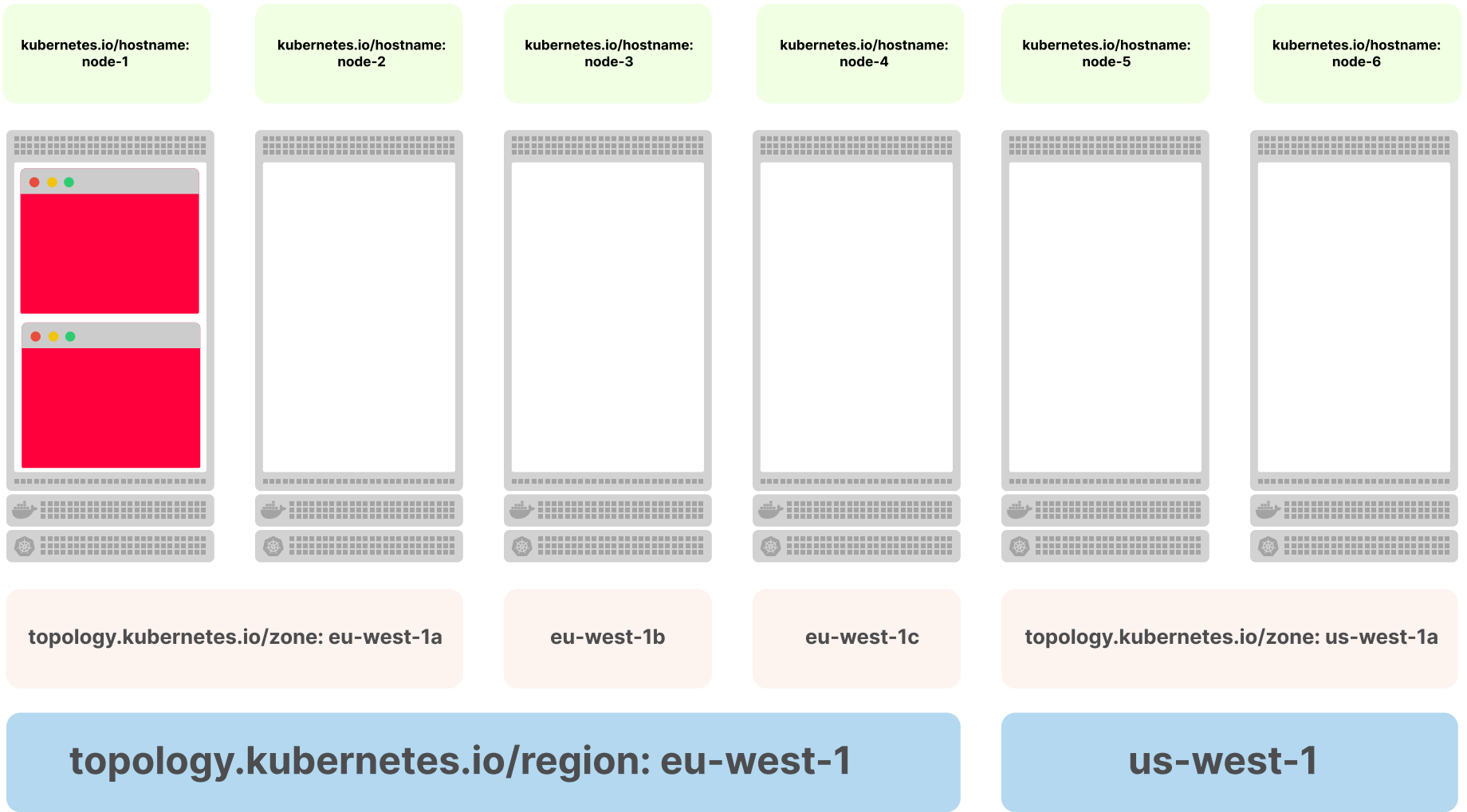


are they "together"?



✅ Region ❌ Zone ❌ Host

are they "together"?



✓ Region ✓ Zone ✓ Host

Pod affinity/anti-affinity

03

Pod labels

Pod affinity/anti-affinity

Pod labels

Namespaces

Pod affinity/anti-affinity

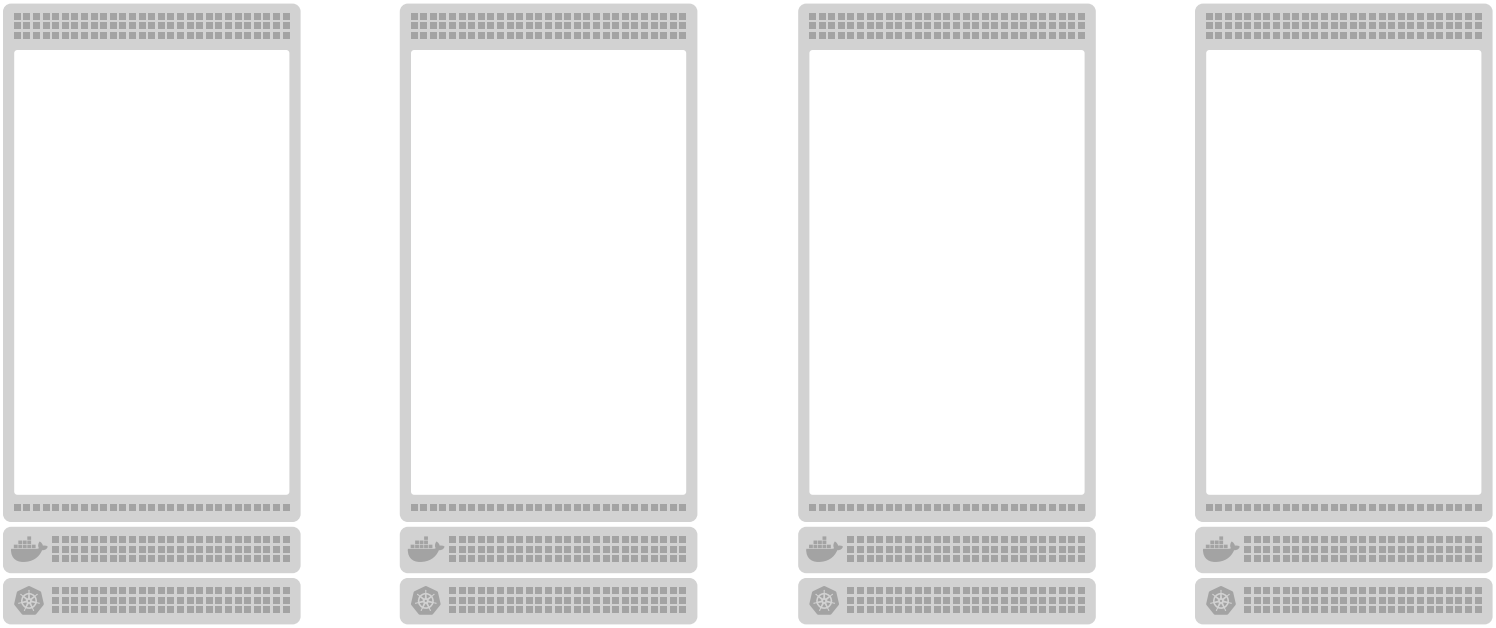
Pod labels

Namespaces

Topology

You can reuse labels from a Service selector

Pod affinity/anti-affinity

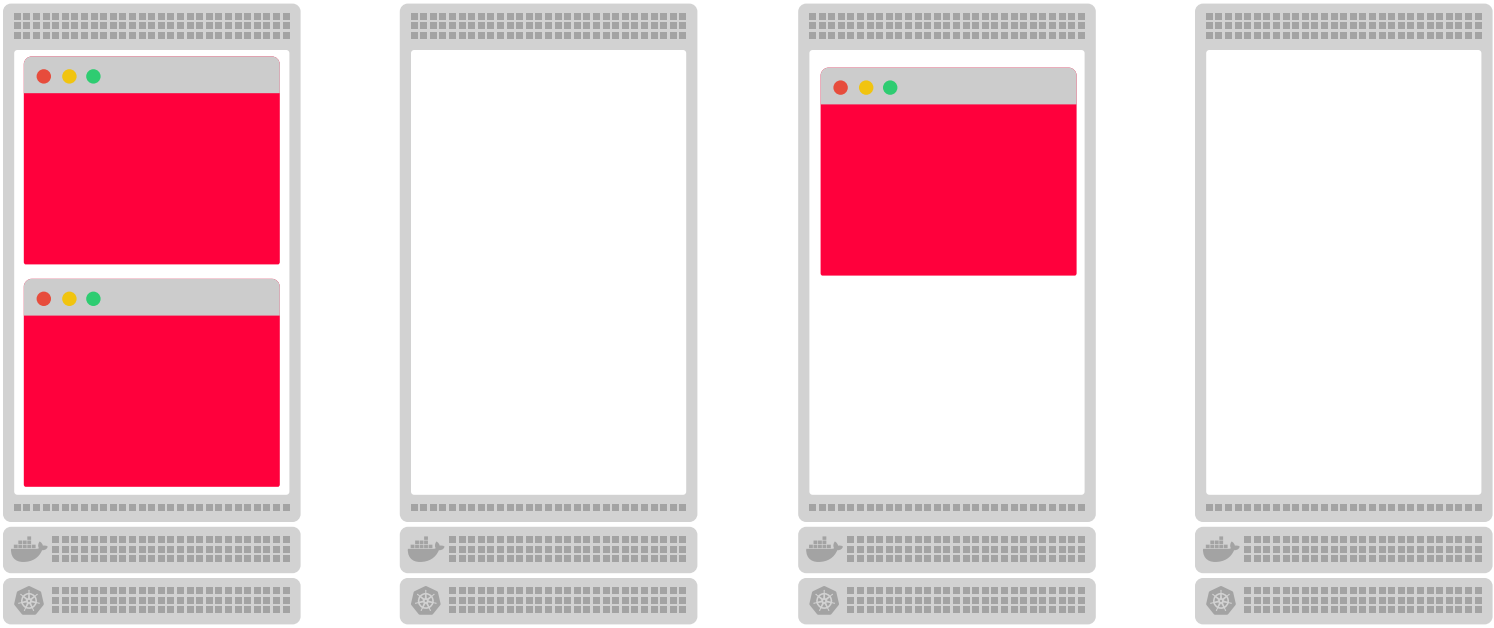


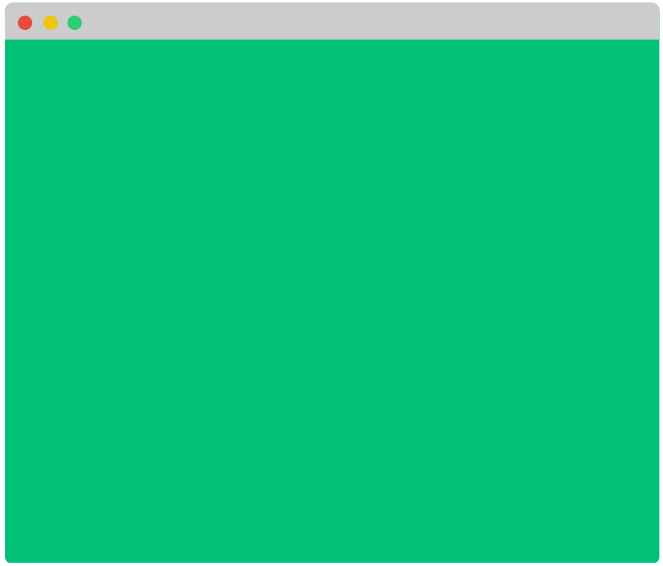


LABELS

app: red

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: red-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: red
  template:
    metadata:
      labels:
        app: red
    spec:
      containers:
      - name: nginx
        image: nginx:1.7.9
```



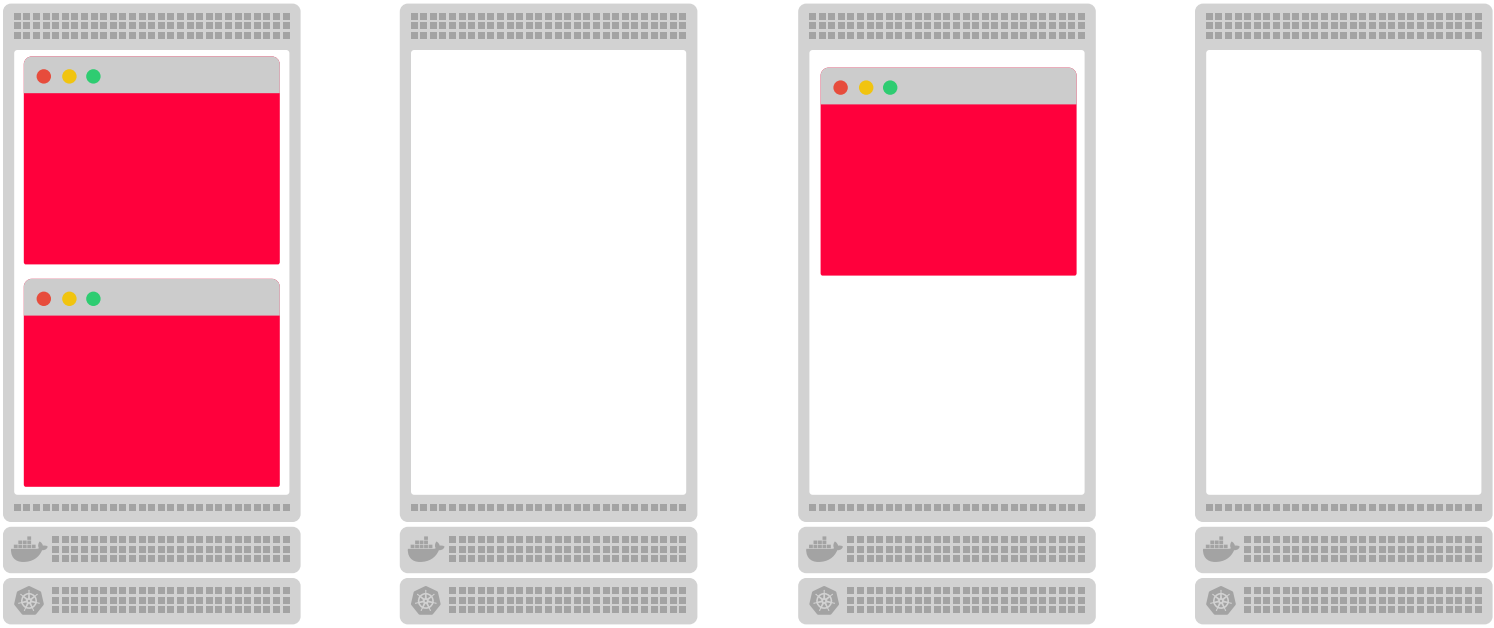


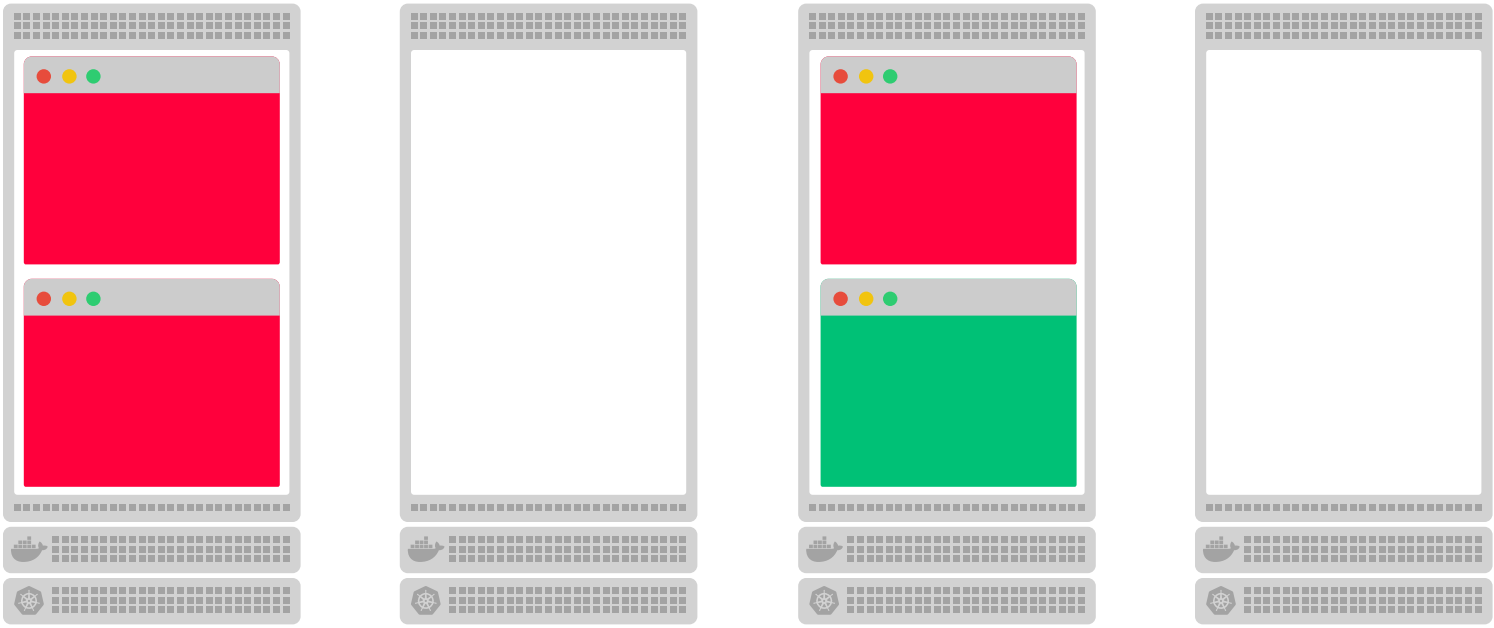
LABELS

app: green

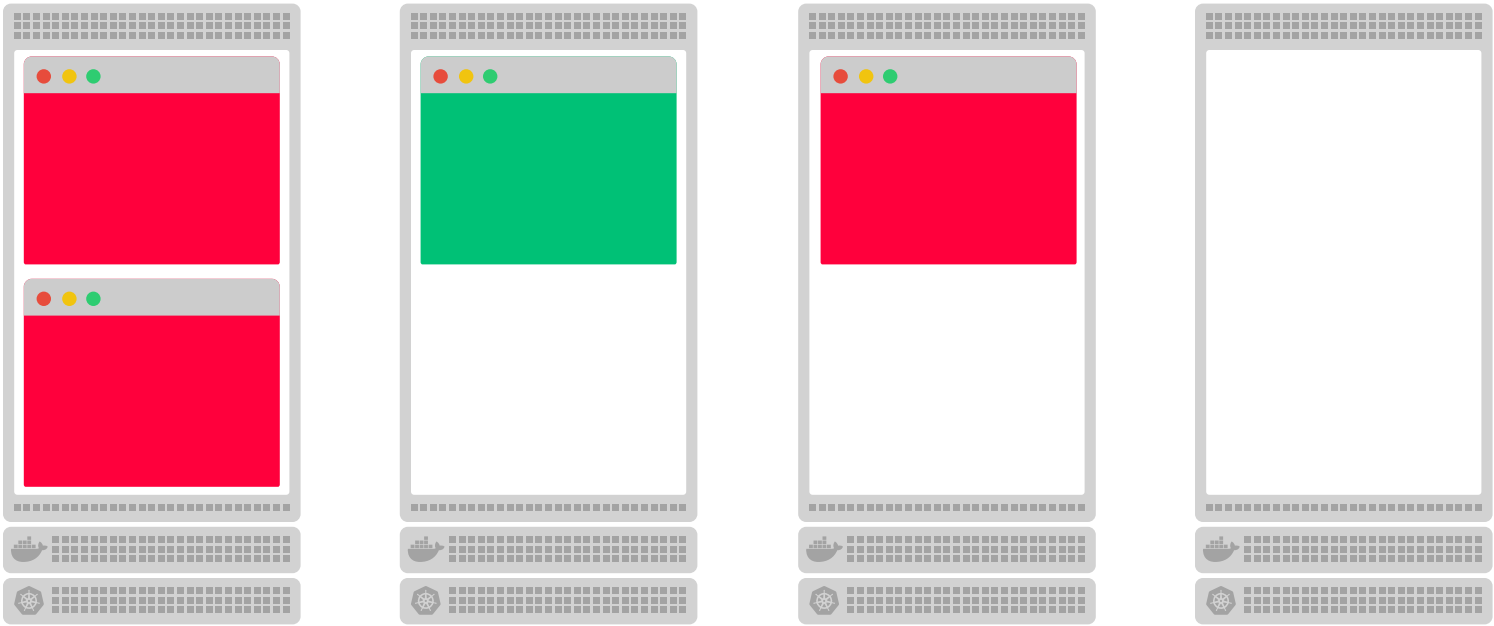
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: green-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: green
  template:
    metadata:
      labels:
        app: green
    spec:
      affinity:
        podAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
              - key: app
                operator: In
                values:
                - red
            topologyKey: kubernetes.io/hostname
      containers:
      - name: nginx
        image: nginx:1.7.9
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: green-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: green
  template:
    metadata:
      labels:
        app: green
    spec:
      affinity:
        podAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
              - key: app
                operator: In
                values:
                - red
            topologyKey: kubernetes.io/hostname
      containers:
      - name: nginx
        image: nginx:1.7.9
```

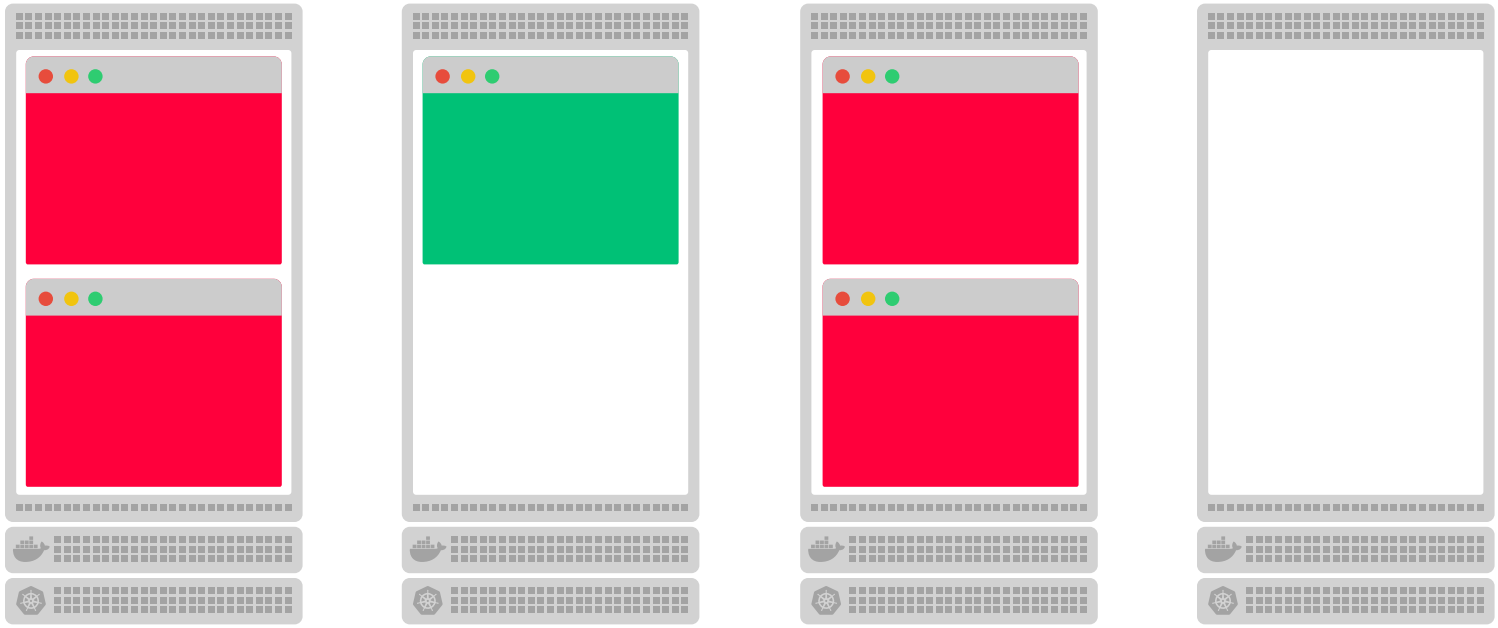




```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: green-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: green
  template:
    metadata:
      labels:
        app: green
    spec:
      affinity:
        podAntiAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
              - key: app
                operator: In
                values:
                - red
            topologyKey: kubernetes.io/hostname
      containers:
      - name: nginx
        image: nginx:1.7.9
```



```
kubectl scale deployment red-deployment --replicas=4
```

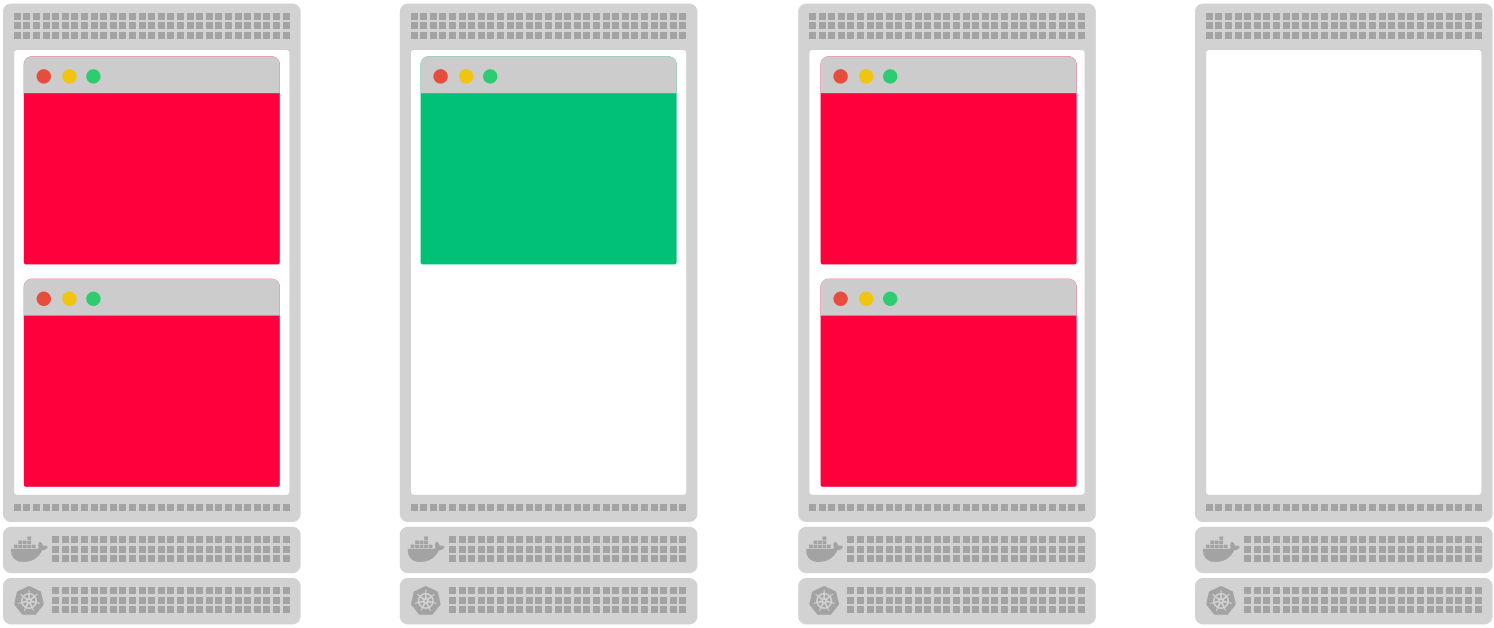


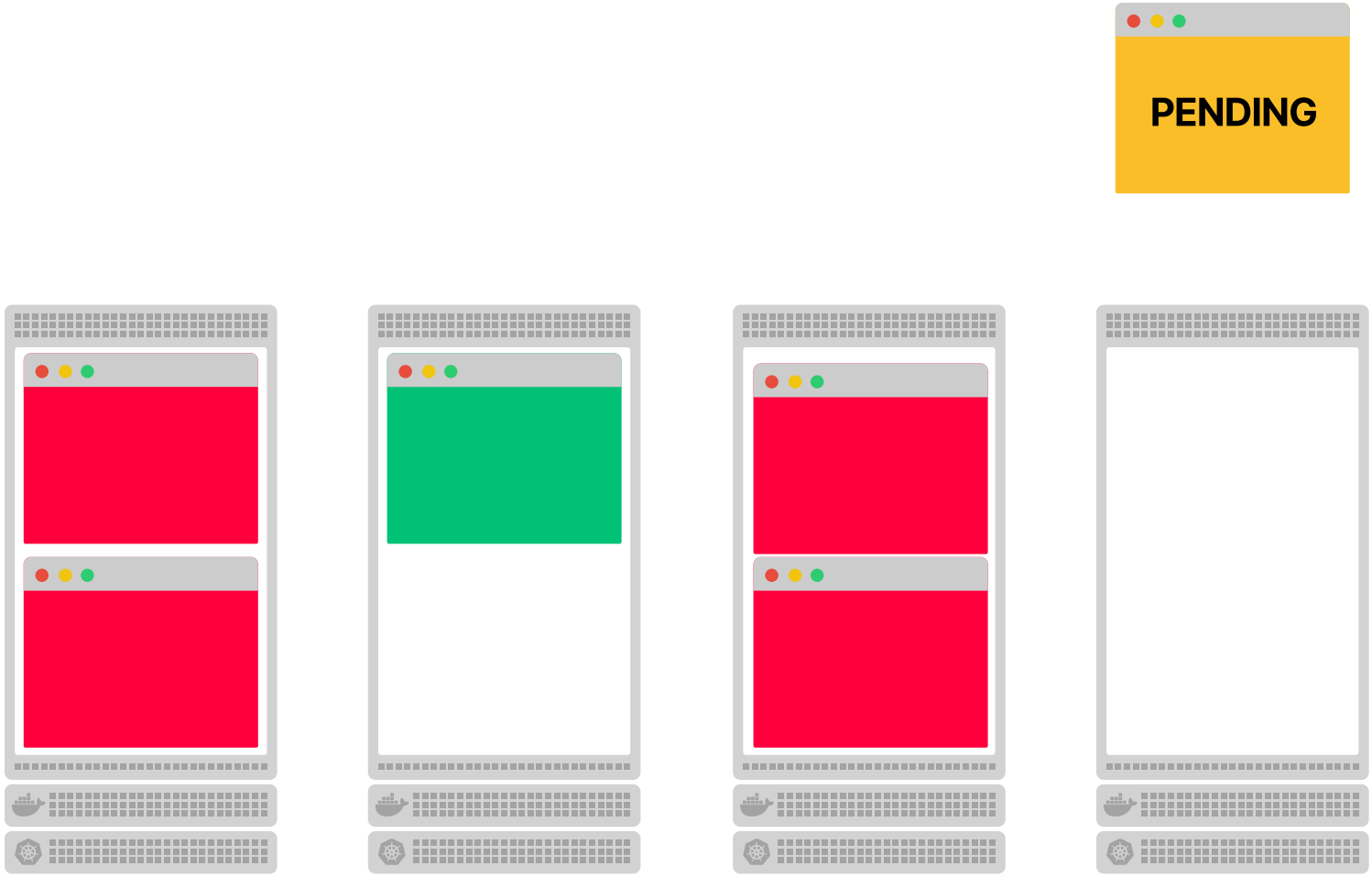


LABELS

app: yellow

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: yellow-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: yellow
  template:
    metadata:
      labels:
        app: yellow
    spec:
      affinity:
        podAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchLabels:
                app: red
            topologyKey: kubernetes.io/hostname
      containers:
      - name: nginx
        image: nginx:1.7.9
```




PENDING

Pod affinity/anti-affinity

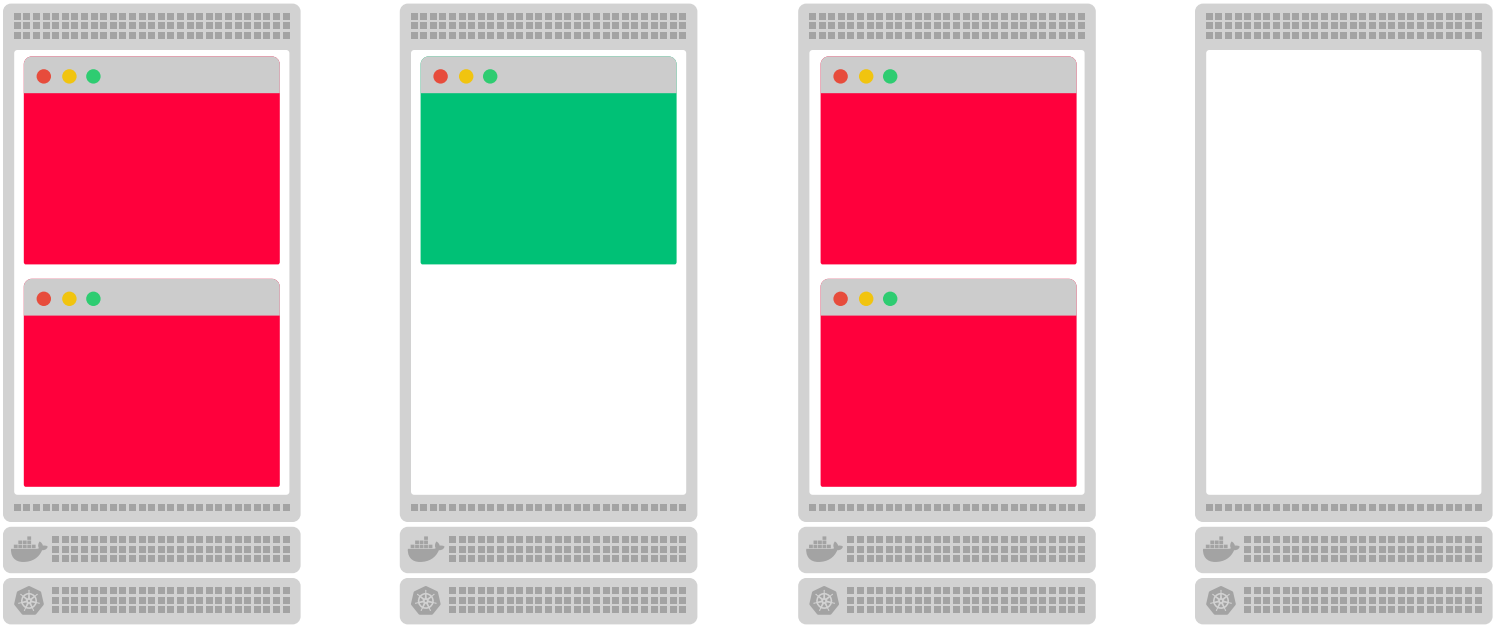
requiredDuringSchedulingIgnoredDuringExecution

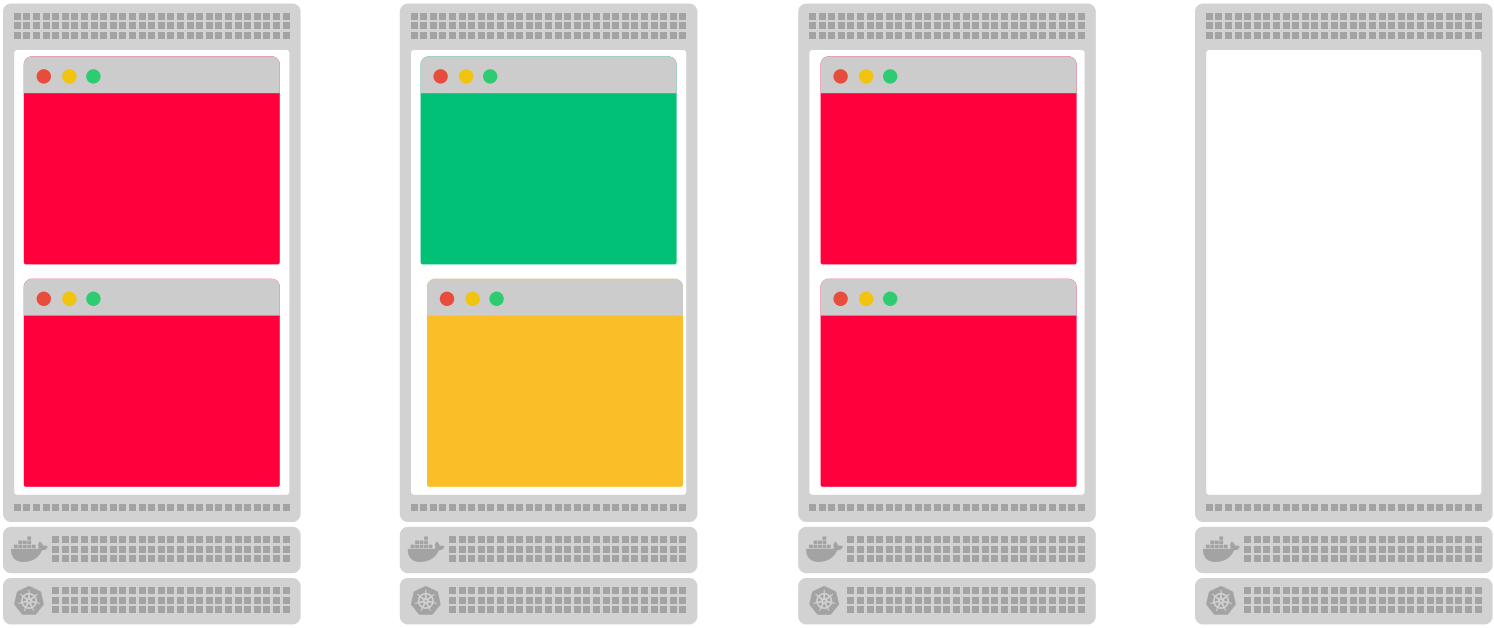
preferredDuringSchedulingIgnoredDuringExecution

Pod affinity/anti-affinity

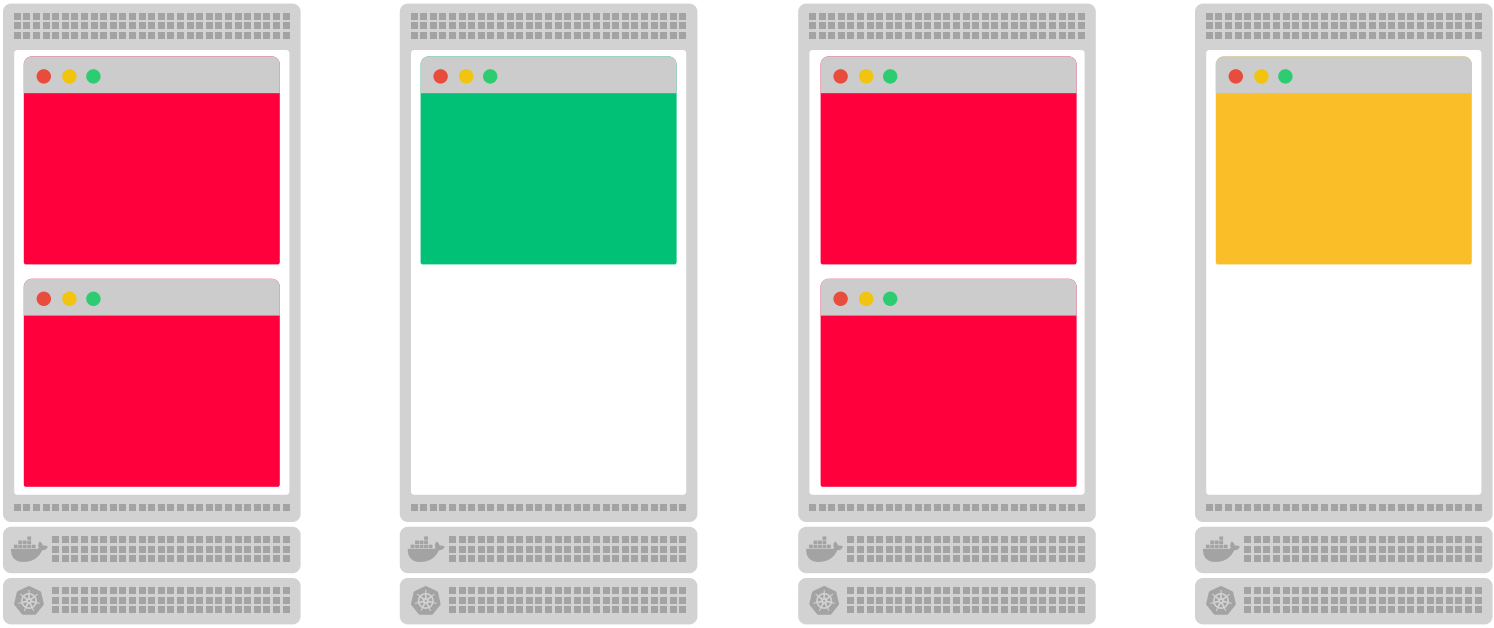
Required
Preferred

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: yellow-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: yellow
  template:
    metadata:
      labels:
        app: yellow
    spec:
      affinity:
        podAffinity:
          preferredDuringSchedulingIgnoredDuringExecution:
          - weight: 100
            podAffinityTerm:
              labelSelector:
                matchLabels:
                  app: red
              topologyKey: kubernetes.io/hostname
      containers:
      - name: nginx
        image: nginx:1.7.9
```





```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: yellow-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: yellow
  template:
    metadata:
      labels:
        app: yellow
    spec:
      affinity:
        podAntiAffinity:
          preferredDuringSchedulingIgnoredDuringExecution:
          - weight: 100
            podAffinityTerm:
              labelSelector:
                matchExpressions:
                - key: app
                  operator: In
                  values:
                  - red
                  - green
              topologyKey: kubernetes.io/hostname
      containers:
      - name: nginx
        image: nginx:1.7.9
```



Pod affinity/anti-affinity

CPU bound

Scheduling Pods on the same Node

Scheduling Pods NOT on the same node

Pod affinity/anti-affinity

CPU bound

Scheduling Pods on the same Node

Scheduling Pods NOT on the same node

Pod affinity/anti-affinity

CPU bound

Scheduling Pods on the same Node

Scheduling Pods NOT on the same node

Pod Affinity plugin

03

PreEnqueue

Queue

PreFilter

Filter

PostFilter

PreScore

Score

NormalizeScore

Reserve

Permit

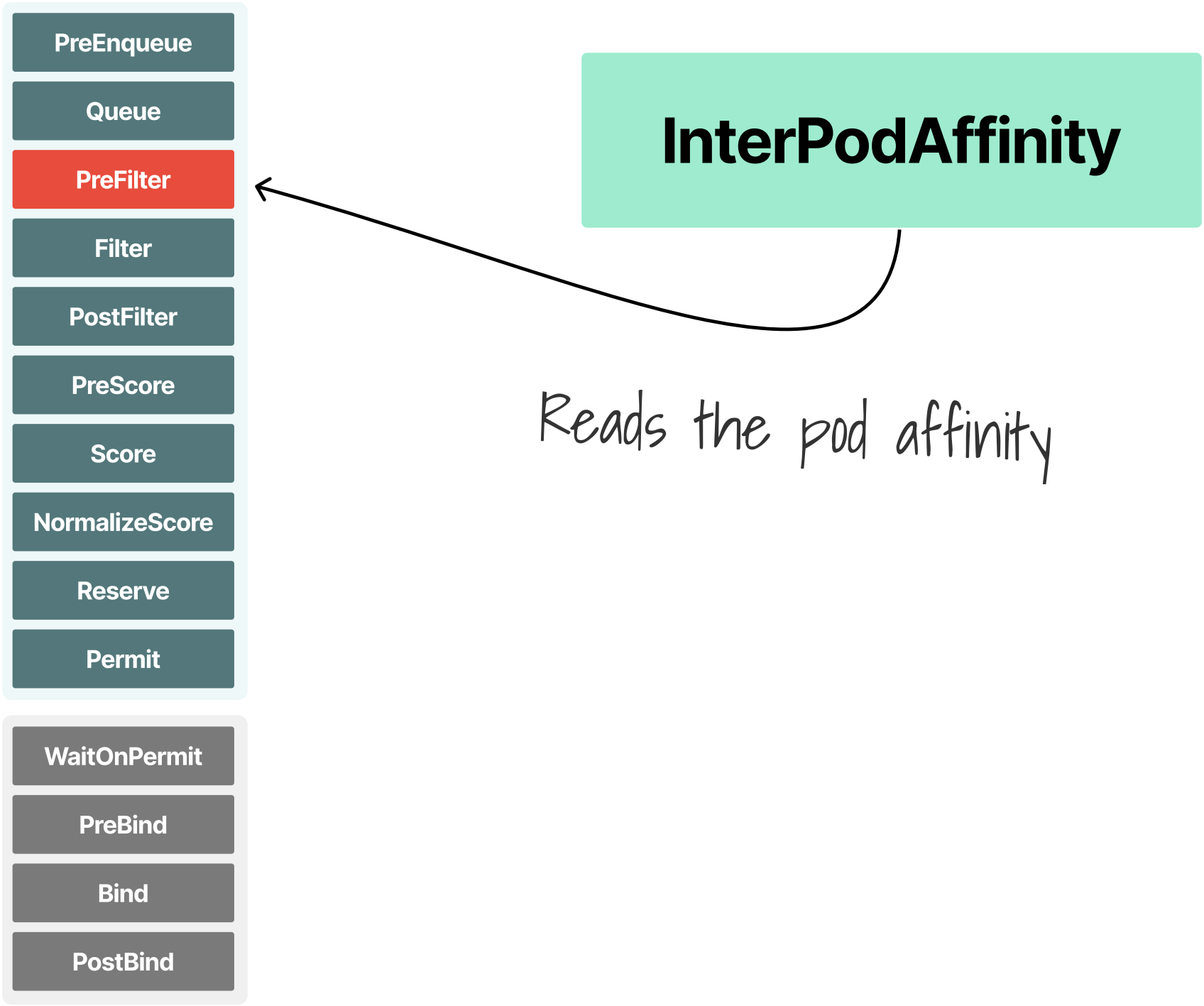
WaitOnPermit

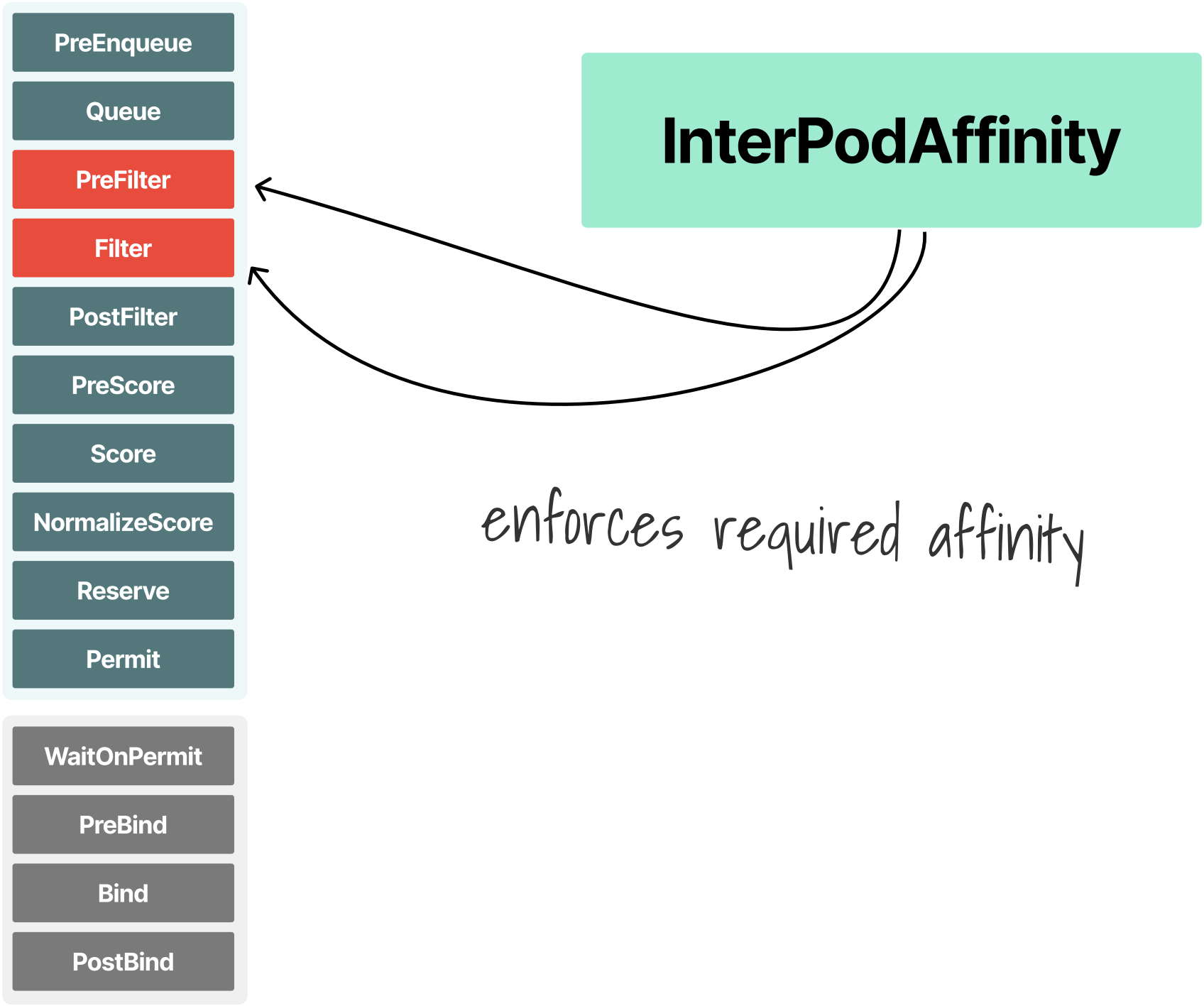
PreBind

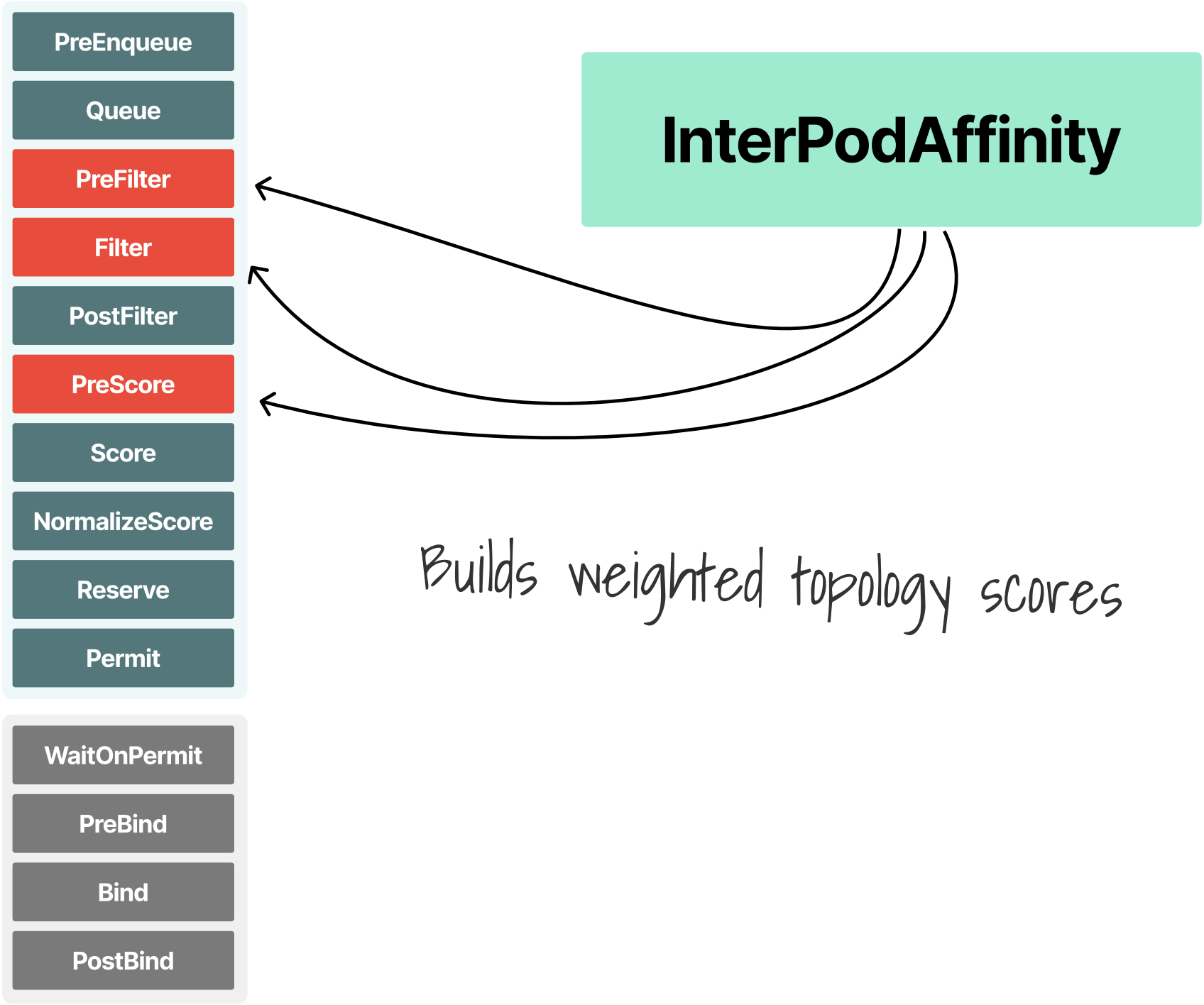
Bind

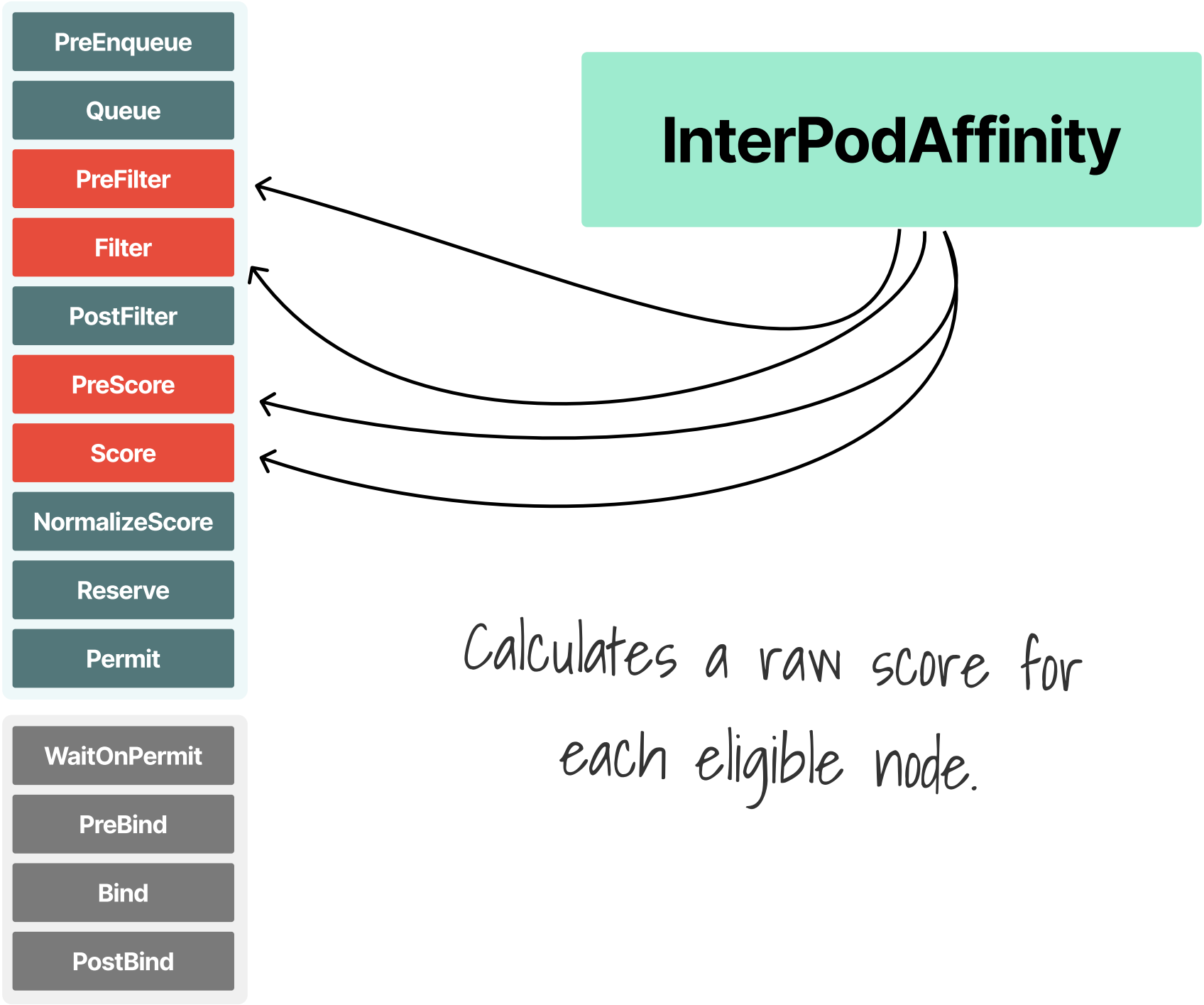
PostBind

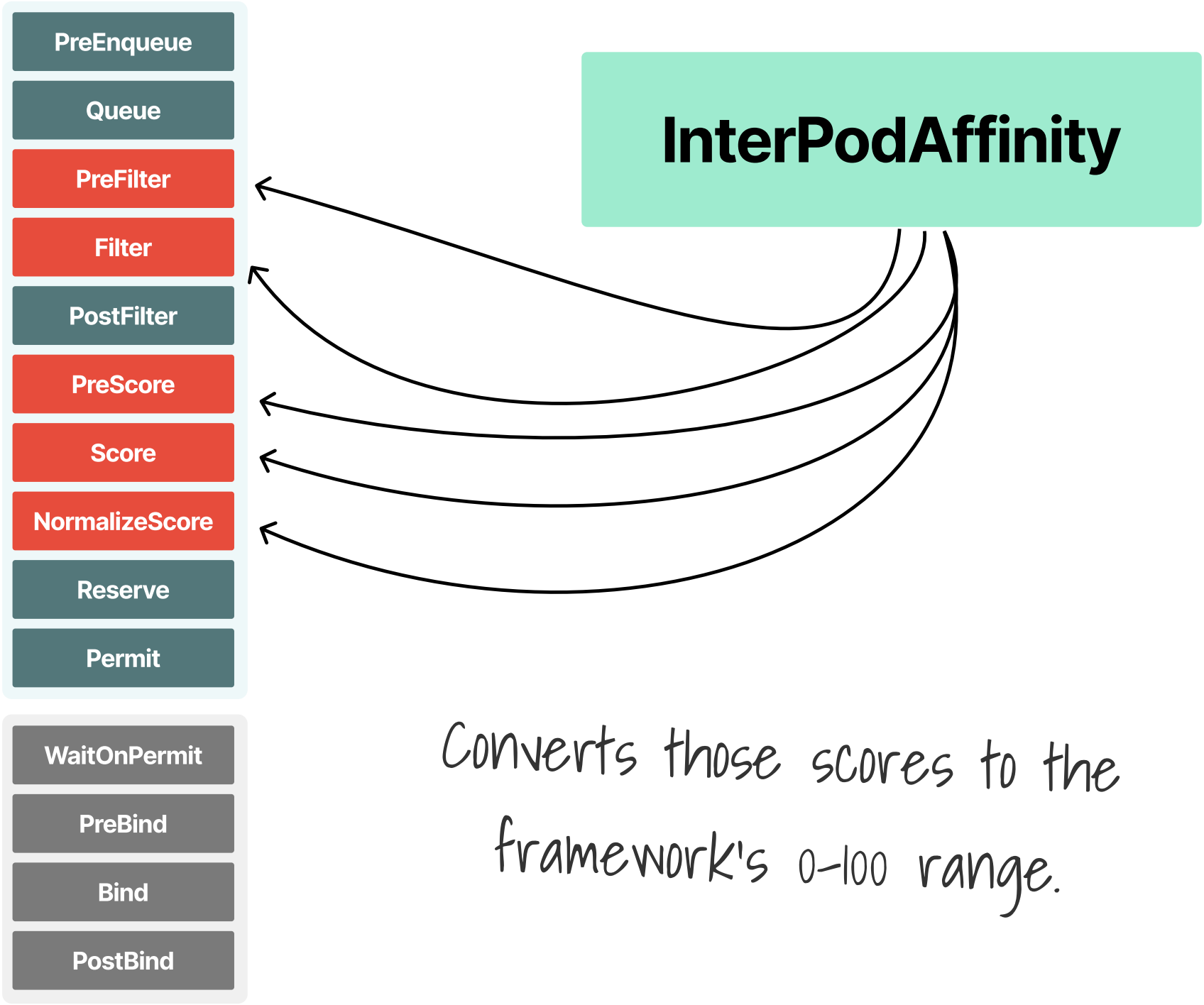
InterPodAffinity


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
InterPodAffinity
Reads the pod affinity


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
InterPodAffinity
enforces required affinity


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
InterPodAffinity
Builds weighted topology scores


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
InterPodAffinity
Calculates a raw score for each eligible node.


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
InterPodAffinity
Converts those scores to the framework's 0-100 range.

Taints and tolerations

04

Nodes can repel Pods

Taints and tolerations

Nodes can repel Pods

Similar to nodeSelector

Taints and tolerations

Nodes can repel Pods

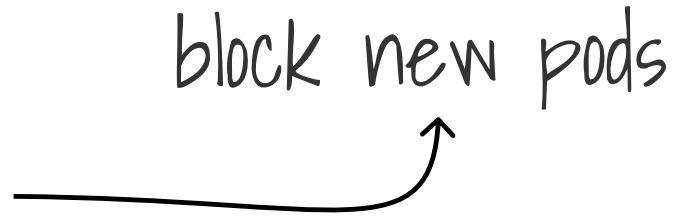
Similar to nodeSelector

Taints are key=value:effect

Taints and tolerations

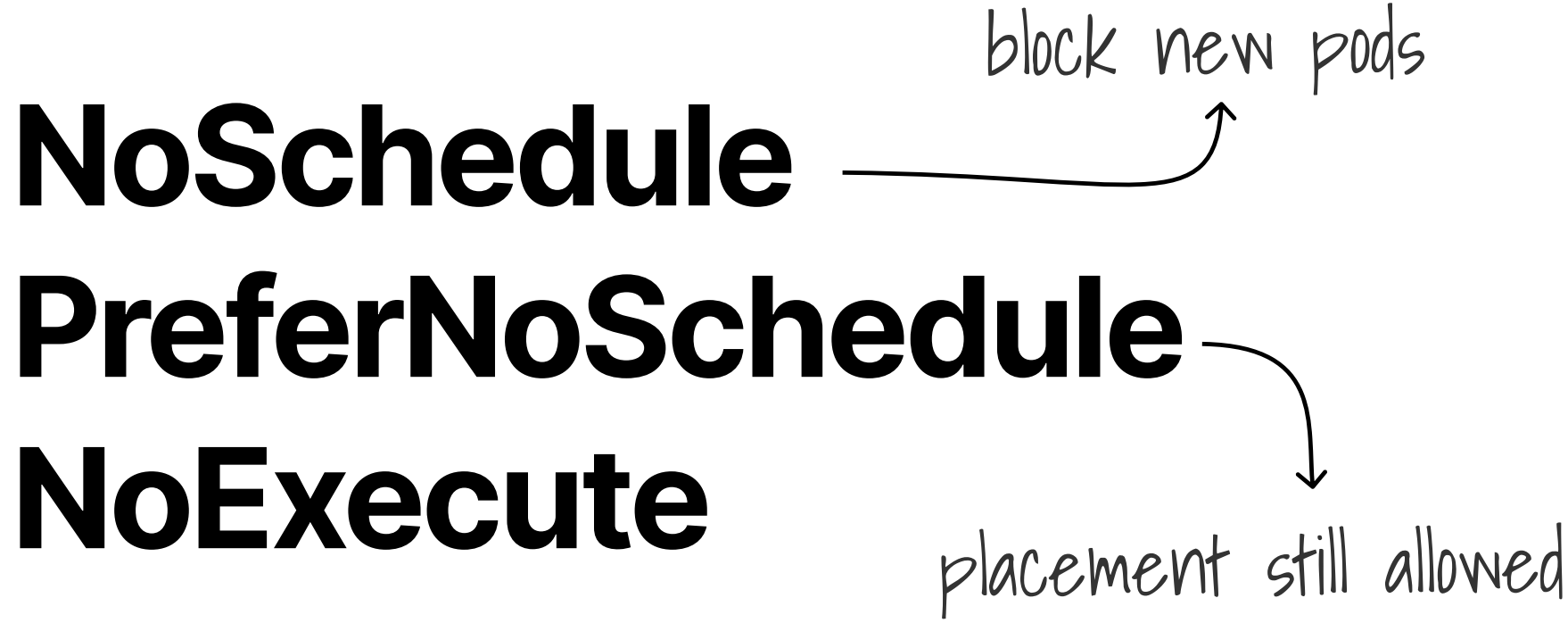
NoSchedule
PreferNoSchedule
NoExecute

Taints and tolerations

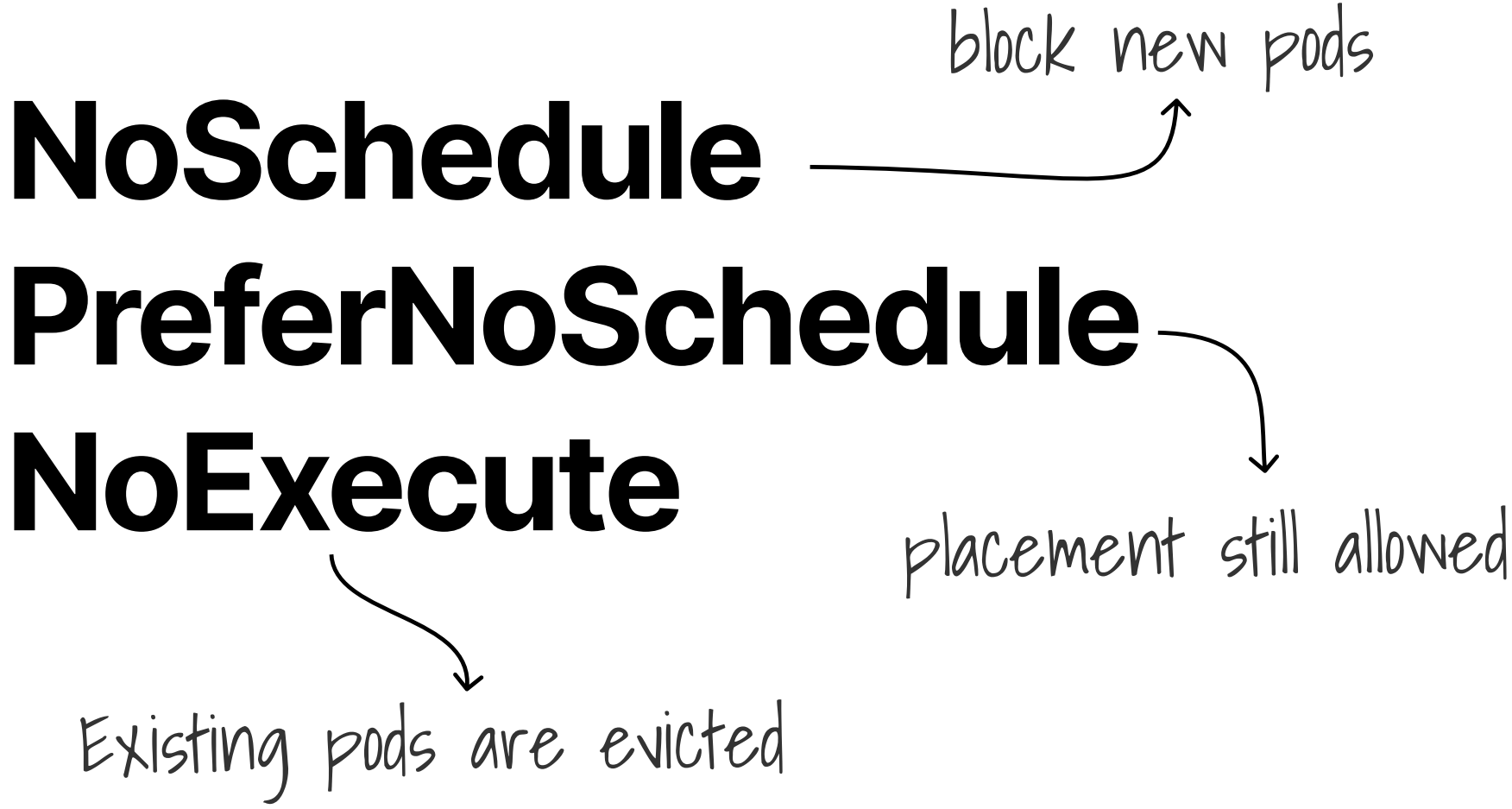


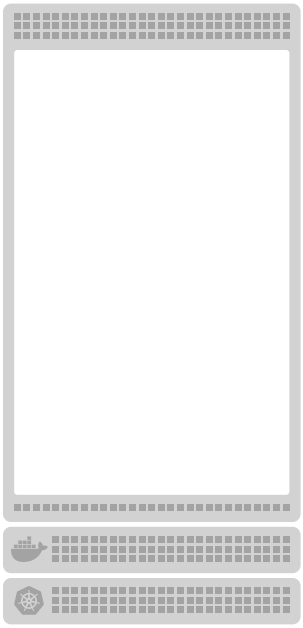
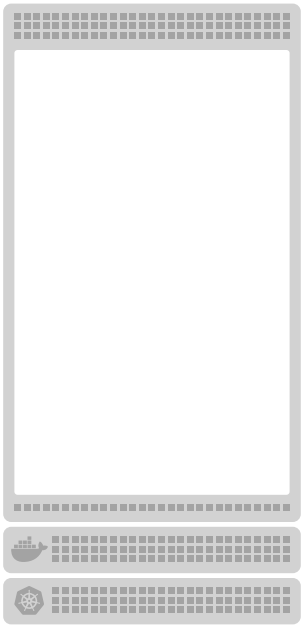
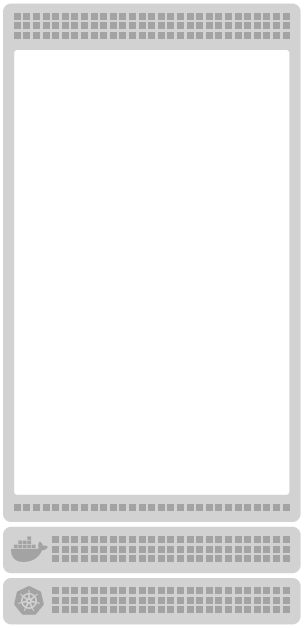
NoSchedule
PreferNoSchedule
NoExecute

Taints and tolerations



Taints and tolerations

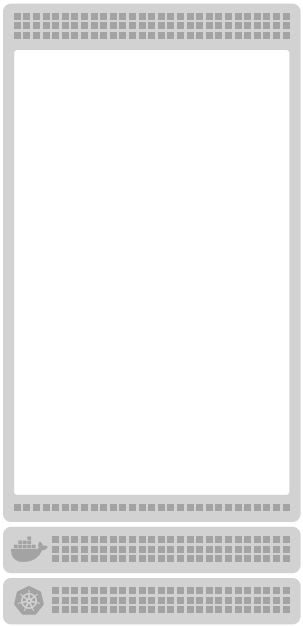


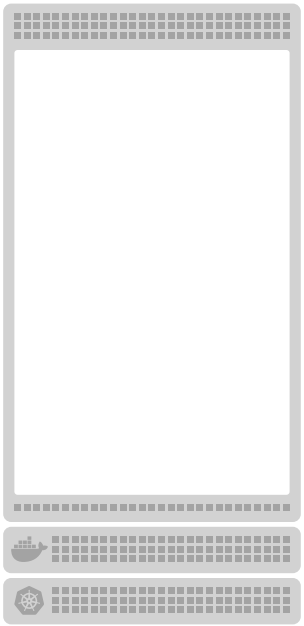


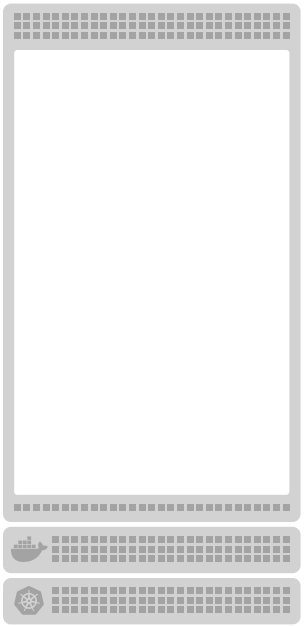
```
~$ kubectl taint nodes \
 worker1 app=yellow:NoSchedule
```

app=yellow:NoSchedule



app=green:PreferNoSchedule

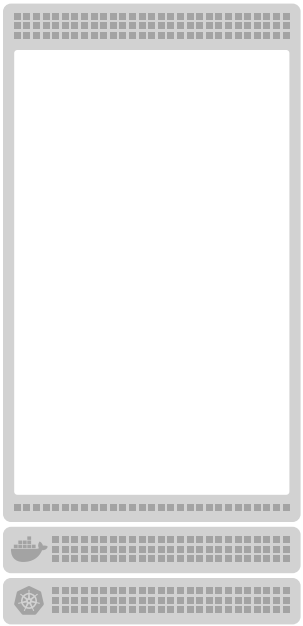




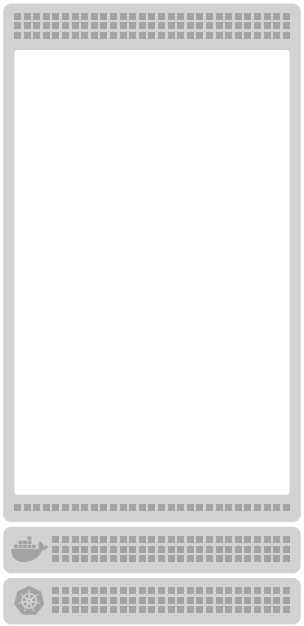


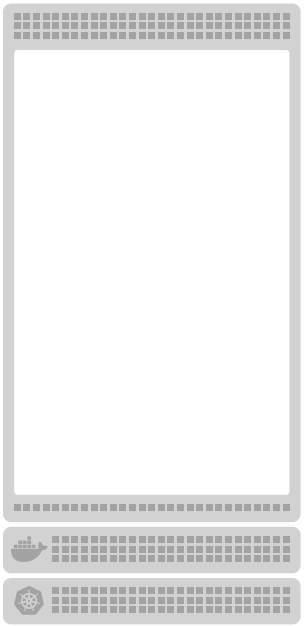
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: red-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: red
  template:
    metadata:
      labels:
        app: red
    spec:
      tolerations:
      - key: app
        operator: Equal
        value: red
      containers:
      - name: nginx
        image: nginx:1.7.9
```

app=yellow:NoSchedule

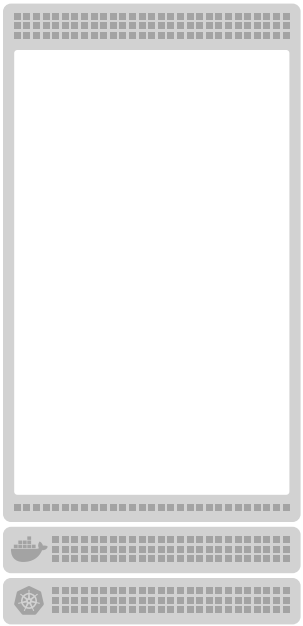


app=green:PreferNoSchedule

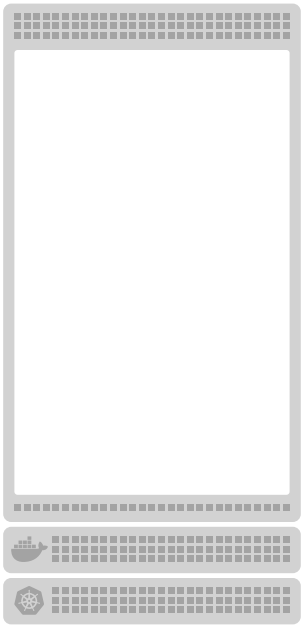


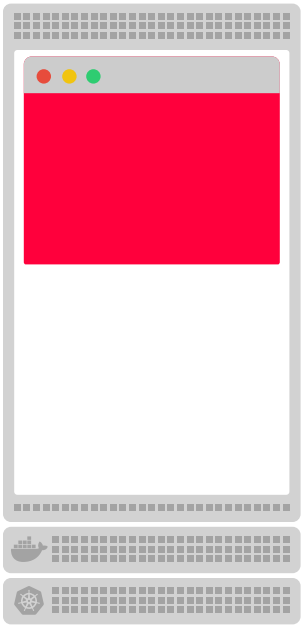


app=yellow:NoSchedule



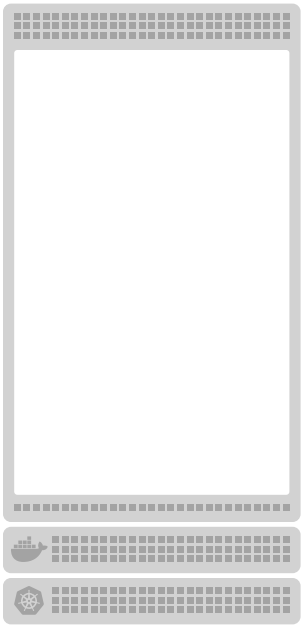
app=green:PreferNoSchedule



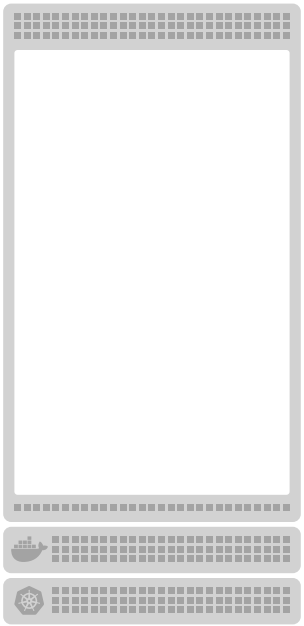


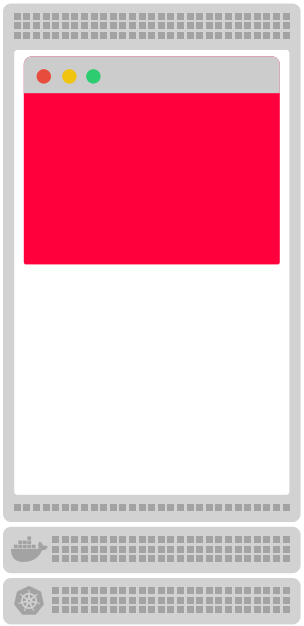
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: red-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: red
  template:
    metadata:
      labels:
        app: red
    spec:
      tolerations:
      - key: app
        operator: Equal
        value: red
      containers:
      - name: nginx
        image: nginx:1.7.9
```

app=yellow:NoSchedule

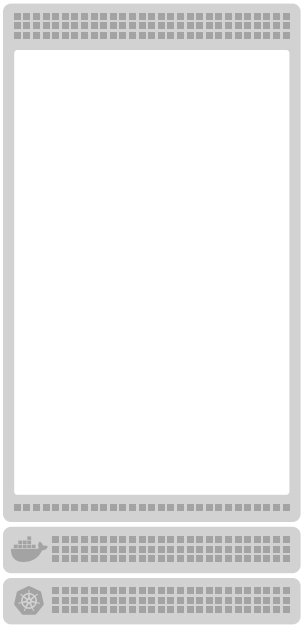


app=green:PreferNoSchedule

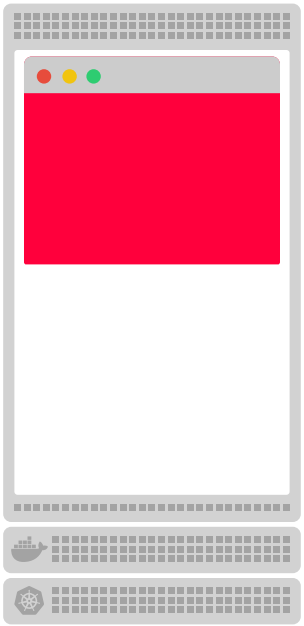


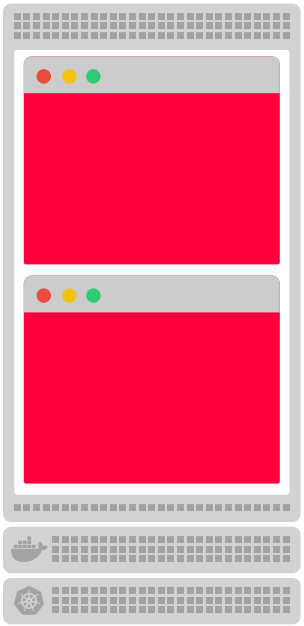


app=yellow:NoSchedule



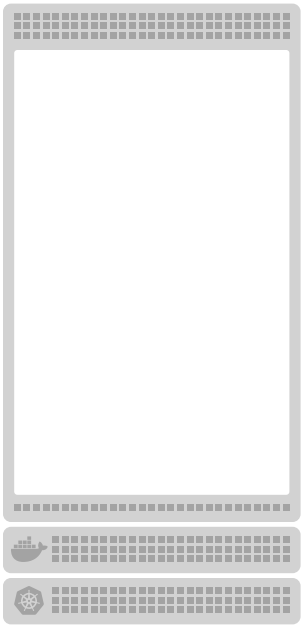
app=green:PreferNoSchedule



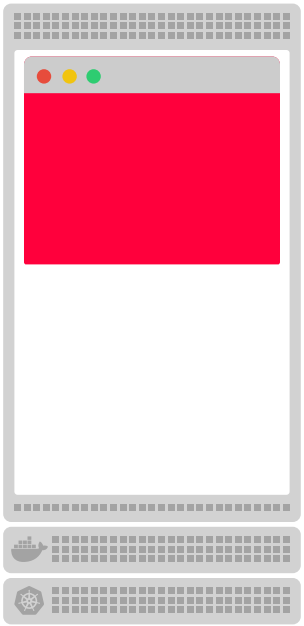


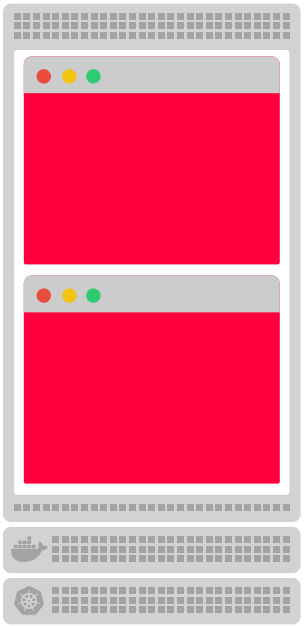
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: red-deployment
spec:
  replicas: 5
  selector:
    matchLabels:
      app: red
  template:
    metadata:
      labels:
        app: red
    spec:
      tolerations:
      - key: app
        operator: Equal
        value: red
      containers:
      - name: nginx
        image: nginx:1.7.9
```

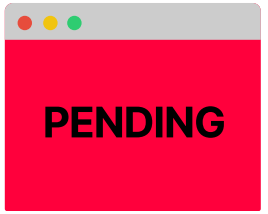
app=yellow:NoSchedule



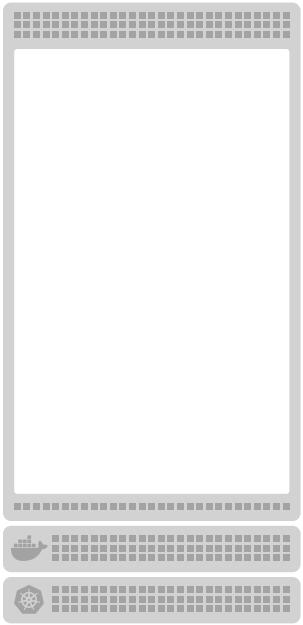
app=green:PreferNoSchedule



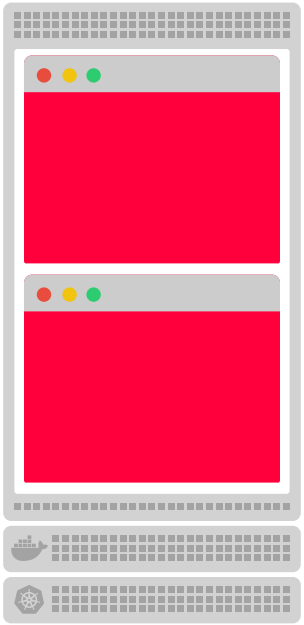


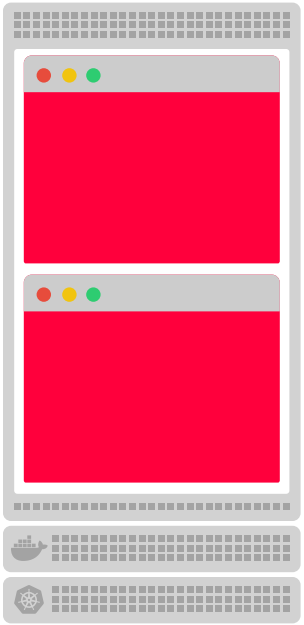


app=yellow:NoSchedule



app=green:PreferNoSchedule

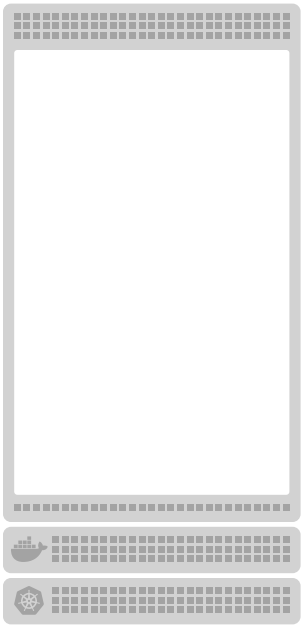


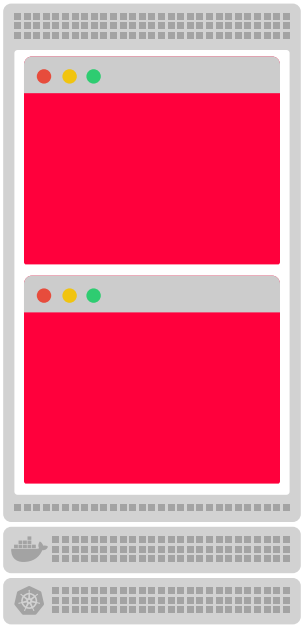




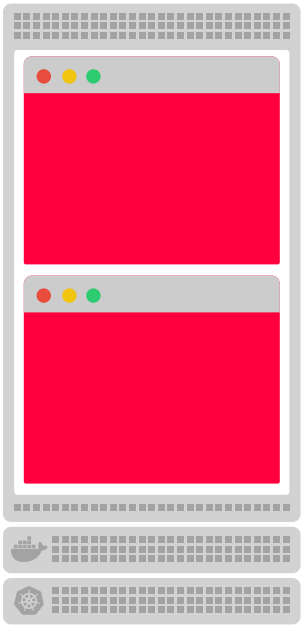
```
~$ kubectl taint nodes \
 worker3 app=green:NoExecute
```

app=yellow:NoSchedule





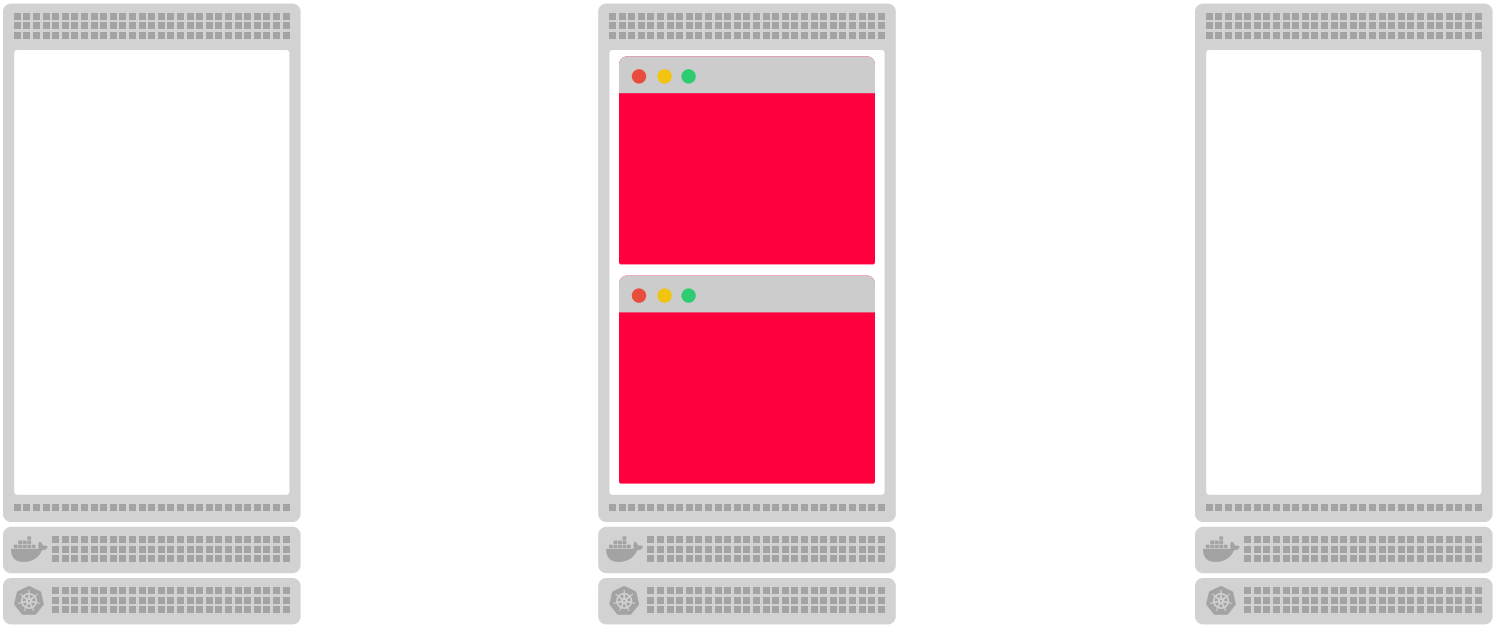
app=green:NoExecute



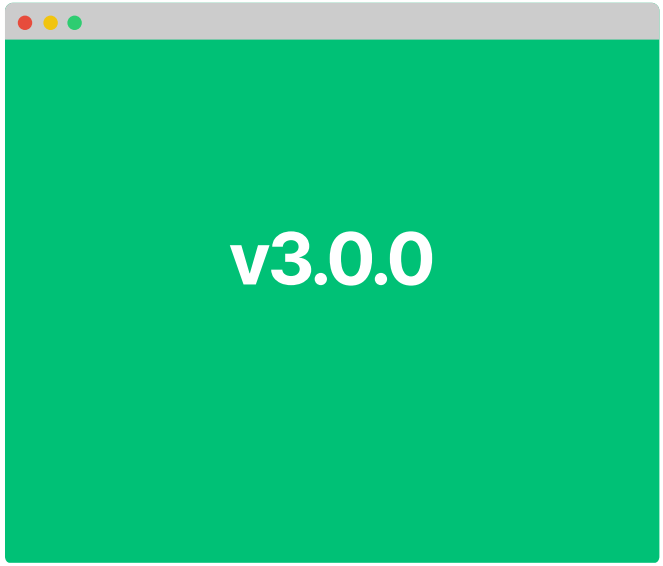
app=green:PreferNoSchedule

app=yellow:NoSchedule

app=green:NoExecute



app=green:PreferNoSchedule



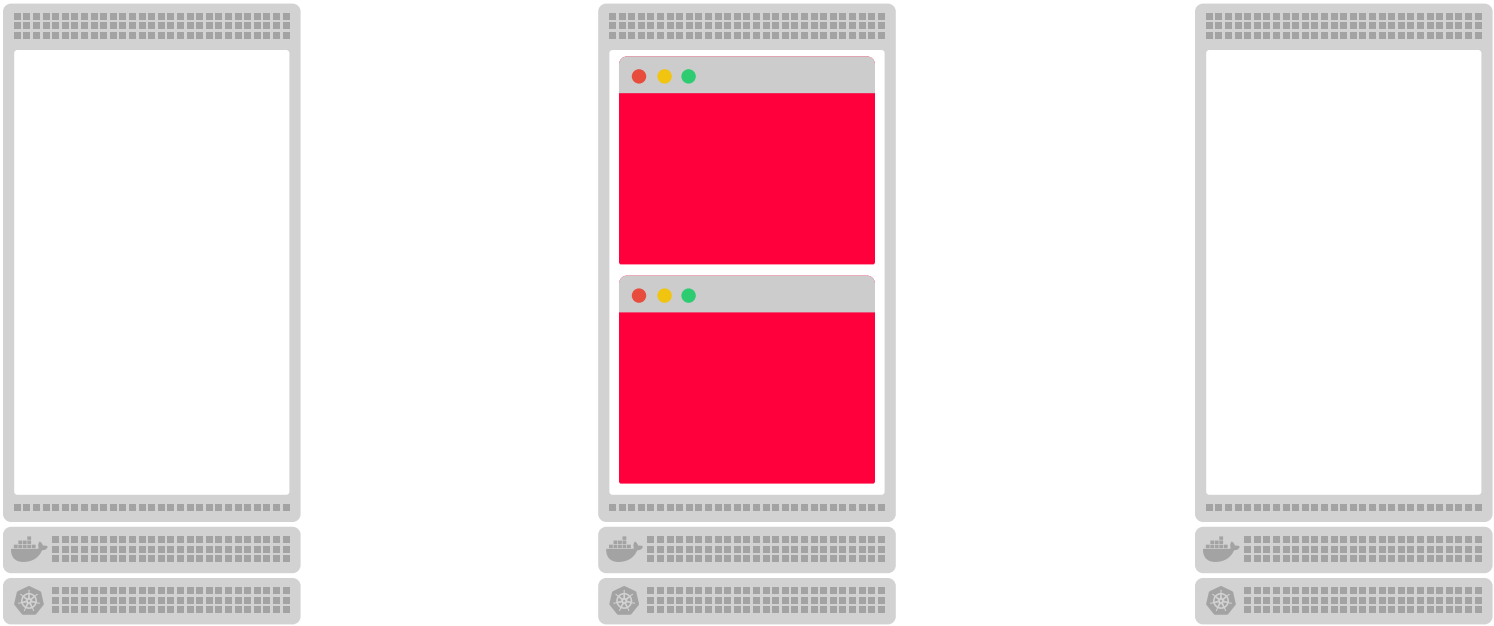
TOLERATE

app: green

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: green-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: green
  template:
    metadata:
      labels:
        app: green
    spec:
      tolerations:
      - key: app
        operator: Equal
        value: green
      containers:
      - name: nginx
        image: nginx:1.7.9
```

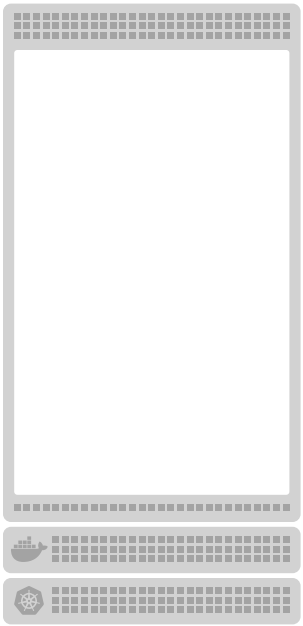
app=yellow:NoSchedule

app=green:NoExecute

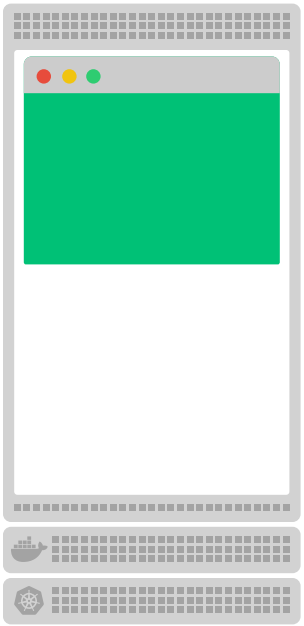


app=green:PreferNoSchedule

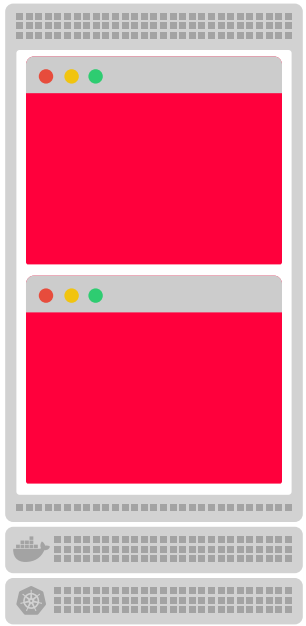
app=yellow:NoSchedule



app=green:NoExecute

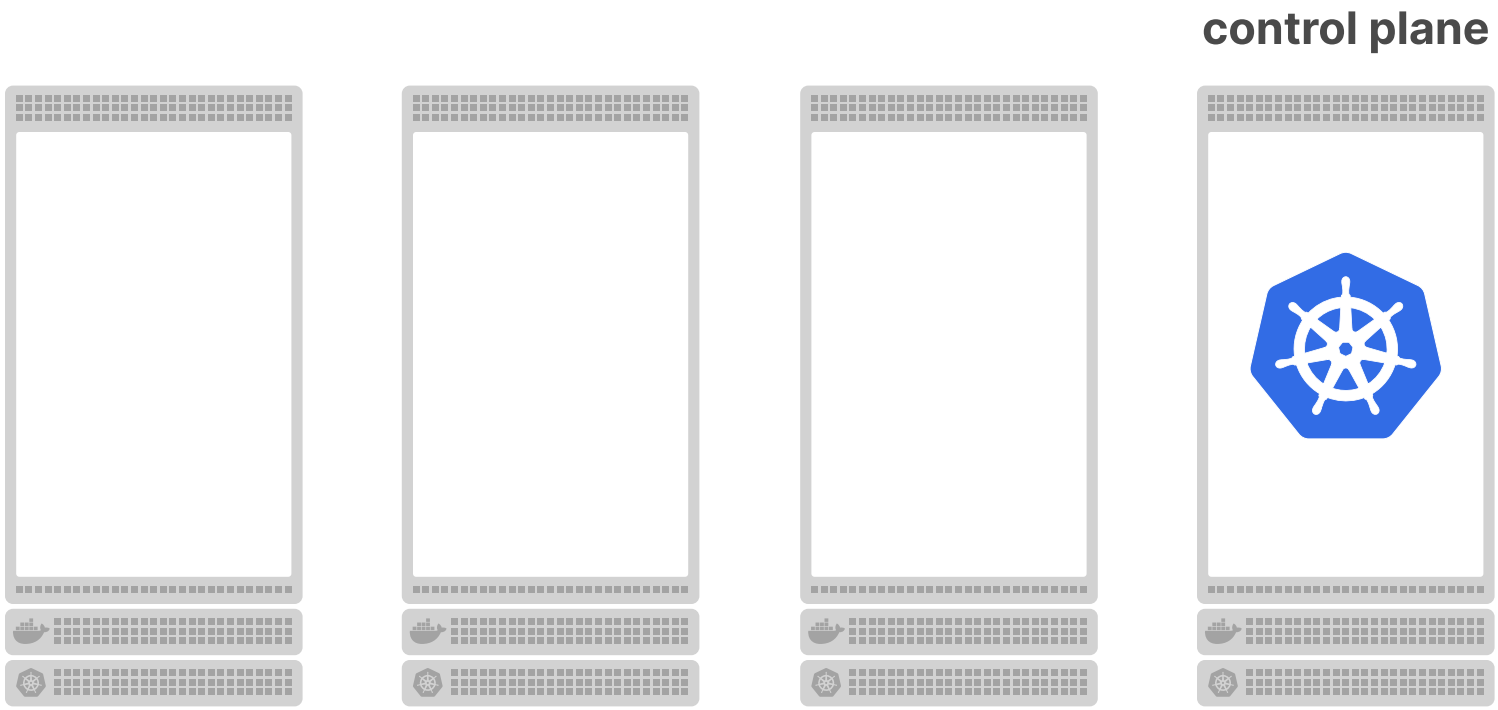


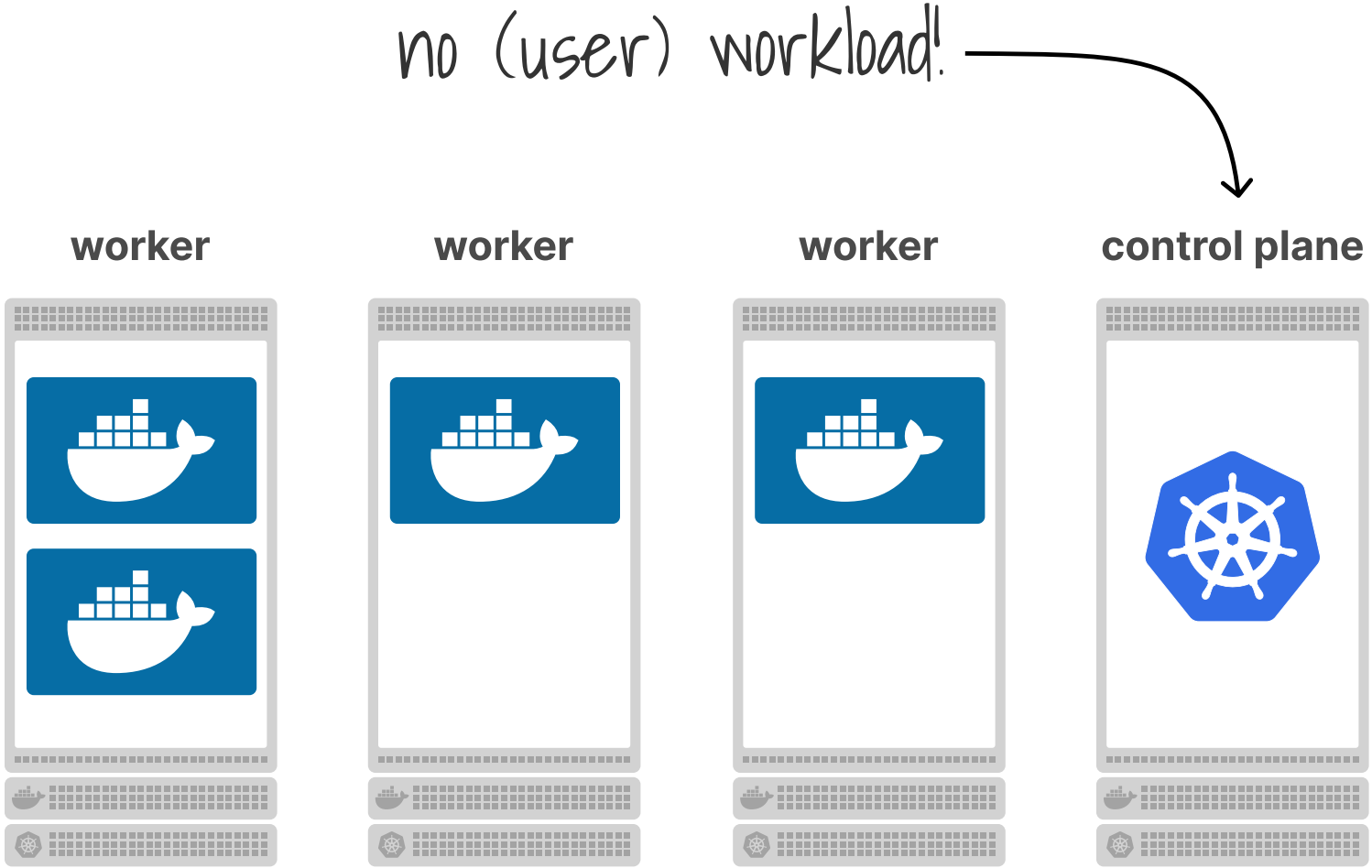
app=green:PreferNoSchedule





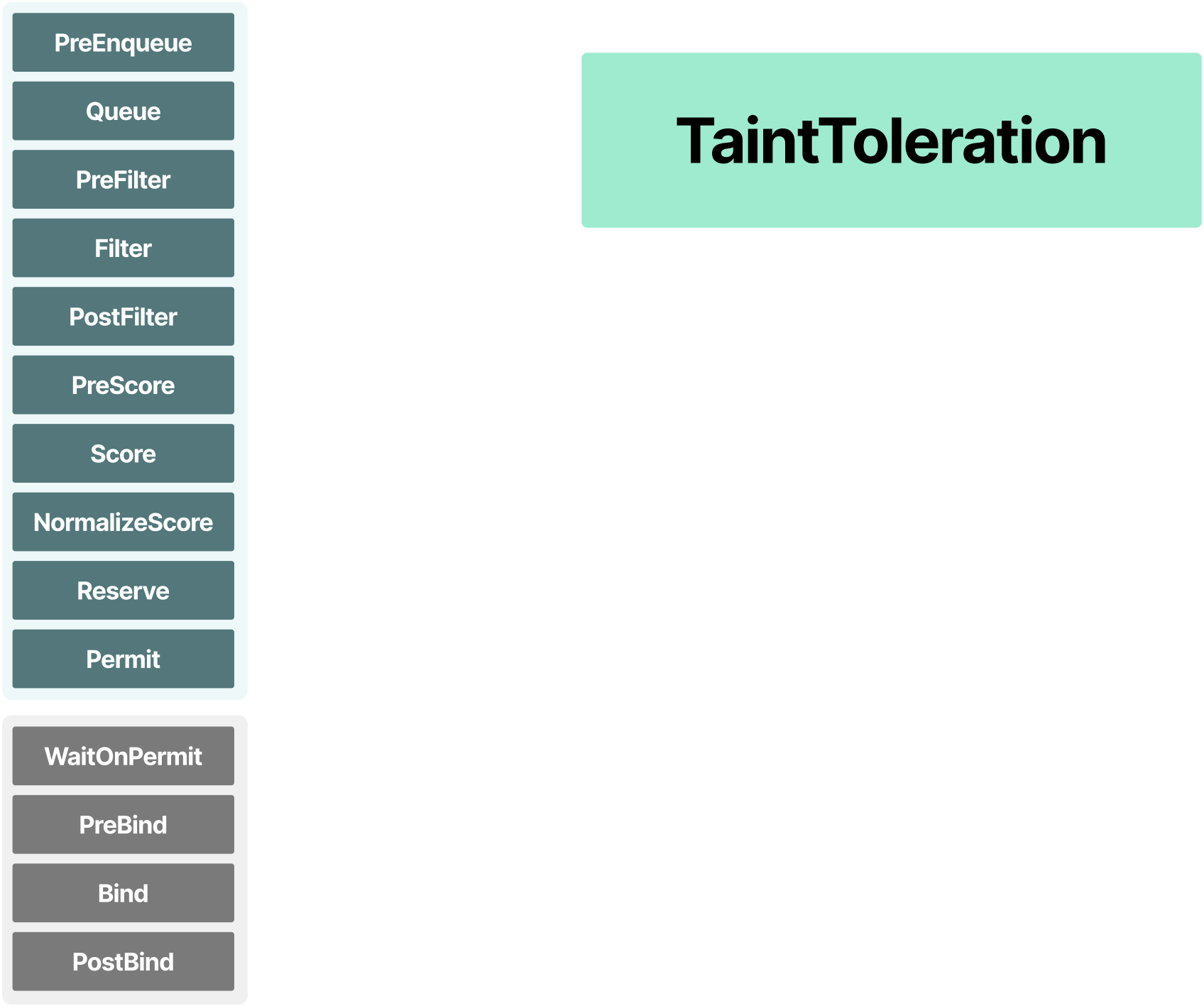
```
~$ kubectl taint nodes master \
node-role.kubernetes.io/control-plane:NoSchedule
```

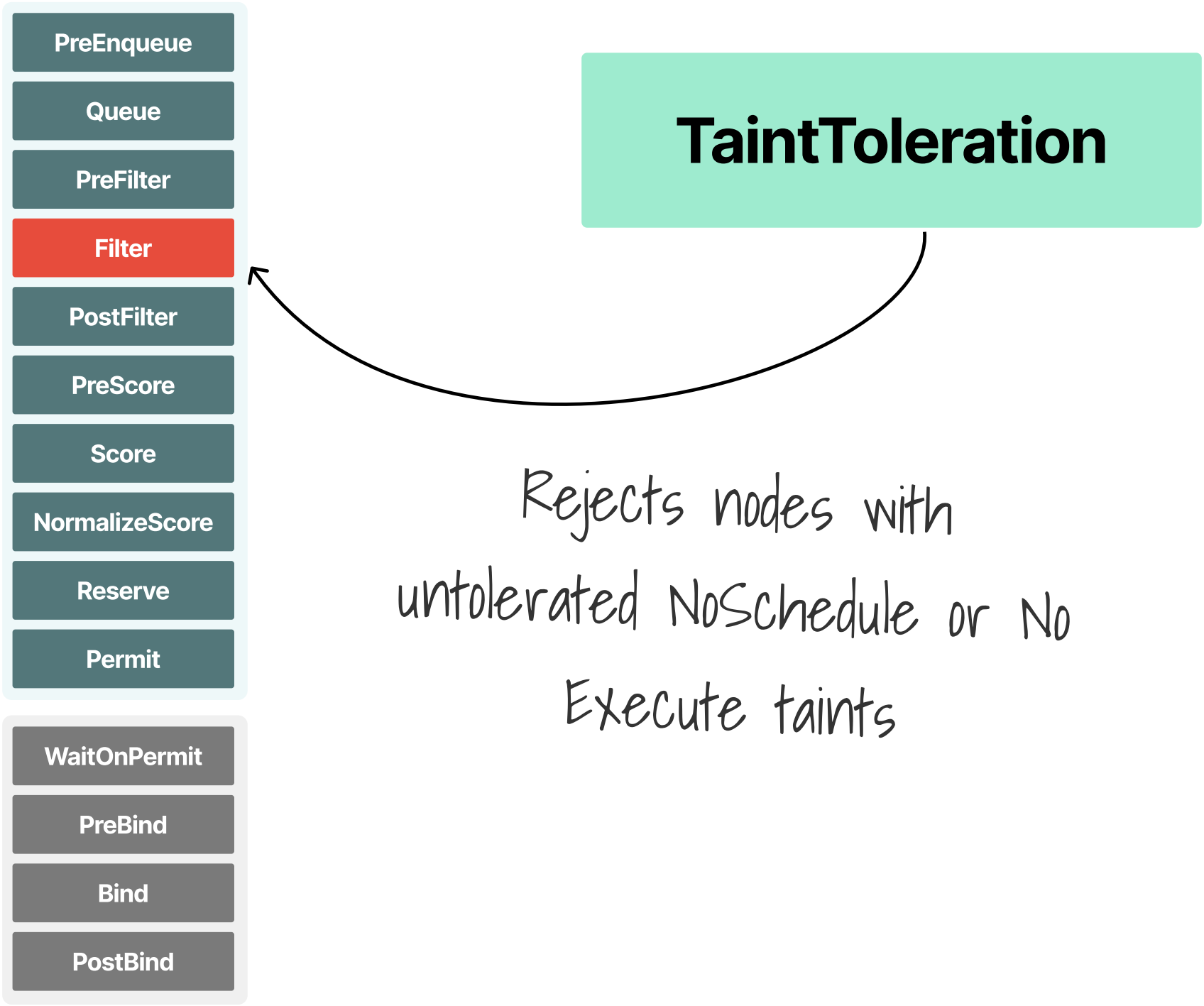

control plane

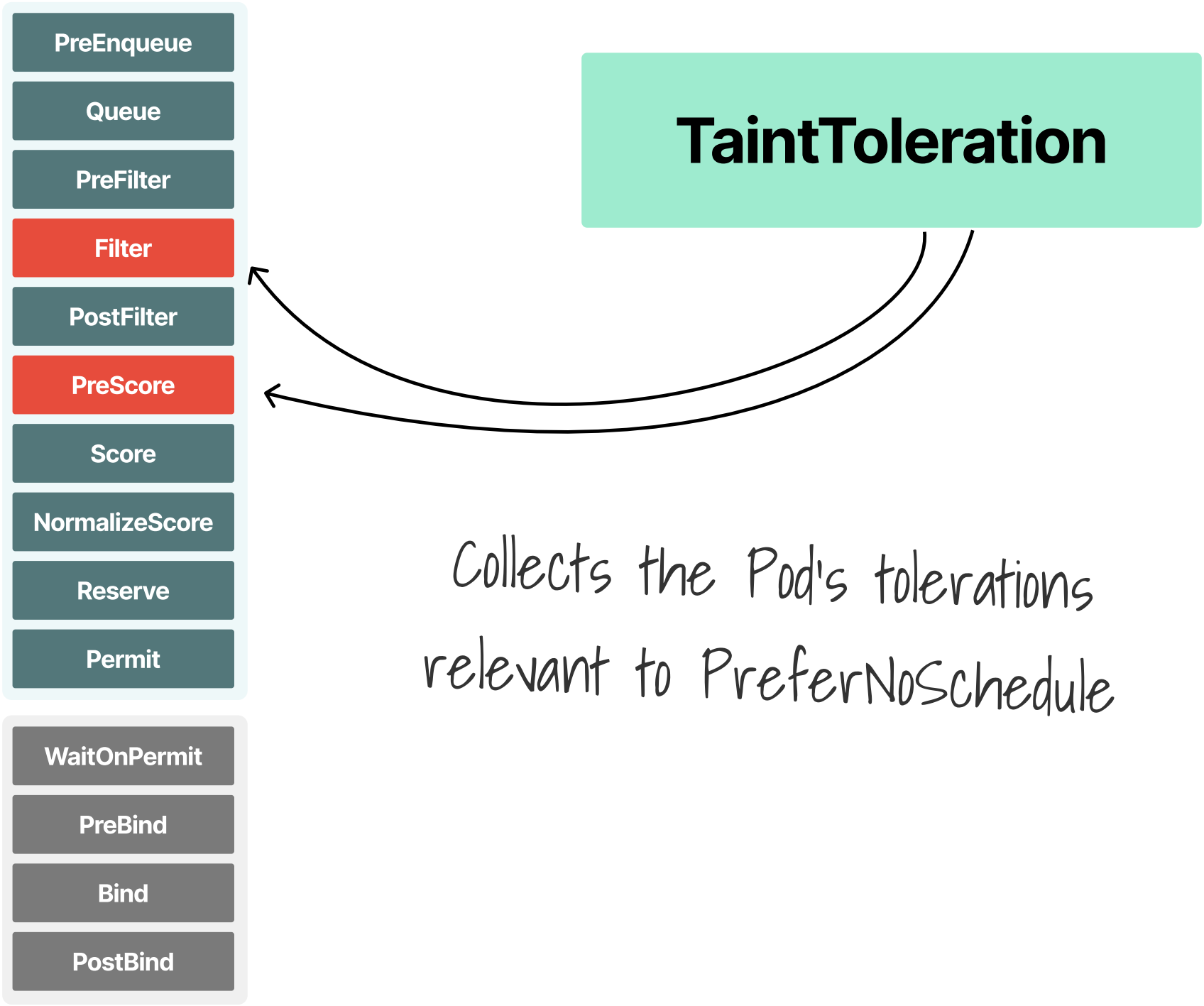

no (user) workload!
worker
worker
worker
control plane

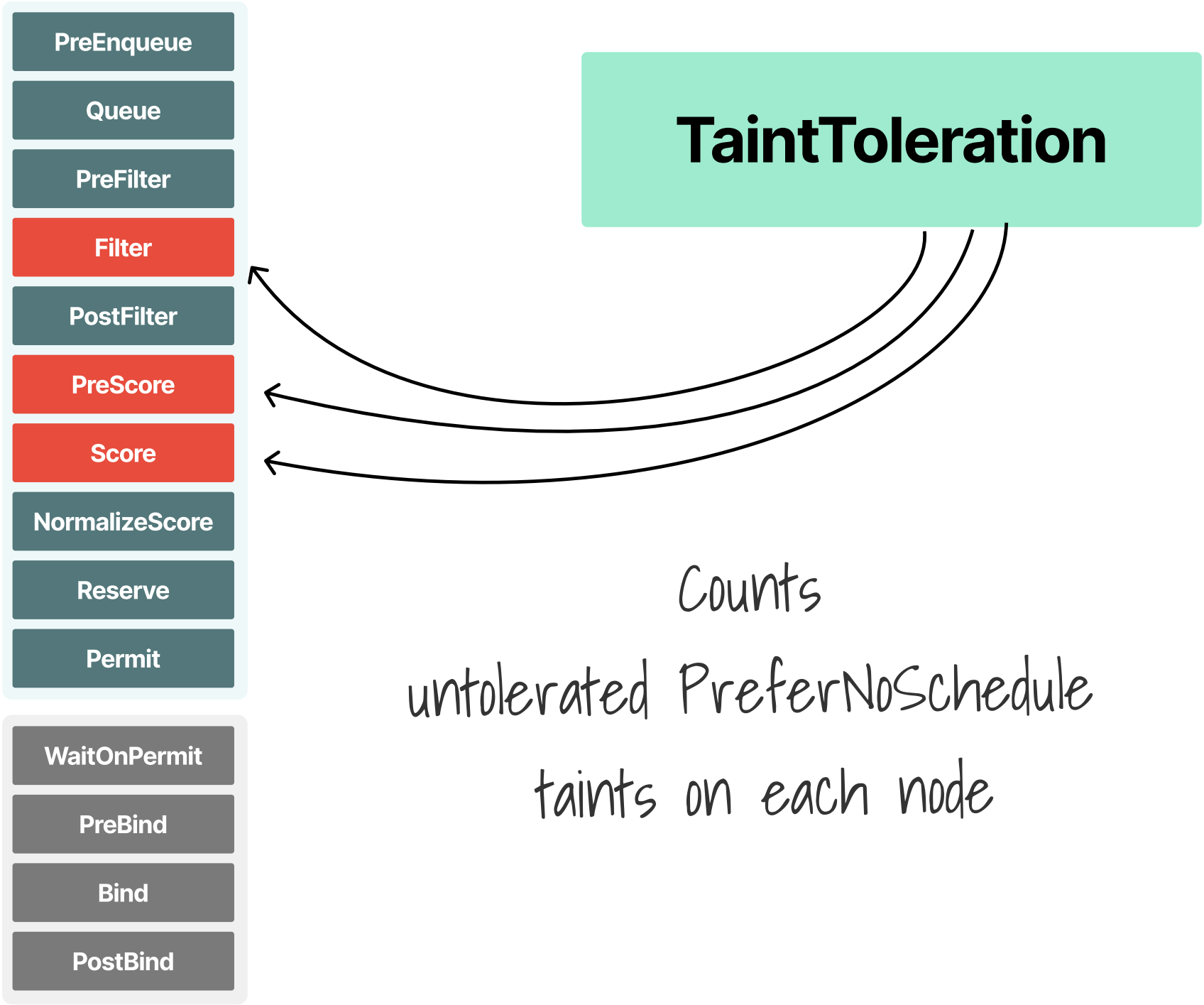
Taints and toleration plugin

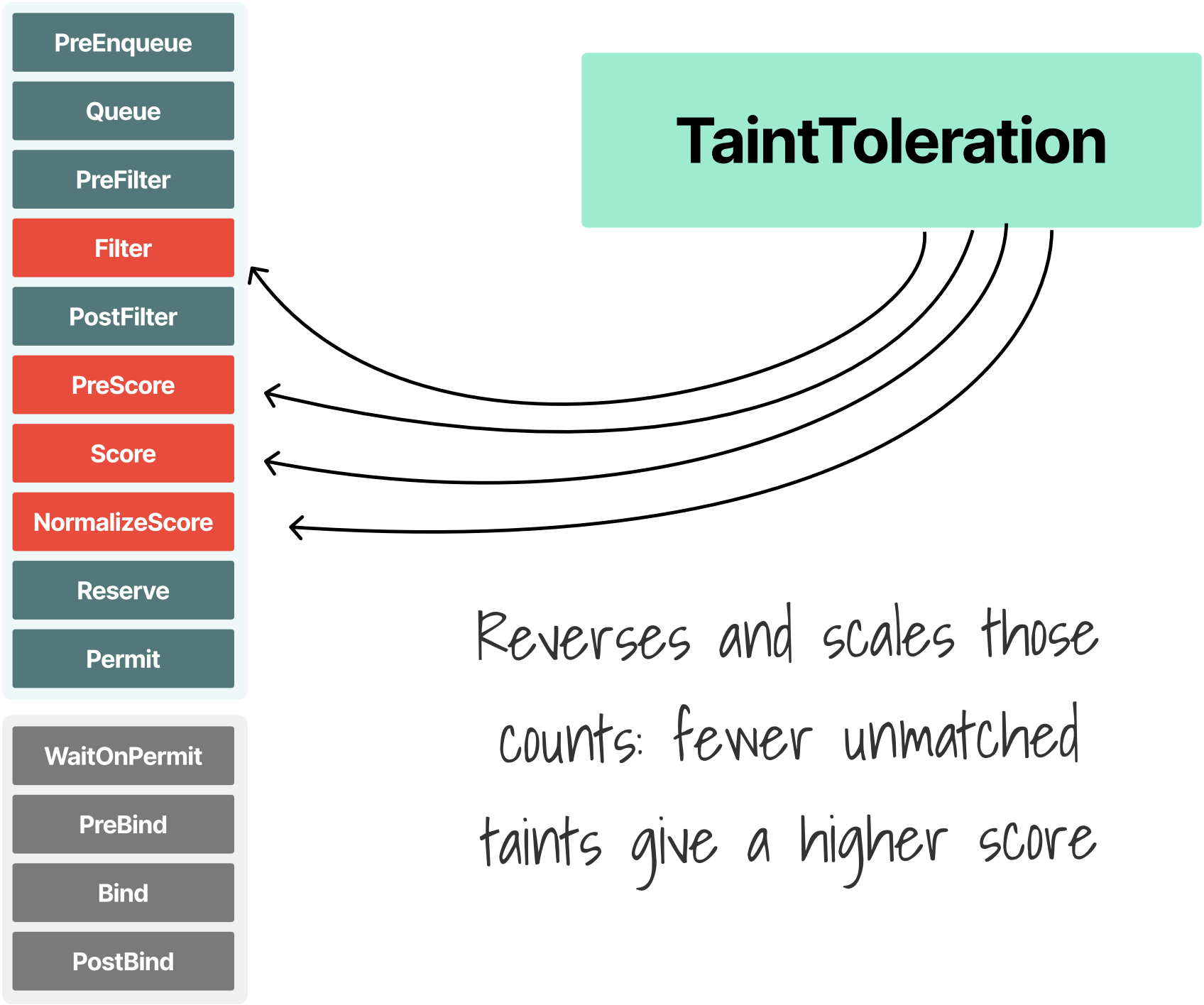
04


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
TaintToleration


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
TaintToleration
Rejects nodes with untolerated NoSchedule or No Execute taints


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
TaintToleration
Collects the Pod's tolerations
relevant to PreferNoSchedule


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
TaintToleration
Counts
untolerated PreferNoSchedule
taints on each node


PreEnqueue
Queue
PreFilter
Filter
PostFilter
PreScore
Score
NormalizeScore
Reserve
Permit
WaitOnPermit
PreBind
Bind
PostBind
TaintToleration
Reverses and scales those counts: fewer unmatched taints give a higher score

Recap

Recap

1. Scheduling and binding

Recap

1. Scheduling and binding

2. Filters and scores

Recap

1. Scheduling and binding

2. Filters and scores

3. **Node Affinity**

Recap

1. Scheduling and binding
2. Filters and scores
3. Node Affinity
4. **Pod affinity/anti-affinity**

Recap

1. Scheduling and binding
2. Filters and scores
3. Node Affinity
4. Pod affinity/anti-affinity
5. **Taints and tolerations**

Thank you.